

Vision Transformer-Based Model can Optimize Curative-Intent Treatment for Patients with Recurrent Hepatocellular Carcinoma

Corresponding Author: Professor Tianan Jiang

This file contains all reviewer reports in order by version, followed by all author rebuttals in order by version.

Version 0:

Reviewer comments:

Reviewer #1

(Remarks to the Author)

The study may be clinically relevant as it attempts to address the critical issue of patient selection for liver surgery or thermal ablation in the context of initial (i) and recurrent (r) hepatocellular carcinoma (HCC). However, several methodological issues need to be addressed to strengthen the study's validity and reliability.

1. It is unclear how observer and internal technical variability regarding ultrasound (US) and contrast-enhanced ultrasound (CEUS) were addressed in the study. Please provide detailed information on how variability among different observers and technical settings was managed. This could include standardization procedures, training protocols for observers, inter-observer reliability assessments, and calibration methods used to ensure consistency across examinations.
2. Provide more detailed descriptions of the preprocessing steps for the image data, including techniques for noise reduction and standardization. Discuss how missing data and artifacts were handled during preprocessing.
3. There is a discrepancy in the clinical settings between the development and validation sets, with the development set focusing on non-treated HCC and the validation set on recurrent HCC. Explain the rationale behind using different clinical settings for development and validation. Address how this difference might affect the model's performance and generalizability. Consider discussing any steps taken to mitigate potential biases introduced by this discrepancy.
4. Description on page 5 lack clarity regarding which datasets were used for development and validation. Reword the description on page 5 to clearly specify which datasets were designated for external testing, particularly lines 121 and 122 are confusing as it seems part of the development cohort was used for external validation. While Figure 5 is very clear, the accompanying text needs to be more precise.
5. A cohort from center A was included as an external validation set, but the patients were treated, and the scans were performed at the same center as the development cohort, which challenges the definition of "external validation." To ensure proper generalization testing, consider including results from only truly external centers (e.g., centers B and C).

(Remarks on code availability)

Reviewer #2

(Remarks to the Author)

Noteworthy results:

- Model Performance and Comparative Advantage: HEROVision, which was developed based on a Vision Transformer (ViT) structure, showed high accuracy in predicting the prognosis for recurrent hepatocellular carcinoma (rHCC). In external testing cohorts, HEROVision-TA (thermal ablation) achieved a 2-year concordance index (C-index) of 0.72, while

HEROVision-SR (surgical resection) reached 0.74. The 2-year area under the curve (AUC) values were 0.78 for TA and 0.82 for SR. These results outperformed the six guideline staging systems, which had 2-year C-index values between 0.51 and 0.58 and 2-year AUC values between 0.51 and 0.61. Additionally, integrating HEROVision into these staging systems improved their 2-year progression-free survival (PFS) prediction accuracy.

- Risk Stratification: HEROVision effectively categorized patients into high- and low-risk groups. In the external testing cohort, the 2-year PFS rates for low-risk patients were 79.0% for TA and 75.8% for SR, compared to 42.2% for TA and 31.9% for SR in the high-risk groups. HEROVision identified that 26.0% of high-risk patients could potentially improve their prognosis by switching from TA to SR.

- Contribution to the Field and Clinical Impact: HEROVision represents a potential advancement in the treatment of HCC by combining a Vision Transformer-based model with multimodal imaging data, including ultrasound (US) and magnetic resonance imaging (MRI), along with clinical variables. This approach enhances risk stratification and treatment decision-making for patients with rHCC and could surpass the current guideline staging systems. By providing personalized treatment recommendations based on preoperative data, HEROVision addresses a potential gap in clinical guidelines. The model's precision in identifying high-risk patients who could benefit from changing their treatment underscores its practical value in enhancing therapeutic strategies for early-stage rHCC.

Significance:

Although superior performance compared to existing staging systems is demonstrated, using deep learning methods in HCC has been described many times. Previous studies have explored machine learning and deep learning techniques to predict early recurrence and optimize treatment for HCC. For instance, other models have successfully used multiparametric MRI and CT imaging combined with clinical data to predict early recurrence and optimize treatment decisions: "Zhan, G., Wang, F., Wang, W., Li, Y., Chen, Q., Hu, H., & Chen, Y.-W. (2023). A transformer-based model for preoperative early recurrence prediction of hepatocellular carcinoma with multi-modality MRI. *Computer Vision – ACCV 2022 Workshops*, 185–194. https://doi.org/10.1007/978-3-031-27066-6_13". The novelty of HEROVision lies in its specific implementation of a specific Vision Transformer-based network and its integration with clinical decision-making tools, which could potentially enhance individualized treatment plans. The paper is therefore moderately innovative. Since the data are not publicly available, benchmarking by other groups is not possible which further dampens enthusiasm.

Methodology:

The paper uses a large multi-institutional dataset and describes models for both thermal ablation and surgical patients, which is a strength. The inclusion of 8946 patients from three geographically distinct centers in China is impressive.

There are issues noted below that may limit the generalizability of the models, which drive the enthusiasm for the paper down.

- The paper fails to compare its results with state-of-the-art (SOTA) Vision Transformer-based models that support multimodal data or publicly available datasets. Vision Transformers rely on the quality and diversity of the dataset because their performance is tied to the patterns they learn during training. Without testing on different publicly available datasets, it is difficult to establish the generalizability and robustness of the proposed approach. Moreover, the diversity of the training data is important for training Vision Transformers. Any bias in the training data, such as race or gender bias, can affect the model's predictions.

- The introduction does not adequately justify the use of Vision Transformers. It does not contrast their advantages over Convolutional Neural Networks (CNNs), despite existing literature highlighting CNNs' benefits in medical image-based diagnosis, such as in the study "Deep Learning Methods in Medical Image-Based Hepatocellular Carcinoma Diagnosis: A Systematic Review and Meta-Analysis" by Wei et al., 2023. Additionally, the Materials and Methods section lacks explanations for some of the dataset division and the rationale behind the specific cohort division ratios. This could impact the experimental setup's validity as well as the model's training process.

- The study also does not provide detailed information on sex and gender considerations, failing to comply with the SAGER guidelines (as recommended in the review instructions).

- The computational complexity of the Vision Transformer model and the potential loss of diagnostic details due to image resizing and strip partitioning limit the model's clinical feasibility. The paper does not address the variability in imaging quality or interpretation (many imaging modalities are used), which are important factors that could affect the accuracy of individualized predictions.

- Moreover, the comparisons with major guideline methods do not address inherent biases or limitations in the staging systems. This could undermine the validity of the results, as these biases can impact the model's performance. There is also a drop in performance metrics when the model is applied to the external testing cohort. This could indicate potential overfitting and limited generalizability. In addition, the statistical analysis lacks detailed justification for the selection of some specific tests and thresholds. The choice of propensity score matching (PSM) and the methods used to evaluate statistical significance are not sufficiently explained.

- The rationale behind the selection of specific variables and the chosen caliper value of 0.02 for PSM should be explicitly

justified. Furthermore, while the section effectively details the re-stratification process using the HEROVision models, a more thorough discussion on the clinical implications of these re-stratifications is necessary. Specifically, elaborating on how the re-categorization from high-risk to low-risk influences patient management and treatment outcomes would provide valuable insights into the practical applications of the findings. Also, the title of this section should be in all capital letters to be consistent with previous headers.

(Remarks on code availability)

The provided GitHub repository is well-documented. The README file offers an overview of the model's purpose, prerequisites, and instructions for installing necessary libraries and running the application. The dataset handling script (dataset.py) details the preprocessing steps, ensuring reproducibility. The model architecture (Model_hovertrans.py) is documented, including hyperparameters and training specifics, which are crucial for replicating the results. The training scripts (train_hover_SR.py and train_hover_TA.py) provide instructions and use established methods such as the Adam optimizer and Cox Loss, with callbacks for early stopping and model checkpointing. Overall, the repository's organization and detailed documentation make it user-friendly and support the reproducibility of the paper's findings.

Reviewer #3

(Remarks to the Author)

I co-reviewed this manuscript with one of the reviewers who provided the listed reports. This is part of the Nature Communications initiative to facilitate training in peer review and to provide appropriate recognition for Early Career Researchers who co-review manuscripts.

(Remarks on code availability)

Reviewer #4

(Remarks to the Author)

Comments on Vision Transformer-Based Biomarker can Optimize Curative Treatment for Patients with Recurrent Hepatocellular Carcinoma

General

This is a very eager and innovative study. The topic is relevant. The cohorts are large and I consider them large enough as they are restricted by the inclusion/exclusion criteria.

But I do have some concerns about the aim and the study design.

By narrowing the in/exclusion criteria so much, it is not surprising that the clinical classification models (guideline systems) are not performing well. By your inclusion restrictions you focus on an indication niche, which was left liberally open for individual treatment decision making based on knowing the specific patient and the possibilities of the specific medical center. It basically does not make sense to use the clinical scores and classification inside this narrow group at all.

The actual goal of your study and the algorithm is not making a decision regarding which local therapy to use or if it should be used at all but to give an outlook on the prognosis of the patient, namely the OS/PFS. So the decision derived from this information would be to treat or not to treat locally – with the alternative being systemic treatment – or TACE/TARE (which we know to perform inferior to local ablation and surgery in suitable cases). Therefore the title is misleading. You do not investigate on how to improve HCC treatment. In order to give aid in the treatment decision, you would have to include patients in whom the therapy decision was wrong or suboptimal. But how do you know in your cohorts that the treatment they received was inferior to any other treatment option? You cant know that and neither your AI algorithm.

So the whole paper should be changed from the title on into “prognostication tool in HCC to be treated by means of local therapy”.

To include just T2w, DWI and CEUS sounds surprising. How did you decide to use these specific parts of the diagnostic work-up? What do you expect the AI to read on these images? We know that T2 signal and diffusion restriction loosely correlates with the grading of the tumors. What if you provided the algorithm with the grading of the tumors instead of MRI? Or to give just SI-values/radiomics from these images instead of letting the algorithm read this on its own? Or if you run the algorithm against sets of (para)clinical parameters, which are known to determine the prognosis and outcome of patients? As said previously, the clinical guideline based categories are not designed to give this fine granulated info on prognosis.

Specific

Aim:

“This approach aimed to optimize individual treatment selection between RHR and TA for patients with early-stage rHCC in real-world scenarios.”

Well, but the algorithm was not product of the study at hand, right? So what was the actual aim of the present study? Please phrase a specific aim or hypothesis.

Methods: Did you perform any tumor segmentations on the MRI images or the US images to let the algorithm know which is

the tumor and which the surrounding liver? Did the algorithm use any anatomic information, which is the most important input a surgeon or radiologist need to make a decision regarding local treatment? For me the plausibility is lacking if these important informations were not used by the algorithm.

Fig 1:

Legend should explain all abbreviations. These abbreviations should then be used in all occasions where they appear inside the table (e.g. not hepatectomy but SR).

Inclusion: clarify by writing "... of PRETHERAPEUTIC contrast enhanced US and liver MRI within a 1-month interval"

Exclusion: Please clarify: "unqualified artefacts" – shouldn't this read: "unqualifying artefacts"?

In the external testing cohort, you include patients with resection of rHCC. How can you end up with a TA testing cohort?

(Remarks on code availability)

Version 1:

Reviewer comments:

Reviewer #1

(Remarks to the Author)

Thank you for addressing my main methodological concerns. The revisions have satisfactorily clarified the issues.

(Remarks on code availability)

I have broadly reviewed the provided code for structure. Although I did not run it, it appears well-documented and aligns with the methodology described in the manuscript.

Reviewer #3

(Remarks to the Author)

I co-reviewed this manuscript with one of the reviewers who provided the listed reports. This is part of the Nature Communications initiative to facilitate training in peer review and to provide appropriate recognition for Early Career Researchers who co-review manuscripts.

(Remarks on code availability)

The provided GitHub repository is well-documented, with the README file offering an overview of the model's purpose, prerequisites, and instructions for installing necessary libraries and running the application. The dataset handling script (dataset.py) details the preprocessing steps, ensuring reproducibility. The model architecture (Model_hovertrans.py) is thoroughly documented, including hyperparameters and training specifics, which are crucial for replicating the results. The training scripts (train_hover_SR.py and train_hover_TA.py) provide clear instructions and utilize established methods such as the Adam optimizer and Cox Loss, with callbacks for early stopping and model checkpointing. Overall, the repository is well-organized, making it user-friendly and supporting the reproducibility of the paper's findings.

Reviewer #5

(Remarks to the Author)

I serve a new reviewer for the second round of review of this article.

I have been able to appreciate the additional work done by the authors in respond to the first round of comments.

The methodology is comprehensive and employs state-of-the-art deep learning techniques. The use of multimodal imaging data and propensity score matching enhances the study's validity. The decision to validate the HEROVision model on external datasets is a significant strength, providing evidence of generalizability. Nonetheless, some concerns include:

- Training on initial hepatocellular carcinoma (iHCC) data for application to rHCC may introduce biases, despite biological similarities between the two.
- The geographic and demographic variability of datasets is noted but not deeply analyzed for potential biases
- While the model architecture is described, details on hyperparameter optimization and model interpretability are limited.
- The explainability of the models is limited. Did the authors attempt to understand why the model classifies patients into low- or high-risk categories? Providing insights into this aspect would enhance interpretability and clinical relevance.
- As noted by a previous reviewer, the study focuses exclusively on patients who are candidates for both surgical resection and thermal ablation, excluding those eligible for other treatment modalities. This limitation should be addressed or acknowledged in the discussion.
- The most stimulating part of the study is the comparison between the two models, which may suggest which treatment is better. However, the models do not provide clear guidance in cases where both classify patients into the same risk group. This limitation should be explored further.
- The authors should focus more on discrepant cases, where one model reclassifies a patient into a different risk group.

Detailed characteristics of these patients and their tumors should be provided to better understand why they were initially allocated to the “wrong” group and how reclassification impacts outcomes.

- Current staging systems, such as the BCLC, are not designed for recurrent disease. Therefore, it is not surprising that they perform less effectively than the proposed models. This limitation should be explicitly discussed or mentioned in the manuscript.

- Minor comment: The term "curative-intent" is preferable to "curative" for accuracy and consistency in describing the therapeutic goals.

- Additionally, CEUS is rarely used as a surveillance tool in clinical practice, whereas CT is more commonly employed. This discrepancy should be addressed in the manuscript, as the lack of discussion on CT may limit the applicability of the study's findings.

Overall, the study provides valuable insights into the use of advanced deep learning models for optimizing rHCC treatment. However, revisions are necessary to address concerns. With these improvements, the manuscript would be a strong candidate for publication.

(Remarks on code availability)

Version 2:

Reviewer comments:

Reviewer #1

(Remarks to the Author)

My comments and suggestions were already addressed in the previous version. In this revision, the authors have also satisfactorily addressed the pending issues raised by Reviewer 2.

I have no further comments.

(Remarks on code availability)

Reviewer #5

(Remarks to the Author)

I would like to thank the authors for their thoughtful and thorough responses to all my questions and comments during the review process. Their detailed explanations and the revisions made have significantly enhanced the clarity and robustness of the manuscript. I greatly appreciate the time and effort invested in addressing these points. While some limitations persist, the authors have acknowledged these and provided a thoughtful discussion of the most important ones.

(Remarks on code availability)

Open Access This Peer Review File is licensed under a Creative Commons Attribution 4.0 International License, which permits use, sharing, adaptation, distribution and reproduction in any medium or format, as long as you give appropriate credit to the original author(s) and the source, provide a link to the Creative Commons license, and indicate if changes were made.

In cases where reviewers are anonymous, credit should be given to 'Anonymous Referee' and the source.

The images or other third party material in this Peer Review File are included in the article's Creative Commons license, unless indicated otherwise in a credit line to the material. If material is not included in the article's Creative Commons license and your intended use is not permitted by statutory regulation or exceeds the permitted use, you will need to obtain permission directly from the copyright holder.

To view a copy of this license, visit <https://creativecommons.org/licenses/by/4.0/>

Response to the Comments of All Reviewers

Revision Report

Manuscript Title: Vision Transformer-Based Biomarker can Optimize Curative Treatment for Patients with Recurrent Hepatocellular Carcinoma

Manuscript Number: NCOMMS-24-36298

Date of this Revision: October 15, 2024

Thank you very much for kindly giving us an opportunity to revise our manuscript. We have revised the manuscript accordingly and gave point-by-point response to all the reviewers. Great thanks to our reviewers. They really helped us a lot to improve this work. We hope the revision has met the standard of *Nature Communications*. Please feel free to contact us if any further concerns should be addressed.

Table of Contents:

●	REVIEWER COMMENTS	2
●	Response to Reviewer 1	8
●	Response to Reviewer 2	31
●	Response to Reviewer 3	63
●	Response to Reviewer 4	64

● REVIEWER COMMENTS

Reviewer #1 (Remarks to the Author): Expert in cancer radiomics, machine learning, and digital pathology

The study may be clinically relevant as it attempts to address the critical issue of patient selection for liver surgery or thermal ablation in the context of initial (i) and recurrent (r)hepatocellular carcinoma (HCC). However, several methodological issues need to be addressed to strengthen the study's validity and reliability.

1. It is unclear how observer and internal technical variability regarding ultrasound (US) and contrast-enhanced ultrasound (CEUS) were addressed in the study. Please provide detailed information on how variability among different observers and technical settings was managed. This could include standardization procedures, training protocols for observers, inter-observer reliability assessments, and calibration methods used to ensure consistency across examinations.
2. Provide more detailed descriptions of the preprocessing steps for the image data, including techniques for noise reduction and standardization. Discuss how missing data and artifacts were handled during preprocessing.
3. There is a discrepancy in the clinical settings between the development and validation sets, with the development set focusing on non-treated HCC and the validation set on recurrent HCC. Explain the rationale behind using different clinical settings for development and validation. Address how this difference might affect the model's performance and generalizability. Consider discussing any steps taken to mitigate potential biases introduced by this discrepancy.
4. Description on page 5 lack clarity regarding which datasets were used for development and validation. Reword the description on page 5 to clearly specify which datasets were designated for external testing, particularly lines 121 and 122 are confusing as it seems part of the development cohort was used for external validation. While Figure 5 is very clear, the accompanying text needs to be more precise.
5. A cohort from center A was included as an external validation set, but the patients were treated, and the scans were performed at the same center as the development cohort, which challenges the definition of "external validation." To ensure proper generalization testing, consider including results from only truly external centers (e.g., centers B and C).

Reviewer #2 (Remarks to the Author): Expert in cancer radiomics, machine learning, and digital pathology

Noteworthy results:

- **Model Performance and Comparative Advantage:** HEROVision, which was developed based on a Vision Transformer (ViT) structure, showed high accuracy in predicting the prognosis for recurrent hepatocellular carcinoma (rHCC). In external testing cohorts, HEROVision-TA (thermal ablation) achieved a 2-year concordance index (C-index) of 0.72, while HEROVision-SR (surgical resection) reached 0.74. The 2-year area under the curve (AUC) values were 0.78 for TA and 0.82 for SR. These results outperformed the six guideline staging systems, which had 2-year C-index values between 0.51 and 0.58 and 2-year AUC values between 0.51 and 0.61. Additionally, integrating HEROVision into these staging systems improved their 2-year progression-free survival (PFS) prediction accuracy.

- **Risk Stratification:** HEROVision effectively categorized patients into high- and low-risk groups. In the external testing cohort, the 2-year PFS rates for low-risk patients were 79.0% for TA and 75.8% for SR, compared to 42.2% for TA and 31.9% for SR in the high-risk groups. HEROVision identified that 26.0% of high-risk patients could potentially improve their prognosis by switching from TA to SR.

- **Contribution to the Field and Clinical Impact:** HEROVision represents a potential advancement in the treatment of HCC by combining a Vision Transformer-based model with multimodal imaging data, including ultrasound (US) and magnetic resonance imaging (MRI), along with clinical variables. This approach enhances risk stratification and treatment decision-making for patients with rHCC and could surpass the current guideline staging systems. By providing personalized treatment recommendations based on preoperative data, HEROVision addresses a potential gap in clinical guidelines. The model's precision in identifying high-risk patients who could benefit from changing their treatment underscores its practical value in enhancing therapeutic strategies for early-stage rHCC.

Significance:

Although superior performance compared to existing staging systems is demonstrated, using deep learning methods in HCC has been described many times. Previous studies have explored machine learning and deep learning techniques to predict early recurrence and optimize treatment for HCC. For instance, other models have successfully used multiparametric MRI and CT imaging combined with clinical data to predict early recurrence and optimize treatment decisions: "Zhan, G., Wang, F., Wang, W., Li, Y., Chen, Q., Hu, H., & Chen, Y.-W. (2023). A

transformer-based model for preoperative early recurrence prediction of hepatocellular carcinoma with multi-modality MRI. Computer Vision – ACCV 2022 Workshops, 185–194. https://doi.org/10.1007/978-3-031-27066-6_13". The novelty of HEROVision lies in its specific implementation of a specific Vision Transformer-based network and its integration with clinical decision-making tools, which could potentially enhance individualized treatment plans. The paper is therefore moderately innovative. Since the data are not publicly available, benchmarking by other groups is not possible which further dampens enthusiasm.

Methodology:

The paper uses a large multi-institutional dataset and describes models for both thermal ablation and surgical patients, which is a strength. The inclusion of 8946 patients from three geographically distinct centers in China is impressive.

There are issues noted below that may limit the generalizability of the models, which drive the enthusiasm for the paper down.

- The paper fails to compare its results with state-of-the-art (SOTA) Vision Transformer-based models that support multimodal data or publicly available datasets. Vision Transformers rely on the quality and diversity of the dataset because their performance is tied to the patterns they learn during training. Without testing on different publicly available datasets, it is difficult to establish the generalizability and robustness of the proposed approach. Moreover, the diversity of the training data is important for training Vision Transformers. Any bias in the training data, such as race or gender bias, can affect the model's predictions.

- The introduction does not adequately justify the use of Vision Transformers. It does not contrast their advantages over Convolutional Neural Networks (CNNs), despite existing literature highlighting CNNs' benefits in medical image-based diagnosis, such as in the study "Deep Learning Methods in Medical Image-Based Hepatocellular Carcinoma Diagnosis: A Systematic Review and Meta-Analysis" by Wei et al., 2023.

Additionally, the Materials and Methods section lacks explanations for some of the dataset division and the rationale behind the specific cohort division ratios. This could impact the experimental setup's validity as well as the model's training process.

- The study also does not provide detailed information on sex and gender considerations, failing to comply with the SAGER guidelines (as recommended in the review instructions).

- The computational complexity of the Vision Transformer model and the potential loss of diagnostic details due to image resizing and strip partitioning limit the model's clinical

feasibility. The paper does not address the variability in imaging quality or interpretation (many imaging modalities are used), which are important factors that could affect the accuracy of individualized predictions.

- Moreover, the comparisons with major guideline methods do not address inherent biases or limitations in the staging systems. This could undermine the validity of the results, as these biases can impact the model's performance. There is also a drop in performance metrics when the model is applied to the external testing cohort. This could indicate potential overfitting and limited generalizability. In addition, the statistical analysis lacks detailed justification for the selection of some specific tests and thresholds. The choice of propensity score matching (PSM) and the methods used to evaluate statistical significance are not sufficiently explained.

- The rationale behind the selection of specific variables and the chosen caliper value of 0.02 for PSM should be explicitly justified. Furthermore, while the section effectively details the re-stratification process using the HEROVision models, a more thorough discussion on the clinical implications of these re-stratifications is necessary. Specifically, elaborating on how the re-categorization from high-risk to low-risk influences patient management and treatment outcomes would provide valuable insights into the practical applications of the findings. Also, the title of this section should be in all capital letters to be consistent with previous headers.

Reviewer #2 (Remarks on code availability):

The provided GitHub repository is well-documented. The README file offers an overview of the model's purpose, prerequisites, and instructions for installing necessary libraries and running the application. The dataset handling script (dataset.py) details the preprocessing steps, ensuring reproducibility. The model architecture (Model_hovertrans.py) is documented, including hyperparameters and training specifics, which are crucial for replicating the results. The training scripts (train_hover_SR.py and train_hover_TA.py) provide instructions and use established methods such as the Adam optimizer and Cox Loss, with callbacks for early stopping and model checkpointing. Overall, the repository's organization and detailed documentation make it user-friendly and support the reproducibility of the paper's findings.

Reviewer #3 (Remarks to the Author): Early-Career Researcher co-reviewer

I co-reviewed this manuscript with one of the reviewers who provided the listed reports. This

is part of the Nature Communications initiative to facilitate training in peer review and to provide appropriate recognition for Early Career Researchers who co-review manuscripts.

Reviewer #4 (Remarks to the Author): Clinical expert in hepatocellular carcinoma and radiology

Comments on Vision Transformer-Based Biomarker can Optimize Curative Treatment for Patients with Recurrent Hepatocellular Carcinoma

General

This is a very eager and innovative study. The topic is relevant. The cohorts are large and I consider them large enough as they are restricted by the inclusion/exclusion criteria.

But I do have some concerns about the aim and the study design.

By narrowing the in/exclusion criteria so much, it is not surprising that the clinical classification models (guideline systems) are not performing well. By your inclusion restrictions you focus on an indication niche, which was left liberally open for individual treatment decision making based on knowing the specific patient and the possibilities of the specific medical center. It basically does not make sense to use the clinical scores and classification inside this narrow group at all.

The actual goal of your study and the algorithm is not making a decision regarding which local therapy to use or if it should be used at all but to give an outlook on the prognosis of the patient, namely the OS/PFS. So the decision derived from this information would be to treat or not to treat locally – with the alternative being systemic treatment – or TACE/TARE (which we know to perform inferior to local ablation and surgery in suitable cases). Therefore the title is misleading. You do not investigate on how to improve HCC treatment. In order to give aid in the treatment decision, you would have to include patients in whom the therapy decision was wrong or suboptimal. But how do you know in your cohorts that the treatment they received was inferior to any other treatment option? You cant know that and neither your AI algorithm. So the whole paper should be changed from the title on into “prognostication tool in HCC to be treated by means of local therapy”.

To include just T2w, DWI and CEUS sounds surprising. How did you decide to use these specific parts of the diagnostic work-up? What do you expect the AI to read on these images? We know that T2 signal and diffusion restriction loosely correlates with the grading of the

tumors. What if you provided the algorithm with the grading of the tumors instead of MRI? Or to give just SI-values/radiomics from these images instead of letting the algorithm read this on its own? Or if you run the algorithm against sets of (para)clinical parameters, which are known to determine the prognosis and outcome of patients?

As said previously, the clinical guideline based categories are not designed to give this fine granulated info on prognosis.

Specific

Aim:

“This approach aimed to optimize individual treatment selection between RHR and TA for patients with early-stage rHCC in real-world scenarios.”

Well, but the algorithm was not product of the study at hand, right? So what was the actual aim of the present study? Please phrase a specific aim or hypothesis.

Methods: Did you perform any tumor segmentations on the MRI images or the US images to let the algorithm know which is the tumor and which the surrounding liver? Did the algorithm use any anatomic information, which is the most important input a surgeon or radiologist need to make a decision regarding local treatment? For me the plausibility is lacking if these important informations were not used by the algorithm.

Fig 1:

Legend should explain all abbreviations. These abbreviations should then be used in all occasions where they appear inside the table (e.g. not hepatectomy but SR).

Inclusion: clarify by writing “... of PRETHERAPEUTIC contrast enhanced US and liver MRI within a 1-month interval“

Exclusion: Please clarify: “unqualified artefacts” – shouldn’t this read: “unqualifying artefacts”?

In the external testing cohort, you include patients with resection of rHCC. How can you end up with a TA testing cohort?

● **Response to Reviewer 1**

1. *[It is unclear how observer and internal technical variability regarding ultrasound (US) and contrast-enhanced ultrasound (CEUS) were addressed in the study. Please provide detailed information on how variability among different observers and technical settings was managed. This could include standardization procedures, training protocols for observers, inter-observer reliability assessments, and calibration methods used to ensure consistency across examinations.]*

Our Response:

Thank you for your insightful comments regarding the management of observer and technical variability in our study. We have implemented several measures to ensure consistency and reliability in our ultrasound (US) and contrast-enhanced ultrasound (CEUS) data. Below is a detailed explanation of how we addressed these issues:

1. Expertise of Radiologists:

Abdominal CEUS examinations were conducted by experienced radiologists, each with over 10-20 years of experience, from the five participating hospitals. Their expertise ensured high-quality image acquisition, thereby minimizing observer variability. Further details are provided in **Appendix S3**.

2. Adherence to Guidelines:

In this study, CEUS examinations strictly followed established guidelines from the China Primary Liver Cancer Study Group[1, 2], the Japanese Society of Hepatology[3], and the American College of Radiology[4]. Radiologists obtained gray-scale ultrasound images of each lesion as follows: (1) one image showing the lesion's maximum diameter; and (2) another image taken by rotating the probe 90° while maintaining the same position. For CEUS, images were acquired during the arterial phase (10-40 s), portal venous phase (60-90 s), and delayed venous phase (90-180 s) for each lesion. More details are provided in **Appendix S3**. Compliance with these guidelines helped standardize imaging procedures and ensured consistent diagnostic practices across different centers.

3. Major manufacturers of US Equipment:

US data were collected using equipment from four major manufacturers in the Chinese market: MyLab90 XVision (Esaote, Florence, Italy), Logiq E9 (GE Healthcare, Milwaukee, WI, USA), and Siemens Acuson Sequoia (Siemens Healthineers, Erlangen, Germany), and Resona 7 (Mindray, Shenzhen, China), each equipped with a 1- to 13-MHz abdominal transducer and contrast-tuned imaging mode. For CEUS, the ultrasound contrast agent used was SonoVue microbubbles. Additional details can be found in **Appendix S3**.

4. Image Preprocessing and Standardization:

To minimize potential variability due to differences in hospitals, equipment, and technical parameters, we implemented a rigorous preprocessing pipeline. This included denoising, image

augmentation, and normalization techniques to mitigate the impact of these variables on AI model analysis. These preprocessing steps were crucial in ensuring that the input data were as uniform as possible. Further details are provided in **Appendix S5**.

5. Subgroup Analyses, Ablation Studies, and Variability Assessment:

We collected additional data from two hospitals in China to include recurrent HCC (rHCC) patients in the external testing cohort. In the external testing cohort, we conducted subgroup analyses to evaluate potential variability between different centers and equipment manufacturers. Specifically, HEROVision's 2-year C-index for the 2ndTA group was 0.72, 0.72, 0.73, 0.76 and 0.75 in Centers A, B, C, D and E, respectively ($P = 0.654-0.996$), with corresponding 2-year AUCs of 0.75, 0.78, 0.78, 0.90 and 0.81 ($P = 0.005-0.483$). For the 2ndSR group, the 2-year C-index was 0.76, 0.69, and 0.73 in Centers A, B, and D ($P = 0.375$ and 0.690), with corresponding 2-year AUCs of 0.80, 0.82, and 0.80 ($P = 0.739$ and 0.939). Additionally, HEROVision's 2-year C-index for 2ndTA across MyLab90 XVision, Logiq E9, Siemens Acuson Sequoia systems, and Resona 7 was 0.73, 0.73, 0.74, and 0.76 ($P = 0.646-0.907$), with corresponding 2-year AUCs of 0.78, 0.79, 0.79, and 0.87 ($P = 0.077-0.740$). For SR, the 2-year C-index across these systems was 0.74, 0.72, 0.75, and 0.68 ($P = 0.445-0.888$), with corresponding 2-year AUCs of 0.79, 0.84, 0.88, and 0.78 ($P = 0.167-0.942$). These results indicate that HEROVision demonstrated no significant variability across different center and manufacturers subgroups. Additionally, we also considered subgroups based on sex, age, tumor size, and the number of tumors in the 2ndTA and 2ndSR datasets. More details are provided in **Figures S14, S15** and **Table S7**. The absence of significant variability across subgroups demonstrates the robustness and consistency of our HEROVision model.

Furthermore, we performed three ablation studies, investigating different training and internal validation cohort splitting ratios, various modality models, and different deep learning (DL) models. These results demonstrate that varying the dataset partitioning ratios had minimal effect on model performance (**Figure S10** and **Table S4**), incorporating information from multiple modalities enhances the prediction of PFS (**Figure S11** and **Table S5**), and the performance of our ViT model was superior than other models (**Figures S12, S13** and **Table S6**) in predicting PFS for rHCC.

Overall, these measures collectively ensure that the observer and technical variability are minimized, thereby enhancing the reliability and generalizability of HEROVision. We hope this detailed explanation addresses your concerns effectively.

References:

- [1] J. Zhou *et al.*, "Guidelines for Diagnosis and Treatment of Primary Liver Cancer in China (2017 Edition)," (in eng), *Liver Cancer*, vol. 7, no. 3, pp. 235-260, Sep 2018, doi: 10.1159/000488035.
- [2] G. O. o. N. H. Commission, "Standard for diagnosis and treatment of primary liver

- cancer (2022 edition)," *J Clin Hepatol*, vol. 38, no. 2, pp. 288-303, 2022, doi: 10.3969/j.issn.1001-5256.2022.02.009.
- [3] M. Kudo, O. Matsui, N. Izumi, H. Iijima, M. Kadoya, and Y. Imai, "Surveillance and diagnostic algorithm for hepatocellular carcinoma proposed by the Liver Cancer Study Group of Japan: 2014 update," (in eng), *Oncology*, vol. 87 Suppl 1, pp. 7-21, 2014, doi: 10.1159/000368141.
- [4] "Liver Imaging Reporting and Data System." ACR American College of Radiology. <https://www.acr.org/Quality-Safety/Resources/LIRADS> (accessed 12 Sep, 2017).

Revised text:

1. Expertise of Radiologists:

(page 4/Appendix S3 section of the revised supplementary material)

[Abdominal CEUS examinations were conducted by an experienced radiologist, with over 10-20 years of experience, from each of the five hospitals.]

2. Adherence to Guidelines:

(page 4-5/Appendix S3 section of the revised supplementary material)

[Radiologists obtained gray-scale ultrasound images of each lesion during each scan as follows: (1) one image showing the lesion's maximum diameter; and (2) another image taken by rotating the probe 90° while maintaining the same position. Both transverse and longitudinal images of the lesion were captured. The ultrasound contrast agent used was SonoVue microbubbles (Bracco, Ivrea, Italy; Bracco, Milan, Italy; Bracco Suisse SA, Geneva, Switzerland; Shanghai Bracco Sine Pharmaceutical Corp. Ltd, Shanghai, China). Each patient received an intravenous bolus injection of 2.4 mL SonoVue, followed by 5 mL of normal saline. A timer was started with the bolus injection, and the region of interest was observed continuously for 3 minutes. Digital image sequences were saved for later analysis, capturing the arterial phase (10-40 s), portal venous phase (60-90 s), and delayed venous phase (90-180 s).]

3. Major manufacturers of US Equipment:

(page 4/Appendix S3 section of the revised supplementary material)

[The real-time ultrasound systems used included the MyLab90 XVision (Esaote, Florence, Italy), Logiq E9 (GE Healthcare, Milwaukee, WI, USA), Siemens Acuson Sequoia (Siemens Healthineers, Erlangen, Germany), and Resona 7 (Mindray, Shenzhen, China), each equipped with a 1- to 13-MHz abdominal transducer and contrast-tuned imaging mode.]

4. Image Preprocessing and Standardization:

(page 6-7/Appendix S5 section of the revised supplementary material)

[To minimize the impact of variability due to different hospitals, equipment, and technical

parameters on subsequent analysis, we employed denoising and image augmentation techniques^{2, 3}. Denoising was performed using advanced filtering methods, such as Gaussian filters, to remove random noise and enhance image quality². Image augmentation techniques, including rotation, scaling, translation, and contrast adjustment, were utilized to diversify the dataset, improve model robustness, and reduce the risk of overfitting³. Then, in order to objectively extract tumor subregions within the tumor and alleviate human bias, we expanded all drawn regions of interest masks outward by 5mm, allowing for consideration of both the internal region of the lesion and the surrounding liver parenchyma, providing more feature information for deep learning (DL) networks. Python was used to extract irregular bounding boxes of tumor subregions from various patterns. In order to improve the processing efficiency of DL models, we transform irregular boundaries into regular square boundaries and then crop them. Additional preprocessing included normalization of size and grayscale values to produce resized and standardized grayscale ROI images for model development. Each subregion image was adjusted to a standardized size of 128 × 128 pixels, with grayscale values normalized to ensure uniformity across the dataset, thereby improving model training.

Gaussian filters:

$$G(x,y)= 1/(2\pi\sigma^2) e^{-(x^2+y^2)/(2\sigma^2)}$$

Among them, x, y represents the position of the pixel; σ is the standard deviation of the pixels in the training cohort.

Normalization:

$$\text{Standardization } (x) = (x-\mu)/(\max(\sigma, 1.0/\sqrt{N}))$$

$$\text{Norm } (x) = (x-\min(x))/(\max(x)-\min(x))$$

Among them, x represents the images; μ is the average value of the pixels in the training cohort; σ is the standard deviation of the pixels in the training cohort; N is the number of pixels; min(x) represents the minimum pixel value and max(x) represents maximum pixel value.]

5. Subgroup Analyses, Ablation Studies, and Variability Assessment:

(page 14-15/Methods section of the revised manuscript)

[Ablation Studies

We performed three ablation studies, investigating different training and internal validation cohort splitting ratios, various modality models, and different DL models. For the splitting ratios of the TA and SR datasets, based on chronological order, we retrained our ViT model under three settings, using 3/4, 4/5, and 5/6 of the training cohort, respectively, and then validated the model in each setting on the internal validation cohort. In exploring different modality models, we assessed the contributions of CEUS and MRI modalities to the final performance of our ViT model on both the training and internal validation cohorts. Lastly, for different DL models, we compared our ViT model with other DL models on both TA and SR datasets using the same experimental setup. This included two CNN models: ResNet^{50 49} and ConvNext⁵⁰, and two Transformer-based models: original ViT⁵¹ and GC-ViT⁵².]

(page 15/Methods section of the revised manuscript)

[Subgroup Analyses

To address potential variability and ensure the robustness of the findings, we conducted subgroup analyses in the 2ndTA and 2ndSR datasets, including sex (female and male), age (≤ 60 years and > 60 years), tumor size (≤ 3 cm and > 3 cm), the number of tumors (solitary and multiple), manufacturers (MyLab90 Xvision, Logiq E9, Siemens Acuson Sequoia, and Resona 7), and centers (Centers A, B, C, D, and E).]

(page 39/Figure S10 of the revised supplementary material)

[

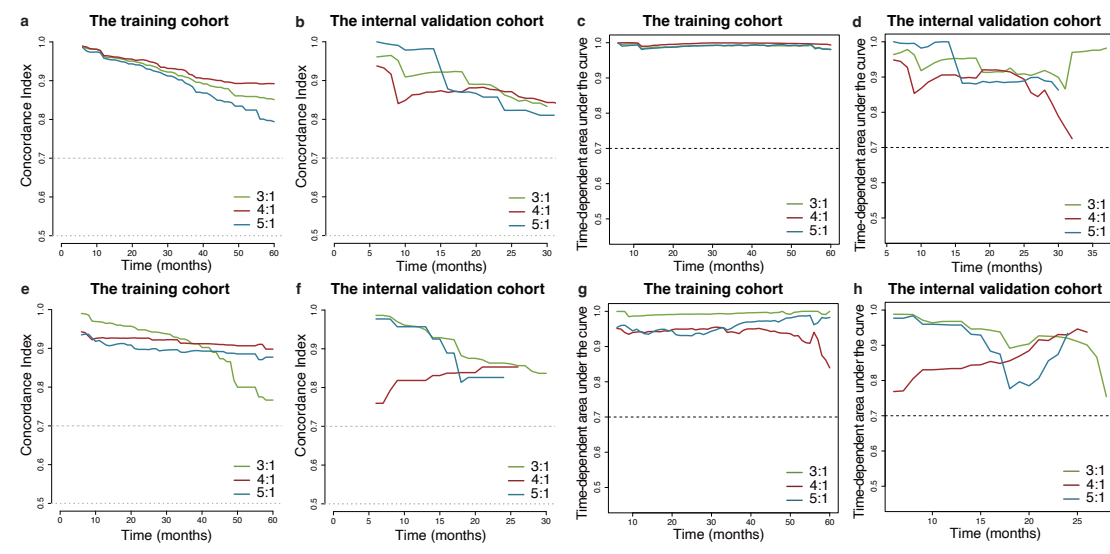


Figure S10. The Discriminative Performances of Different Training and Internal Validation Cohort Splitting Ratios

For TA dataset (a-d) and SR dataset (e-h), the C-index in the training and internal validation cohorts (a, b, e and f), and the time-dependent AUC in both cohorts (c, d, g and h) reflected the models' discriminatory performance at various points.

TA, thermal ablation; SR, surgical resection; C-index, concordance index; AUC, area under the curve.]

(page 40/Figure S11 of the revised supplementary material)

[

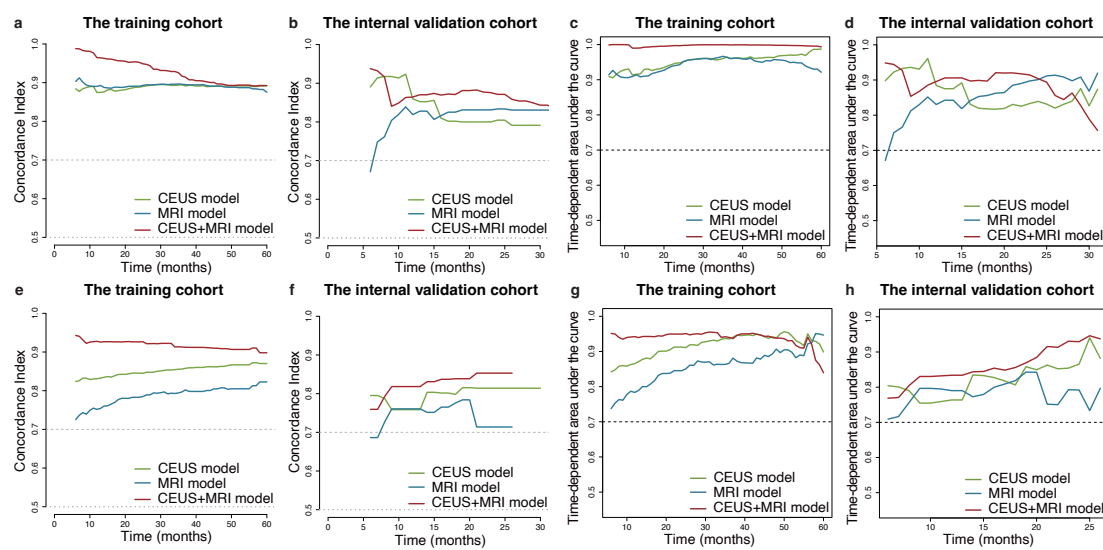


Figure S11. The Discriminative Performances of Different Modality Models in the Training and Internal Validation Cohorts

For TA dataset (a-d) and SR dataset (e-h), the C-index in the training and internal validation cohorts (a, b, e and f), and the time-dependent AUC in both cohorts (c, d, g and h) reflected the models' discriminatory performance at various points.

TA, thermal ablation; SR, surgical resection; C-index, concordance index; AUC, area under the curve; MRI, magnetic resonance imaging; CEUS, contrast-enhanced ultrasound.]

(page 41/Figure S12 of the revised supplementary material)

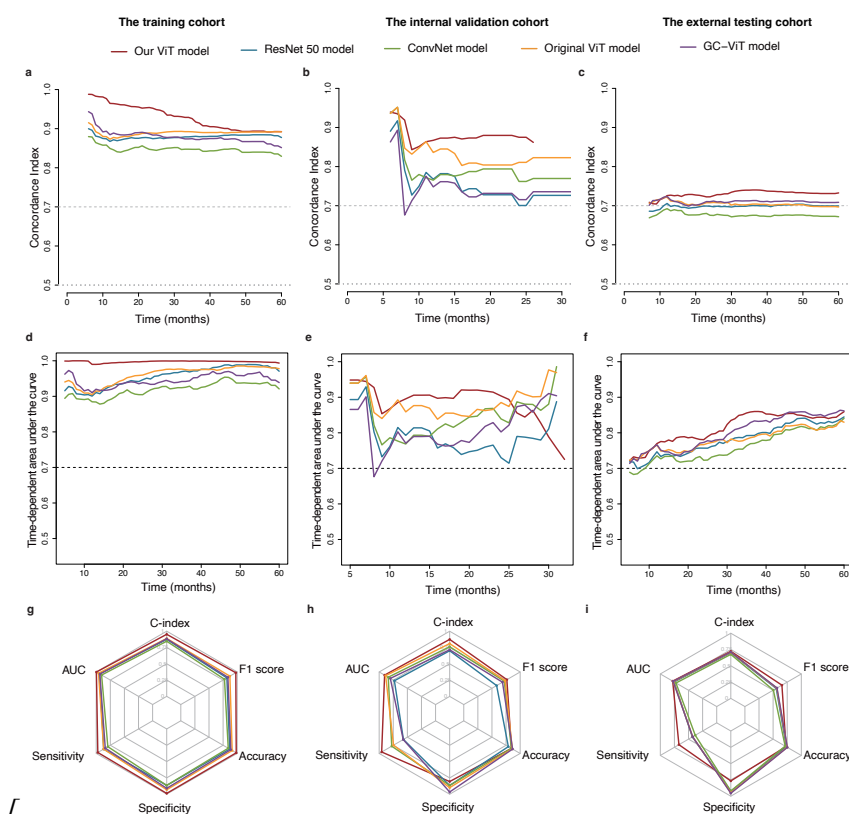


Figure S12. The Discriminative Performances of Different Deep Learning Models for the TA Dataset

The C-index (a-c) and time-dependent AUC (d-f) of different deep learning models demonstrated the models' discriminatory performance at various points in the training, internal validation, and external testing cohorts. (g-i) Radar charts illustrated the C-index, AUC, specificity, sensitivity, accuracy, and F1 score for 2-year PFS across these three cohorts. TA, thermal ablation; ViT, Vision Transformer; C-index, concordance index; AUC, area under the curve; PFS, progression-free survival.]

(page 42/ Figure S13 of the revised supplementary material)

[

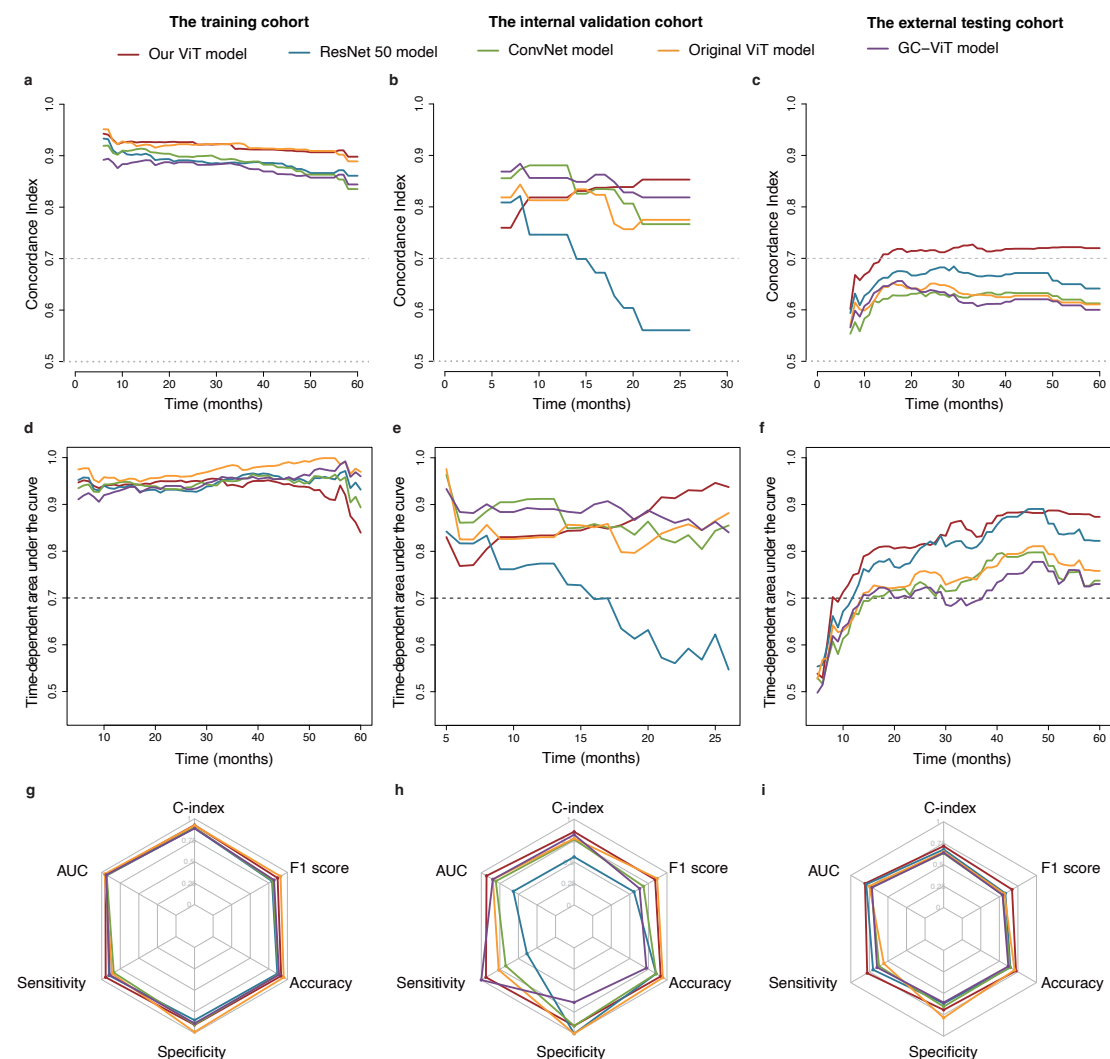


Figure S13. The Discriminative Performances of Different Deep Learning Models for the SR Dataset

The C-index (a-c) and time-dependent AUC (d-f) of different deep learning models demonstrated the models' discriminatory performance at various points in the training, internal validation, and external testing cohorts. (g-i) Radar charts illustrated the C-index, AUC, specificity, sensitivity, accuracy, and F1 score for 2-year PFS across these three cohorts.

specificity, sensitivity, accuracy, and F1 score for 2-year PFS across these three cohorts.
SR, surgical resection; ViT, Vision Transformer; C-index, concordance index; AUC, area under
the curve; PFS, progression-free survival.]

(page 43-44/Figure S14 of the revised supplementary material)

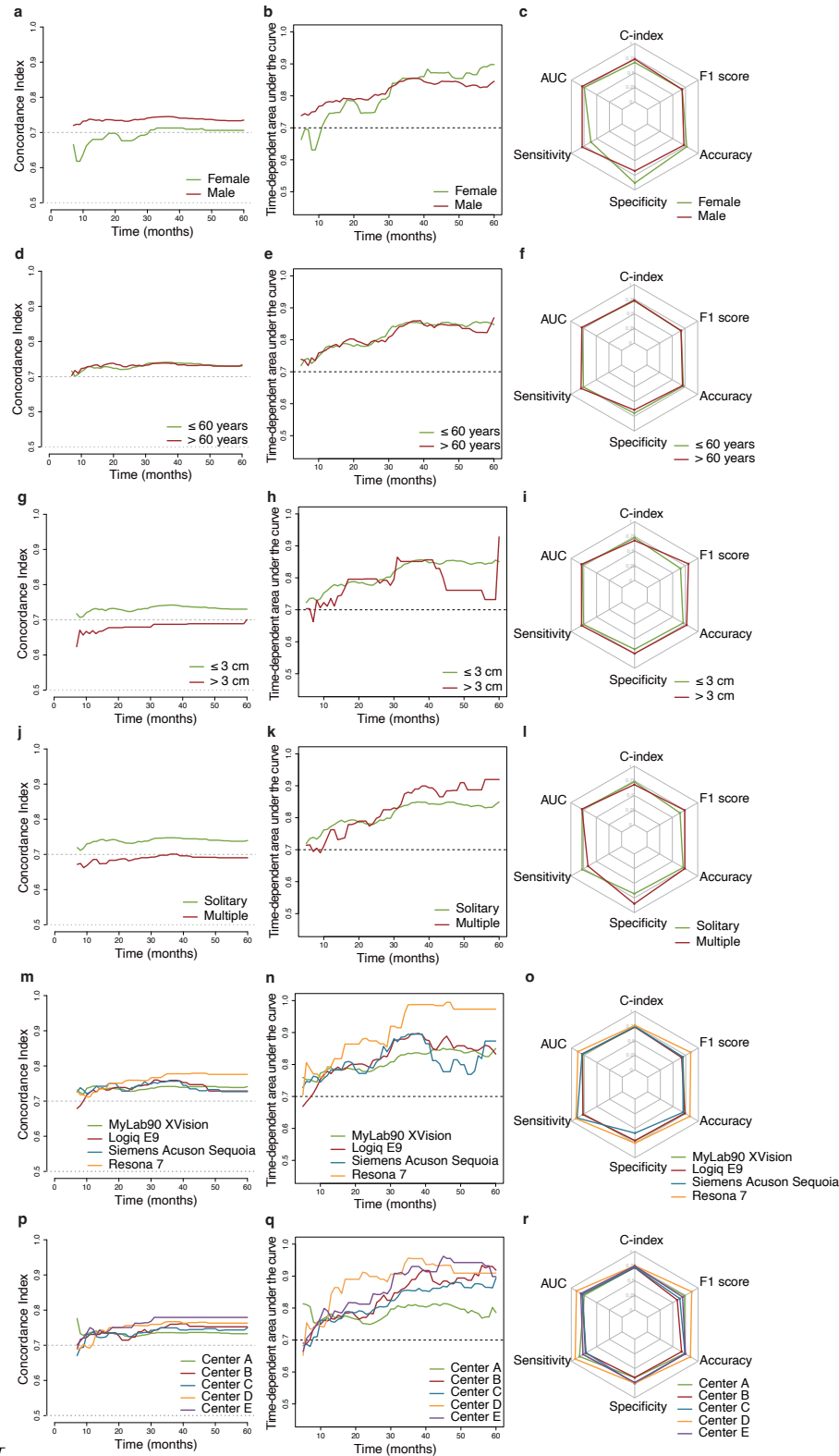


Figure S14. The Discriminative Performances of Different Subgroups for the 2ndTA Dataset
The C-index (a, d, g, j, m, and p) and time-dependent AUC (b, e, h, k, n, and q) for different subgroups, including sex, age, tumor size, the number of tumors, manufacturers, and centers, demonstrated the models' discriminatory performance at various points in the 2ndTA cohort. (c, f, i, l, o, and r) Radar charts illustrated the C-index, AUC, specificity, sensitivity, accuracy, and F1 score for 2-year PFS across these subgroups.
TA, thermal ablation; ViT, Vision Transformer; C-index, concordance index; AUC, area under the curve; PFS, progression-free survival.]

(page 45-46/Figure S15 of the revised supplementary material)

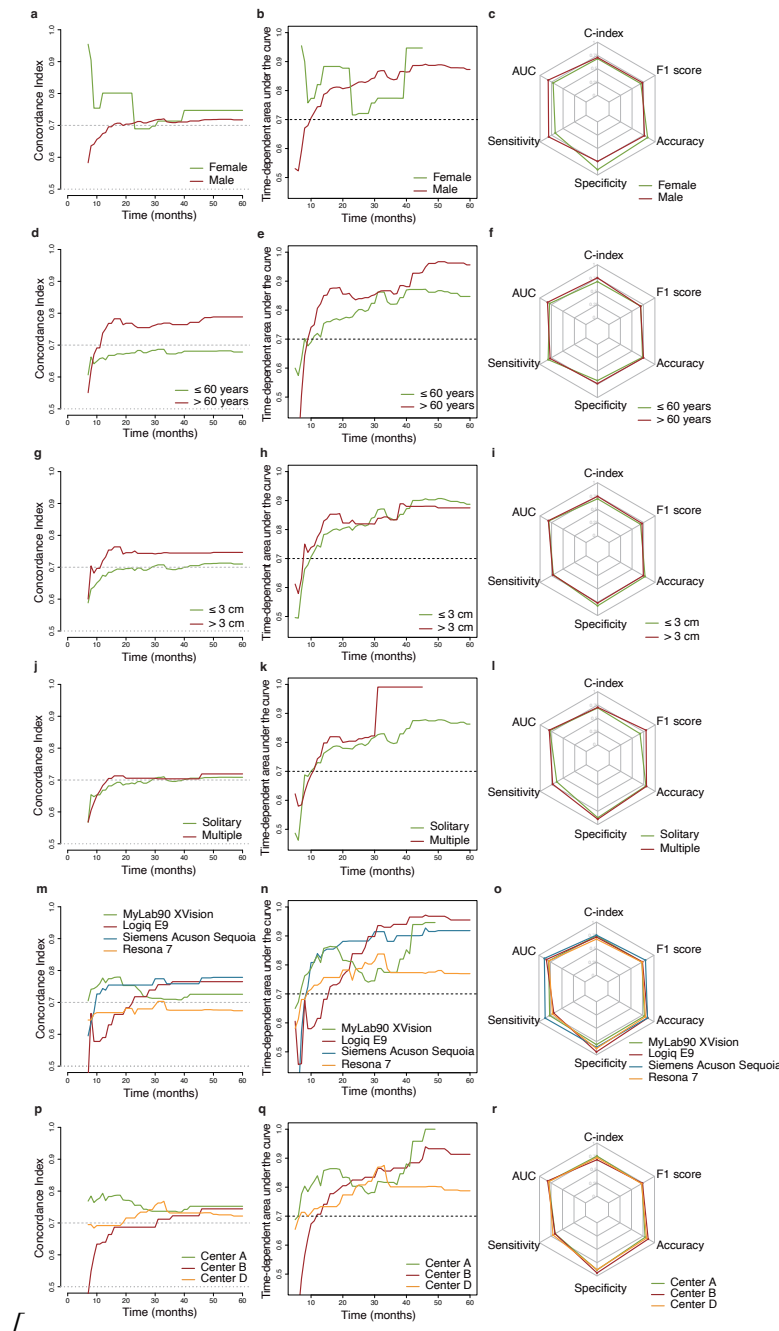


Figure S15. The Discriminative Performances of Different Subgroups for the 2ndSR Dataset
The C-index (a, d, g, j, m, and p) and time-dependent AUC (b, e, h, k, n, and q) for different
subgroups, including sex, age, tumor size, the number of tumors, manufacturers, and centers,
demonstrated the models' discriminatory performance at various points in the 2ndSR cohort.
(c, f, i, l, o, and r) Radar charts illustrated the C-index, AUC, specificity, sensitivity, accuracy,
and F1 score for 2-year PFS across these subgroups.
TA, thermal ablation; ViT, Vision Transformer; C-index, concordance index; AUC, area under
the curve; PFS, progression-free survival.]

(page 21/Table S4 of the revised supplementary material)

[Table S4. Performance Comparison of Different Training and Internal Validation Cohort Splitting Ratios for 2-Year PFS

Cohorts	Ratios	TA dataset				SR dataset			
		2-year C-index	P Value	2-year AUC	P Value	2-year C-index	P Value	2-year AUC	P Value
The training cohort	3:1	0.94 (0.91 - 0.97)	Ref.	0.99 (0.97 - 1.00)	Ref.	0.95 (0.92 - 0.97)	Ref.	0.99 (0.98 - 1.00)	Ref.
	4:1	0.95 (0.93 - 0.98)	0.496	1.00 (0.99 - 1.00)	0.341	0.93 (0.90 - 0.95)	0.266	0.95 (0.92 - 0.98)	0.001
	5:1	0.93 (0.90 - 0.96)	0.634	0.99 (0.97 - 1.00)	0.913	0.90 (0.86 - 0.93)	0.016	0.93 (0.90 - 0.96)	<0.001
The internal validation cohort	3:1	0.86 (0.79 - 0.93)	Ref.	0.91 (0.82 - 1.00)	Ref.	0.86 (0.79 - 0.94)	Ref.	0.92 (0.83 - 1.00)	Ref.
	4:1	0.87 (0.80 - 0.94)	0.866	0.90 (0.81 - 1.00)	0.932	0.85 (0.76 - 0.94)	0.866	0.93 (0.82 - 1.00)	0.835
	5:1	0.82 (0.73 - 0.92)	0.502	0.89 (0.74 - 1.00)	0.663	0.83 (0.72 - 0.94)	0.589	0.93 (0.80 - 1.00)	0.772

TA, thermal ablation; SR, surgical resection; PFS, progression-free survival; AUC, area under the curve; Ref, reference.

(page 22/Table S5 of the revised supplementary material)

[Table S5. Performance Comparison of the Different Modality Models for 2-year PFS in the Training and Internal Validation Cohorts

Cohorts	Models	TA dataset				SR dataset			
		2-year C-index	P Value	2-year AUC	P Value	2-year C-index	P Value	2-year AUC	P Value
The training cohort	CEUS+MRI	0.95 (0.93 - 0.98)	Ref.	1.00 (0.99 - 1.00)	Ref.	0.93 (0.90 - 0.95)	Ref.	0.95 (0.92 - 0.98)	Ref.
	CEUS	0.89 (0.85 - 0.92)	0.003	0.95 (0.92 - 0.98)	<0.001	0.85 (0.81 - 0.88)	0.001	0.91 (0.88 - 0.94)	0.063
	MRI	0.89 (0.86 - 0.93)	0.004	0.94 (0.92 - 0.96)	<0.001	0.78 (0.74 - 0.83)	<0.001	0.85 (0.80 - 0.89)	<0.001
The internal validation cohort	CEUS+MRI	0.87 (0.80 - 0.94)	Ref.	0.90 (0.81 - 1.00)	Ref.	0.85 (0.76 - 0.94)	Ref.	0.93 (0.82 - 1.00)	Ref.
	CEUS	0.80 (0.72 - 0.89)	0.254	0.84 (0.72 - 0.96)	0.224	0.81 (0.71 - 0.91)	0.570	0.87 (0.74 - 0.99)	0.250
	MRI	0.83 (0.75 - 0.92)	0.504	0.90 (0.80 - 1.00)	0.956	0.71 (0.60 - 0.83)	0.063	0.79 (0.58 - 1.00)	0.028

TA, thermal ablation; SR, surgical resection; PFS, progression-free survival; AUC, area under the curve; MRI, magnetic resonance imaging; CEUS, contrast-enhanced ultrasound; Ref, reference.

(page 23/Table S6 of the revised supplementary material)

[Table S6. Performance Comparison of the Different Deep Learning Models for 2-year PFS in the Training, Internal Validation, and External Testing Cohorts]

<i>Datasets</i>	<i>Cohorts</i>	<i>Models</i>	<i>2-year C-index</i>	<i>P value</i>	<i>2-year AUC</i>	<i>P value</i>	<i>Sensitivity</i>	<i>Specificity</i>	<i>Accuracy</i>	<i>F1 score</i>
<i>TA dataset</i>	<i>The training cohort</i>	<i>Our ViT model</i>	0.95 (0.93 - 0.98)	<i>Ref.</i>	1.00 (0.99 - 1.00)	<i>Ref.</i>	0.97 (0.95 - 0.99)	0.99 (0.98 - 1.00)	0.98 (0.97 - 1.00)	0.98 (0.96 - 0.99)
		<i>ResNet 50 model</i>	0.87 (0.84 - 0.91)	0.001	0.94 (0.92 - 0.97)	<0.001	0.87 (0.83 - 0.90)	0.87 (0.83 - 0.90)	0.87 (0.83 - 0.90)	0.82 (0.77 - 0.86)
		<i>ConvNext model</i>	0.85 (0.81 - 0.89)	<0.001	0.90 (0.87 - 0.94)	<0.001	0.79 (0.75 - 0.84)	0.87 (0.83 - 0.90)	0.84 (0.80 - 0.88)	0.77 (0.73 - 0.82)
		<i>Original ViT model</i>	0.89 (0.85 - 0.92)	0.003	0.96 (0.94 - 0.98)	0.001	0.87 (0.83 - 0.91)	0.93 (0.91 - 0.96)	0.91 (0.88 - 0.94)	0.87 (0.83 - 0.91)
		<i>GC-ViT model</i>	0.88 (0.85 - 0.92)	0.002	0.94 (0.91 - 0.96)	<0.001	0.84 (0.80 - 0.88)	0.91 (0.88 - 0.94)	0.89 (0.85 - 0.92)	0.84 (0.79 - 0.88)
	<i>The internal validation cohort</i>	<i>Our ViT model</i>	0.87 (0.80 - 0.95)	<i>Ref.</i>	0.90 (0.81 - 1.00)	<i>Ref.</i>	0.96 (0.91 - 1.00)	0.81 (0.72 - 0.89)	0.85 (0.77 - 0.92)	0.76 (0.67 - 0.86)
		<i>ResNet 50 model</i>	0.70 (0.60 - 0.80)	0.006	0.73 (0.58 - 0.88)	0.004	0.57 (0.46 - 0.68)	0.87 (0.79 - 0.94)	0.79 (0.70 - 0.88)	0.58 (0.47 - 0.69)
		<i>ConvNext model</i>	0.76 (0.67 - 0.86)	0.063	0.84 (0.71 - 0.96)	0.223	0.78 (0.68 - 0.87)	0.87 (0.79 - 0.94)	0.84 (0.76 - 0.92)	0.72 (0.62 - 0.82)
		<i>Original ViT model</i>	0.81 (0.72 - 0.90)	0.266	0.89 (0.79 - 0.98)	0.706	0.75 (0.65 - 0.84)	0.90 (0.83 - 0.97)	0.86 (0.78 - 0.94)	0.73 (0.63 - 0.83)
		<i>GC-ViT model</i>	0.72 (0.62 - 0.81)	0.011	0.80 (0.67 - 0.93)	0.070	0.57 (0.46 - 0.68)	0.97 (0.93 - 1.00)	0.87 (0.79 - 0.94)	0.69 (0.58 - 0.79)
	<i>The external testing cohort</i>	<i>Our ViT model</i>	0.72 (0.69 - 0.76)	<i>Ref.</i>	0.78 (0.74 - 0.82)	<i>Ref.</i>	0.67 (0.63 - 0.71)	0.77 (0.73 - 0.80)	0.73 (0.69 - 0.76)	0.66 (0.62 - 0.69)
		<i>ResNet 50 model</i>	0.70 (0.66 - 0.73)	0.355	0.76 (0.71 - 0.80)	0.325	0.43 (0.39 - 0.47)	0.92 (0.90 - 0.95)	0.74 (0.70 - 0.77)	0.56 (0.52 - 0.59)
		<i>ConvNext model</i>	0.68 (0.64 - 0.71)	0.073	0.72 (0.68 - 0.77)	0.019	0.38 (0.34 - 0.42)	0.92 (0.90 - 0.94)	0.71 (0.68 - 0.75)	0.50 (0.46 - 0.54)
		<i>Original ViT model</i>	0.71 (0.67 - 0.74)	0.562	0.77 (0.73 - 0.81)	0.560	0.43 (0.39 - 0.47)	0.95 (0.93 - 0.97)	0.75 (0.71 - 0.78)	0.57 (0.53 - 0.60)
		<i>GC-ViT model</i>	0.71 (0.67 - 0.75)	0.633	0.77 (0.73 - 0.81)	0.744	0.43 (0.39 - 0.47)	0.96 (0.94 - 0.97)	0.75 (0.72 - 0.79)	0.57 (0.53 - 0.61)
<i>SR dataset</i>	<i>The training cohort</i>	<i>Our ViT model</i>	0.93 (0.90 - 0.95)	<i>Ref.</i>	0.95 (0.92 - 0.98)	<i>Ref.</i>	0.95 (0.92 - 0.97)	0.90 (0.87 - 0.94)	0.92 (0.89 - 0.95)	0.87 (0.83 - 0.91)
		<i>ResNet 50 model</i>	0.89 (0.86 - 0.93)	0.120	0.93 (0.90 - 0.97)	0.333	0.90 (0.87 - 0.94)	0.85 (0.81 - 0.89)	0.86 (0.83 - 0.90)	0.79 (0.75 - 0.84)
		<i>ConvNext model</i>	0.90 (0.87 - 0.93)	0.207	0.93 (0.90 - 0.97)	0.368	0.84 (0.80 - 0.88)	0.90 (0.87 - 0.93)	0.88 (0.85 - 0.92)	0.80 (0.76 - 0.85)
		<i>Original ViT model</i>	0.92 (0.89 - 0.95)	0.851	0.96 (0.93 - 0.99)	0.479	0.86 (0.82 - 0.90)	0.99 (0.97 - 1.00)	0.95 (0.93 - 0.97)	0.91 (0.88 - 0.94)
		<i>GC-ViT model</i>	0.89 (0.85 - 0.92)	0.088	0.94 (0.91 - 0.97)	0.581	0.89 (0.86 - 0.93)	0.89 (0.85 - 0.92)	0.89 (0.85 - 0.92)	0.82 (0.78 - 0.86)
	<i>The internal cohort</i>	<i>Our ViT model</i>	0.85 (0.76 - 0.94)	<i>Ref.</i>	0.93 (0.82 - 1.00)	<i>Ref.</i>	0.94 (0.88 - 1.00)	0.91 (0.84 - 0.98)	0.92 (0.84 - 0.99)	0.84 (0.75 - 0.93)

<i>Datasets</i>	<i>Cohorts</i>	<i>Models</i>	<i>2-year C-index</i>	<i>P value</i>	<i>2-year AUC</i>	<i>P value</i>	<i>Sensitivity</i>	<i>Specificity</i>	<i>Accuracy</i>	<i>F1 score</i>
	<i>validation cohort</i>	<i>ResNet 50 model</i>	0.56 (0.43 - 0.69)	<0.001	0.57 (0.34 - 0.80)	<0.001	0.39 (0.26 - 0.51)	1.00 (1.00 - 1.00)	0.85 (0.76 - 0.94)	0.56 (0.43 - 0.68)
		<i>ConvNext model</i>	0.77 (0.66 - 0.87)	0.228	0.80 (0.63 - 0.98)	0.042	0.67 (0.55 - 0.79)	0.91 (0.84 - 0.98)	0.85 (0.76 - 0.94)	0.68 (0.56 - 0.80)
		<i>Original ViT model</i>	0.77 (0.67 - 0.88)	0.273	0.85 (0.67 - 1.00)	0.145	0.77 (0.66 - 0.87)	1.00 (1.00 - 1.00)	0.94 (0.89 - 1.00)	0.87 (0.78 - 0.95)
		<i>GC-ViT model</i>	0.82 (0.72 - 0.92)	0.612	0.85 (0.69 - 1.00)	0.145	1.00 (1.00 - 1.00)	0.64 (0.51 - 0.76)	0.72 (0.61 - 0.84)	0.63 (0.51 - 0.75)
		<i>Our ViT model</i>	0.72 (0.66 - 0.78)	Ref.	0.81 (0.74 - 0.88)	Ref.	0.78 (0.72 - 0.83)	0.69 (0.63 - 0.75)	0.72 (0.67 - 0.78)	0.67 (0.61 - 0.73)
	<i>The external testing cohort</i>	<i>ResNet 50 model</i>	0.68 (0.62 - 0.74)	0.381	0.79 (0.72 - 0.86)	0.647	0.70 (0.64 - 0.76)	0.62 (0.55 - 0.68)	0.65 (0.58 - 0.71)	0.58 (0.51 - 0.65)
		<i>ConvNext model</i>	0.63 (0.57 - 0.70)	0.064	0.73 (0.65 - 0.81)	0.043	0.62 (0.55 - 0.68)	0.65 (0.59 - 0.72)	0.64 (0.58 - 0.71)	0.55 (0.48 - 0.61)
		<i>Original ViT model</i>	0.65 (0.59 - 0.71)	0.144	0.75 (0.68 - 0.82)	0.138	0.56 (0.49 - 0.62)	0.79 (0.73 - 0.84)	0.71 (0.64 - 0.77)	0.57 (0.50 - 0.63)
		<i>GC-ViT model</i>	0.64 (0.58 - 0.70)	0.084	0.72 (0.63 - 0.80)	0.022	0.65 (0.58 - 0.71)	0.61 (0.54 - 0.67)	0.62 (0.56 - 0.69)	0.54 (0.48 - 0.61)

[TA, thermal ablation; SR, surgical resection; PFS, progression-free survival; AUC, area under the curve; Ref, reference.]

(page 25-26/Table S7 of the revised supplementary material)

[Table S7. Performance Comparison of HEROVision Across Different Subgroups for 2-Year PFS in the External Testing Cohorts]

<i>Datasets</i>	<i>Subgroups</i>	<i>Number</i>	<i>2-year C-index</i>	<i>P value</i>	<i>2-year AUC</i>	<i>P value</i>	<i>Sensitivity</i>	<i>Specificity</i>	<i>Accuracy</i>	<i>F1 score</i>
<i>Sex</i>										
	<i>Female</i>	<i>n=85</i>	0.68 (0.58 - 0.78)	Ref.	0.75 (0.63 - 0.86)	Ref.	0.61 (0.51 - 0.72)	0.88 (0.81 - 0.95)	0.77 (0.69 - 0.86)	0.69 (0.59 - 0.78)
	<i>Male</i>	<i>n=526</i>	0.73 (0.69 - 0.77)	0.307	0.79 (0.75 - 0.83)	0.426	0.79 (0.75 - 0.82)	0.68 (0.64 - 0.72)	0.72 (0.68 - 0.76)	0.68 (0.64 - 0.72)
<i>Age</i>										
<i>2ndTA dataset</i>	<i>≤60 years</i>	<i>n=366</i>	0.72 (0.68 - 0.77)	Ref.	0.78 (0.73 - 0.83)	Ref.	0.75 (0.71 - 0.80)	0.70 (0.65 - 0.74)	0.72 (0.67 - 0.76)	0.67 (0.63 - 0.72)
	<i>>60 years</i>	<i>n=245</i>	0.73 (0.67 - 0.78)	0.852	0.78 (0.72 - 0.85)	0.929	0.80 (0.75 - 0.85)	0.64 (0.58 - 0.70)	0.70 (0.64 - 0.76)	0.67 (0.61 - 0.73)
<i>Tumor size</i>										
	<i>≤3 cm</i>	<i>n=563</i>	0.72 (0.69 - 0.76)	Ref.	0.78 (0.74 - 0.82)	Ref.	0.76 (0.72 - 0.79)	0.68 (0.64 - 0.72)	0.71 (0.67 - 0.74)	0.66 (0.62 - 0.69)
	<i>>3 cm</i>	<i>n=48</i>	0.68 (0.55 - 0.81)	0.525	0.80 (0.64 - 0.95)	0.745	0.80 (0.68 - 0.91)	0.75 (0.63 - 0.87)	0.78 (0.66 - 0.90)	0.81 (0.70 - 0.92)
<i>The number of tumors</i>										

<i>Datasets</i>	<i>Subgroups</i>	<i>Number</i>	<i>2-year C-index</i>	<i>P value</i>	<i>2-year AUC</i>	<i>P value</i>	<i>Sensitivity</i>	<i>Specificity</i>	<i>Accuracy</i>	<i>F1 score</i>
<i>2ndSR dataset</i>	<i>Solitary</i>	<i>n=488</i>	<i>0.73 (0.69 - 0.77)</i>	<i>Ref.</i>	<i>0.78 (0.73 - 0.82)</i>	<i>Ref.</i>	<i>0.78 (0.74 - 0.82)</i>	<i>0.68 (0.64 - 0.72)</i>	<i>0.71 (0.67 - 0.75)</i>	<i>0.65 (0.61 - 0.69)</i>
	<i>Multiple</i>	<i>n=123</i>	<i>0.68 (0.60 - 0.77)</i>	<i>0.308</i>	<i>0.78 (0.69 - 0.87)</i>	<i>0.951</i>	<i>0.66 (0.58 - 0.75)</i>	<i>0.85 (0.79 - 0.91)</i>	<i>0.75 (0.67 - 0.82)</i>	<i>0.74 (0.67 - 0.82)</i>
	<i>Manufacturers</i>									
	<i>MyLab90 Xvision</i>	<i>n=334</i>	<i>0.73 (0.68 - 0.78)</i>	<i>Ref.</i>	<i>0.78 (0.72 - 0.83)</i>	<i>Ref.</i>	<i>0.77 (0.73 - 0.82)</i>	<i>0.71 (0.66 - 0.76)</i>	<i>0.74 (0.69 - 0.78)</i>	<i>0.69 (0.64 - 0.74)</i>
	<i>Logiq E9</i>	<i>n=156</i>	<i>0.73 (0.66 - 0.80)</i>	<i>0.907</i>	<i>0.79 (0.71 - 0.87)</i>	<i>0.740</i>	<i>0.77 (0.71 - 0.84)</i>	<i>0.71 (0.64 - 0.78)</i>	<i>0.73 (0.66 - 0.80)</i>	<i>0.67 (0.60 - 0.74)</i>
	<i>Siemens Acuson Sequoia</i>	<i>n=76</i>	<i>0.74 (0.64 - 0.84)</i>	<i>0.852</i>	<i>0.79 (0.69 - 0.90)</i>	<i>0.777</i>	<i>0.89 (0.82 - 0.96)</i>	<i>0.58 (0.47 - 0.69)</i>	<i>0.70 (0.59 - 0.80)</i>	<i>0.69 (0.59 - 0.80)</i>
	<i>Resona 7</i>	<i>n=45</i>	<i>0.76 (0.63 - 0.88)</i>	<i>0.646</i>	<i>0.87 (0.76 - 0.98)</i>	<i>0.077</i>	<i>0.91 (0.83 - 1.00)</i>	<i>0.75 (0.62 - 0.88)</i>	<i>0.83 (0.72 - 0.94)</i>	<i>0.85 (0.74 - 0.95)</i>
	<i>Centers</i>									
	<i>Center A</i>	<i>n=231</i>	<i>0.72 (0.67 - 0.78)</i>	<i>Ref.</i>	<i>0.75 (0.68 - 0.82)</i>	<i>Ref.</i>	<i>0.84 (0.79 - 0.89)</i>	<i>0.66 (0.60 - 0.72)</i>	<i>0.74 (0.68 - 0.79)</i>	<i>0.73 (0.67 - 0.79)</i>
	<i>Center B</i>	<i>n=102</i>	<i>0.72 (0.64 - 0.81)</i>	<i>0.972</i>	<i>0.78 (0.68 - 0.88)</i>	<i>0.483</i>	<i>0.72 (0.63 - 0.81)</i>	<i>0.65 (0.56 - 0.74)</i>	<i>0.67 (0.58 - 0.76)</i>	<i>0.59 (0.49 - 0.68)</i>
	<i>Center C</i>	<i>n=161</i>	<i>0.73 (0.66 - 0.79)</i>	<i>0.996</i>	<i>0.78 (0.70 - 0.86)</i>	<i>0.419</i>	<i>0.71 (0.64 - 0.78)</i>	<i>0.76 (0.69 - 0.82)</i>	<i>0.74 (0.67 - 0.81)</i>	<i>0.64 (0.56 - 0.71)</i>
	<i>Center D</i>	<i>n=46</i>	<i>0.76 (0.63 - 0.88)</i>	<i>0.654</i>	<i>0.90 (0.81 - 0.99)</i>	<i>0.005</i>	<i>0.92 (0.85 - 1.00)</i>	<i>0.75 (0.62 - 0.88)</i>	<i>0.85 (0.74 - 0.95)</i>	<i>0.87 (0.78 - 0.97)</i>
	<i>Center E</i>	<i>n=71</i>	<i>0.75 (0.65 - 0.85)</i>	<i>0.663</i>	<i>0.81 (0.71 - 0.92)</i>	<i>0.244</i>	<i>0.77 (0.67 - 0.87)</i>	<i>0.74 (0.63 - 0.84)</i>	<i>0.75 (0.65 - 0.85)</i>	<i>0.69 (0.59 - 0.80)</i>
	<i>Sex</i>									
	<i>Female</i>	<i>n=26</i>	<i>0.69 (0.51 - 0.87)</i>	<i>Ref.</i>	<i>0.72 (0.41 - 1.00)</i>	<i>Ref.</i>	<i>0.67 (0.49 - 0.85)</i>	<i>0.90 (0.78 - 1.00)</i>	<i>0.84 (0.70 - 0.98)</i>	<i>0.69 (0.51 - 0.87)</i>
	<i>Male</i>	<i>n=196</i>	<i>0.71 (0.65 - 0.78)</i>	<i>0.812</i>	<i>0.82 (0.76 - 0.89)</i>	<i>0.246</i>	<i>0.81 (0.76 - 0.87)</i>	<i>0.74 (0.68 - 0.80)</i>	<i>0.77 (0.71 - 0.83)</i>	<i>0.72 (0.66 - 0.79)</i>
	<i>Age</i>									
	<i>≤60 years</i>	<i>n=131</i>	<i>0.68 (0.60 - 0.76)</i>	<i>Ref.</i>	<i>0.79 (0.70 - 0.88)</i>	<i>Ref.</i>	<i>0.82 (0.76 - 0.89)</i>	<i>0.68 (0.60 - 0.76)</i>	<i>0.73 (0.66 - 0.81)</i>	<i>0.69 (0.61 - 0.77)</i>
	<i>>60 years</i>	<i>n=91</i>	<i>0.75 (0.67 - 0.84)</i>	<i>0.238</i>	<i>0.84 (0.73 - 0.94)</i>	<i>0.404</i>	<i>0.78 (0.69 - 0.86)</i>	<i>0.74 (0.65 - 0.83)</i>	<i>0.75 (0.66 - 0.84)</i>	<i>0.69 (0.59 - 0.78)</i>
	<i>Tumor size</i>									
	<i>≤3 cm</i>	<i>n=166</i>	<i>0.70 (0.63 - 0.77)</i>	<i>Ref.</i>	<i>0.81 (0.73 - 0.89)</i>	<i>Ref.</i>	<i>0.71 (0.64 - 0.78)</i>	<i>0.82 (0.76 - 0.88)</i>	<i>0.78 (0.72 - 0.85)</i>	<i>0.68 (0.61 - 0.76)</i>
	<i>>3 cm</i>	<i>n=56</i>	<i>0.74 (0.63 - 0.86)</i>	<i>0.503</i>	<i>0.82 (0.70 - 0.94)</i>	<i>0.876</i>	<i>0.72 (0.60 - 0.84)</i>	<i>0.76 (0.65 - 0.88)</i>	<i>0.75 (0.63 - 0.86)</i>	<i>0.72 (0.60 - 0.83)</i>
	<i>The number of tumors</i>									
	<i>Solitary</i>	<i>n=180</i>	<i>0.69 (0.63 - 0.76)</i>	<i>Ref.</i>	<i>0.79 (0.71 - 0.87)</i>	<i>Ref.</i>	<i>0.64 (0.57 - 0.71)</i>	<i>0.87 (0.82 - 0.92)</i>	<i>0.80 (0.74 - 0.86)</i>	<i>0.67 (0.60 - 0.74)</i>

<i>Datasets</i>	<i>Subgroups</i>	<i>Number</i>	<i>2-year C-index</i>	<i>P value</i>	<i>2-year AUC</i>	<i>P value</i>	<i>Sensitivity</i>	<i>Specificity</i>	<i>Accuracy</i>	<i>F1 score</i>
	<i>Multiple</i>	<i>n=42</i>	<i>0.71 (0.57 - 0.84)</i>	<i>0.892</i>	<i>0.80 (0.61 - 1.00)</i>	<i>0.814</i>	<i>0.73 (0.59 - 0.86)</i>	<i>0.90 (0.81 - 0.99)</i>	<i>0.81 (0.69 - 0.93)</i>	<i>0.80 (0.68 - 0.92)</i>
	<i>Manufacturers</i>									
	<i>MyLab90 Xvision</i>	<i>n=91</i>	<i>0.74 (0.65 - 0.83)</i>	<i>Ref.</i>	<i>0.79 (0.67 - 0.91)</i>	<i>Ref.</i>	<i>0.77 (0.68 - 0.85)</i>	<i>0.81 (0.73 - 0.89)</i>	<i>0.79 (0.71 - 0.88)</i>	<i>0.74 (0.65 - 0.83)</i>
	<i>Logiq E9</i>	<i>n=46</i>	<i>0.72 (0.59 - 0.85)</i>	<i>0.758</i>	<i>0.84 (0.70 - 0.98)</i>	<i>0.478</i>	<i>0.68 (0.54 - 0.81)</i>	<i>0.94 (0.88 - 1.00)</i>	<i>0.86 (0.76 - 0.96)</i>	<i>0.75 (0.63 - 0.88)</i>
	<i>Siemens Acuson Sequoia</i>	<i>n=39</i>	<i>0.75 (0.62 - 0.89)</i>	<i>0.888</i>	<i>0.88 (0.74 - 1.00)</i>	<i>0.167</i>	<i>0.87 (0.76 - 0.97)</i>	<i>0.86 (0.75 - 0.97)</i>	<i>0.86 (0.75 - 0.97)</i>	<i>0.82 (0.69 - 0.94)</i>
	<i>Resona 7</i>	<i>n=46</i>	<i>0.68 (0.54 - 0.81)</i>	<i>0.445</i>	<i>0.78 (0.63 - 0.94)</i>	<i>0.942</i>	<i>0.71 (0.58 - 0.84)</i>	<i>0.88 (0.79 - 0.97)</i>	<i>0.82 (0.70 - 0.93)</i>	<i>0.74 (0.61 - 0.87)</i>
	<i>Centers</i>									
	<i>Center A</i>	<i>n=120</i>	<i>0.76 (0.68 - 0.83)</i>	<i>Ref.</i>	<i>0.80 (0.70 - 0.90)</i>	<i>Ref.</i>	<i>0.66 (0.58 - 0.75)</i>	<i>0.89 (0.83 - 0.95)</i>	<i>0.80 (0.72 - 0.87)</i>	<i>0.73 (0.65 - 0.81)</i>
	<i>Center B</i>	<i>n=48</i>	<i>0.69 (0.56 - 0.82)</i>	<i>0.375</i>	<i>0.82 (0.68 - 0.97)</i>	<i>0.739</i>	<i>0.66 (0.53 - 0.80)</i>	<i>0.94 (0.88 - 1.00)</i>	<i>0.86 (0.76 - 0.96)</i>	<i>0.74 (0.61 - 0.86)</i>
	<i>Center D</i>	<i>n=54</i>	<i>0.73 (0.61 - 0.85)</i>	<i>0.690</i>	<i>0.80 (0.66 - 0.94)</i>	<i>0.939</i>	<i>0.71 (0.59 - 0.83)</i>	<i>0.88 (0.80 - 0.97)</i>	<i>0.83 (0.73 - 0.93)</i>	<i>0.72 (0.60 - 0.84)</i>

TA, thermal ablation; SR, surgical resection; PFS, progression-free survival; AUC, area under the curve; Ref, reference.]

2. *[Provide more detailed descriptions of the preprocessing steps for the image data, including techniques for noise reduction and standardization. Discuss how missing data and artifacts were handled during preprocessing.]*

Our Response:

Thank you for suggesting us to provide additional details on our preprocessing steps for image data. Due to word limit, we can only briefly describe this part in the main text, but we have included a more comprehensive description of our techniques in **Appendix S5**. The related content is as following:

“To minimize the impact of variability due to different hospitals, equipment, and technical parameters on subsequent analysis, we employed denoising and image augmentation techniques. Denoising was performed using advanced filtering methods, such as Gaussian filters, to remove random noise and enhance image quality. Image augmentation techniques, including rotation, scaling, translation, and contrast adjustment, were utilized to diversify the dataset, improve model robustness, and reduce the risk of overfitting. Then, in order to objectively extract tumor subregions within the tumor and alleviate human bias, we expanded all drawn regions of interest masks outward by 5mm, allowing for consideration of both the internal region of the lesion and the surrounding liver parenchyma, providing more feature information for deep learning (DL) networks. Python was used to extract irregular bounding boxes of tumor subregions from various patterns. In order to improve the processing efficiency of DL models, we transform irregular boundaries into regular square boundaries and then crop them. Additional preprocessing included normalization of size and grayscale values to produce resized and standardized grayscale ROI images for model development. Each subregion image was adjusted to a standardized size of 128×128 pixels, with grayscale values normalized to ensure uniformity across the dataset, thereby improving model training.

Gaussian filters:

$$G(x, y) = \frac{1}{2\pi\sigma^2} e^{-\frac{x^2+y^2}{2\sigma^2}}$$

Among them, x, y represents the position of the pixel; σ is the standard deviation of the pixels in the training cohort.

Normalization:

$$\text{Standardization (x)} = \frac{x-\mu}{\max(\sigma, \frac{1.0}{\sqrt{N}})}$$

$$\text{Norm (x)} = \frac{x-\min(x)}{\max(x)-\min(x)}$$

Among them, x represents the images; μ is the average value of the pixels in the training cohort; σ is the standard deviation of the pixels in the training cohort; N is the number of pixels; $\min(x)$ represents the minimum pixel value and $\max(x)$ represents maximum pixel value.”

All patients included in the training and validation datasets had complete data, and there were no missing data after applying our inclusion criteria. This is because in this study, our priority is to verify whether our hypothesis is true, and whether the model is effective. For future development, we are exploring more powerful AI methods to handle patients with partial missing data, so that the inclusion criteria can be less strict and more consistent with clinical practice. In this updated version, we added three ablation studies, investigating different training and internal validation cohort splitting ratios, various modality models, and different deep learning (DL) models. The results showed that HEROVision already had a certain ability to handle missing data, but it needs further improvement.

Images with severe artifacts or technical issues, such as significant distortion or inadequate quality, were excluded based on our exclusion criteria, as detailed in the **Appendix S1** of supplementary material. To the best of our knowledge, all AI methods perform poorly on low-quality medical imaging data right now. This is a great challenge for everyone, but it is not the subject of our study.

Revised text:

(page 3/Appendix S1 of supplementary material)

[For initial hepatocellular carcinoma (iHCC)...The exclusion criteria were: ... (6) presence of unqualified image artifacts.]

[For recurrent hepatocellular carcinoma (rHCC)...The exclusion criteria were: ... (6) presence of unqualified image artifacts.]

(page 6-7/Appendix S5 section of the revised supplementary material)

[To minimize the impact of variability due to different hospitals, equipment, and technical parameters on subsequent analysis, we employed denoising and image augmentation techniques^{2, 3}. Denoising was performed using advanced filtering methods, such as Gaussian filters, to remove random noise and enhance image quality². Image augmentation techniques, including rotation, scaling, translation, and contrast adjustment, were utilized to diversify the dataset, improve model robustness, and reduce the risk of overfitting³. Then, in order to objectively extract tumor subregions within the tumor and alleviate human bias, we expanded all drawn regions of interest masks outward by 5mm, allowing for consideration of both the internal region of the lesion and the surrounding liver parenchyma, providing more feature information for deep learning (DL) networks. Python was used to extract irregular bounding boxes of tumor subregions from various patterns. In order to improve the processing efficiency of DL models, we transform irregular boundaries into regular square boundaries and then crop them. Additional preprocessing included normalization of size and grayscale values to produce resized and standardized grayscale ROI images for model development. Each subregion image was adjusted to a standardized size of 128 × 128 pixels, with grayscale values normalized to ensure uniformity across the dataset, thereby improving model training.]

Gaussian filters:

$$G(x,y) = 1/(2\pi\sigma^2) e^{-(x^2+y^2)/(2\sigma^2)}$$

Among them, x, y represents the position of the pixel; σ is the standard deviation of the pixels in the training cohort.

Normalization:

$$\text{Standardization } (x) = (x-\mu)/(\max(\sigma, 1.0/\sqrt{N}))$$

$$\text{Norm } (x) = (x-\min(x))/(\max(x)-\min(x))$$

Among them, x represents the images; μ is the average value of the pixels in the training cohort; σ is the standard deviation of the pixels in the training cohort; N is the number of pixels; $\min(x)$ represents the minimum pixel value and $\max(x)$ represents maximum pixel value.]

3. [There is a discrepancy in the clinical settings between the development and validation sets, with the development set focusing on non-treated HCC and the validation set on recurrent HCC. Explain the rationale behind using different clinical settings for development and validation. Address how this difference might affect the model's performance and generalizability. Consider discussing any steps taken to mitigate potential biases introduced by this discrepancy.]

Our Response:

Thank you for raising this important issue. We acknowledge the discrepancy between the clinical settings of the development and validation datasets and appreciate the opportunity to clarify our approach.

1. Rationale for Using Different Clinical Settings:

The strategy to utilize non-treated initial HCC (iHCC) data for model development and recurrent HCC (rHCC) data for validation was primarily driven by practical constraints. In real-world settings, collecting a sufficient number of rHCC patients who have undergone repeated surgical resection or thermal ablation, with comprehensive follow-up data and no additional treatments between the initial and subsequent interventions, proved to be challenging. Despite working with these major tertiary hospitals located in northern, central, southern, and western China, we faced difficulties in assembling a sufficient rHCC dataset for both training and validation. When the ideal condition cannot be achieved, we can only choose the sub-ideal condition. Consequently, we trained the model by using iHCC data. We hypothesize that iHCCs are likely to have high similarities with rHCCs, in terms of tumor biology and liver environment. If this is true, our model should perform well in the external testing using rHCC data exclusively. Our results proved that the hypothesis was indeed correct.

2. Biological Behavior:

From a pathological and tumor microenvironment perspective, the biological behavior of rHCC closely mirrors that of iHCC. Time to recurrence has been widely recognized as a biological surrogate marker: early recurrence, typically within 2 years post-surgery, is believed

to share the same clonal origin as the iHCC, while late recurrence, occurring after more than 2 years, is thought to arise *de novo*, driven by underlying liver damage that persists despite tumor resection[1-3]. Furthermore, Ding et al.[4] compared genetic features and found that 52% of rHCCs shared the clonal lineage of the initial tumor. Both conditions are characterized by a similar liver environment due to underlying cirrhosis. This biological similarity supports our decision to use iHCC data for model development, offering a practical solution to the challenge of obtaining sufficient rHCC data. Although this approach may not be perfect, it effectively addresses data scarcity and enables meaningful model development.

3. Impact on Model Performance and Acceptability:

We recognize that the differences between iHCC and rHCC could lead to performance variations when applying a model trained on iHCC data to rHCC data. Our study found that the 2-year AUC was 0.90 for the internal validation cohort and 0.78 for the external testing cohort in the TA dataset, and 0.92 and 0.81, respectively, in the SR dataset. The observed differences of 0.12 and 0.11 in AUC between internal and external validation are statistically significant but are within an acceptable range given the context of the study.

4. Innovation and Model Utility:

Despite the discrepancy, our approach represents a novel method for developing AI models specifically for rHCC, addressing a significant gap in current clinical guidelines and staging systems. Our model has demonstrated superior prognostic accuracy compared to the six existing guidelines and staging methods, offering a substantial improvement in predictive performance. We think this may not be the best approach, and the model did not achieve the highest accuracy, but it solved the problem from scratch. No AI model for rHCC has ever been effectively built before, probably because of the lack of data.

5. Generalizability and Future Validation:

To further assess the model's generalizability, we included data from two additional centers involving repeated surgical resection and thermal ablation. These analyses confirmed the model's robustness and applicability across various clinical settings. However, we acknowledge the need for further independent validation, particularly with prospective datasets, to fully establish the model's efficacy and generalizability. This limitation is recognized as a key area for future research.

Overall, while the strategy may have limitations, it provides a feasible approach to model development in the absence of sufficient data, and has yielded promising results in enhancing prognostic prediction for rHCC. We are committed to continuing validation and refinement of the model through future prospective studies to further validate its practical effectiveness. We have supplemented the discussion in the **Discussion section** of the revised manuscript.

References:

- [1] T. Yoh *et al.*, "Surgery for Recurrent Hepatocellular Carcinoma: Achieving Long-term

- Survival," (in eng), *Ann Surg*, vol. 273, no. 4, pp. 792-799, Apr 1 2021, doi: 10.1097/sla.0000000000003358.
- [2] H. Imamura *et al.*, "Risk factors contributing to early and late phase intrahepatic recurrence of hepatocellular carcinoma after hepatectomy," (in eng), *J Hepatol*, vol. 38, no. 2, pp. 200-7, Feb 2003, doi: 10.1016/s0168-8278(02)00360-4.
- [3] J. C. Wu *et al.*, "Risk factors for early and late recurrence in hepatitis B-related hepatocellular carcinoma," (in eng), *J Hepatol*, vol. 51, no. 5, pp. 890-7, Nov 2009, doi: 10.1016/j.jhep.2009.07.009.
- [4] X. Ding *et al.*, "Genomic and Epigenomic Features of Primary and Recurrent Hepatocellular Carcinomas," (in eng), *Gastroenterology*, May 16 2020, doi: 10.1053/j.gastro.2019.09.056.

Revised text:

(page 11/Discussion section of the revised manuscript)

[In our study, we developed HEROVision using iHCC data and validated it with rHCC data for several reasons. On one hand, the strategy to use iHCC for model development and rHCC for validation was primarily driven by practical constraints. Despite our collaboration with hospitals across northern, central, southern, and western China, assembling a large rHCC dataset for training was challenging due to difficulties in collecting a sufficient number of rHCC patients with comprehensive follow-up data and no additional treatments between interventions. On the other hand, from a pathological and tumor microenvironment (TME) perspective, the biological behavior of rHCC closely mirrors that of iHCC. Time to recurrence has been widely recognized as a biological surrogate marker: early recurrence, typically within 2 years post-surgery, is believed to share the same clonal origin as the iHCC, while late recurrence, occurring after more than 2 years, is thought to arise from a de novo origin, driven by underlying liver damage that persists despite tumor resection^{27, 28, 29}. Furthermore, Ding *et al.*³⁰ compared genetic features and found that 52% of rHCCs share the clonal lineage of the initial tumor. Both conditions are characterized by a similar liver environment due to underlying cirrhosis. This biological similarity supports our decision to use iHCC data for model development, offering a practical solution to the challenge of obtaining sufficient rHCC data. Although this approach may not be perfect, it effectively addresses data scarcity and enables meaningful model development. HEROVision demonstrated effectiveness in rHCC patients, as evidenced by our study: the 2-year AUC was 0.90 for the internal validation cohort and 0.78 for the external testing cohort in the TA dataset, and 0.92 and 0.81, respectively, in the SR dataset.]

4. [Description on page 5 lack clarity regarding which datasets were used for development and validation. Reword the description on page 5 to clearly specify which datasets were designated for external testing, particularly lines 121 and 122 are confusing as it seems part of the development cohort was used for external validation. While Figure 5 is very clear, the accompanying text needs to be more precise.]

Our Response:

Thank you for your comment. We have revised the description in the **Methods** section of the revised manuscript to clearly specify which datasets were designated for external testing.

Revised text:

(page 13/Methods section of the revised manuscript)

[Finally, the study comprised 1,605 patients with HCC. Among the patients with iHCC, 772 underwent first-line treatment between January 2015 and December 2022 at Center A, including 391 cases of TA (1stTA dataset) and 381 cases of SR (1stSR dataset). The 1stTA and 1stSR datasets were divided into a training cohort ($n = 311$ and $n = 322$, respectively) and an internal validation cohort ($n = 80$ and $n = 59$, respectively) in a 4:1 ratio, based on chronological order. Among the patients with rHCC, 833 received second-line therapies between May 2010 and August 2023 from three centers, including 611 cases of TA (2ndTA dataset) and 222 cases of SR (2ndSR dataset), which were designated as the external testing cohorts. PSM (1:1) was performed using binary logistic regression to reduce the effects of selection bias and confounding factors between the 2ndTA and 2ndSR datasets³⁹. After PSM, 214 patients from the 2ndTA dataset and 214 patients from the 2ndSR dataset were successfully matched.]

5. [A cohort from center A was included as an external validation set, but the patients were treated, and the scans were performed at the same center as the development cohort, which challenges the definition of "external validation." To ensure proper generalization testing, consider including results from only truly external centers (e.g., centers B and C).]

Our Response:

Thank you for your comment.

We acknowledge the limitation that some patients from center A were included in both the training/internal validation and external testing cohorts. We have addressed this issue by supplementing the **Discussion and Limitations** sections of the revised manuscript to reflect this concern.

However, we would like to clarify that the patients from center A's development cohort (used for training and internal validation) and those in the external testing cohort were independent of each other. The training/internal validation cohort from center A consisted

entirely of iHCC patients, while the rHCC patients from center A were exclusively used for external testing. There was no overlap between these groups (**Figure 1**). Additionally, the data collection periods for iHCC and rHCC patients at center A differed, further minimizing any potential overlap or bias.

As previously mentioned, the number of rHCC patients who underwent repeated surgical resection with available prognostic data is extremely limited. Therefore, we included all rHCC patients from center A in the external testing cohort to ensure robust validation of our model's effectiveness.

Revised text:

(page 11/Discussion section of the revised manuscript)

[In our study, we developed HEROVision using iHCC data and validated it with rHCC data for several reasons. On one hand, the strategy to use iHCC for model development and rHCC for validation was primarily driven by practical constraints. Despite our collaboration with hospitals across northern, central, southern, and western China, assembling a large rHCC dataset for training was challenging due to difficulties in collecting a sufficient number of rHCC patients with comprehensive follow-up data and no additional treatments between interventions. On the other hand, from a pathological and tumor microenvironment (TME) perspective, the biological behavior of rHCC closely mirrors that of iHCC. Time to recurrence has been widely recognized as a biological surrogate marker: early recurrence, typically within 2 years post-surgery, is believed to share the same clonal origin as the iHCC, while late recurrence, occurring after more than 2 years, is thought to arise from a de novo origin, driven by underlying liver damage that persists despite tumor resection^{27, 28, 29}. Furthermore, Ding et al.³⁰ compared genetic features and found that 52% of rHCCs share the clonal lineage of the initial tumor. Both conditions are characterized by a similar liver environment due to underlying cirrhosis. This biological similarity supports our decision to use iHCC data for model development, offering a practical solution to the challenge of obtaining sufficient rHCC data. Although this approach may not be perfect, it effectively addresses data scarcity and enables meaningful model development. HEROVision demonstrated effectiveness in rHCC patients, as evidenced by our study: the 2-year AUC was 0.90 for the internal validation cohort and 0.78 for the external testing cohort in the TA dataset, and 0.92 and 0.81, respectively, in the SR dataset.]

(page 12/Discussion section of the revised manuscript)

[Our study has several limitations. ... Second, because the number of patients with rHCC underwent RHR is limited in clinical practice, we were forced to train HEROVision by employing iHCC data. Although the results proved that our strategy was correct, the performance of HEROVision is likely to be enhanced, if it was sufficiently trained by real rHCC data. Finally, the HEROVision-TA and -SR models were developed for patients experiencing

early-stage recurrence after SR, and thus may not be applicable to advanced-stage recurrence.]

(page 12/Discussion section of the revised manuscript)

L

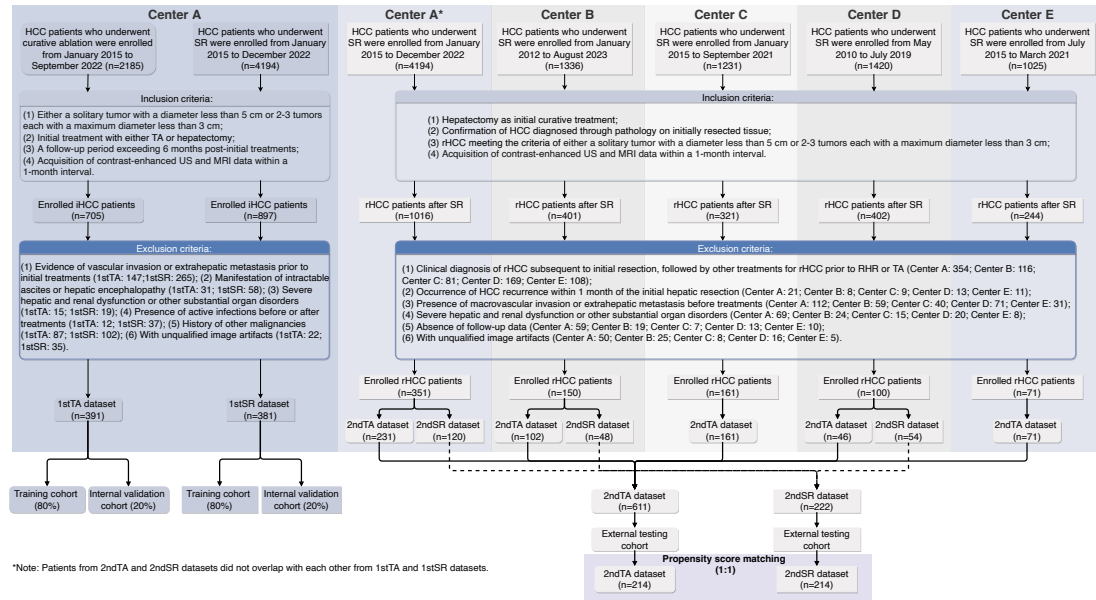


Figure 1. Flowchart of Patient Inclusion and Exclusion

iHCC, initial hepatocellular carcinoma; rHCC, recurrent hepatocellular carcinoma; TA, thermal ablation; SR, surgical resection.]

● **Response to Reviewer 2**

1. [Noteworthy results:]

- Model Performance and Comparative Advantage: HEROVision, which was developed based on a Vision Transformer (ViT) structure, showed high accuracy in predicting the prognosis for recurrent hepatocellular carcinoma (rHCC). In external testing cohorts, HEROVision-TA (thermal ablation) achieved a 2-year concordance index (C-index) of 0.72, while HEROVision-SR (surgical resection) reached 0.74. The 2-year area under the curve (AUC) values were 0.78 for TA and 0.82 for SR. These results outperformed the six guideline staging systems, which had 2-year C-index values between 0.51 and 0.58 and 2-year AUC values between 0.51 and 0.61. Additionally, integrating HEROVision into these staging systems improved their 2-year progression-free survival (PFS) prediction accuracy.

- Risk Stratification: HEROVision effectively categorized patients into high- and low-risk groups. In the external testing cohort, the 2-year PFS rates for low-risk patients were 79.0% for TA and 75.8% for SR, compared to 42.2% for TA and 31.9% for SR in the high-risk groups. HEROVision identified that 26.0% of high-risk patients could potentially improve their prognosis by switching from TA to SR.

- Contribution to the Field and Clinical Impact: HEROVision represents a potential advancement in the treatment of HCC by combining a Vision Transformer-based model with multimodal imaging data, including ultrasound (US) and magnetic resonance imaging (MRI), along with clinical variables. This approach enhances risk stratification and treatment decision-making for patients with rHCC and could surpass the current guideline staging systems. By providing personalized treatment recommendations based on preoperative data, HEROVision addresses a potential gap in clinical guidelines. The model's precision in identifying high-risk patients who could benefit from changing their treatment underscores its practical value in enhancing therapeutic strategies for early-stage rHCC.]

Our Response:

Thank you for your insightful comments on our HEROVision model. We are pleased that you found the model's performance, risk stratification capabilities, and potential clinical impact noteworthy. We appreciate your recognition of our work's contribution to enhancing patient outcomes through a Vision Transformer-based approach combined with multimodal imaging and clinical data. Your comment reinforces the value of our model and motivates us to continue refining and validating this approach to maximize its impact on clinical practice.

2. [Significance:]

Although superior performance compared to existing staging systems is demonstrated, using deep learning methods in HCC has been described many times. Previous studies have explored machine learning and deep learning techniques to predict early recurrence and optimize

treatment for HCC. For instance, other models have successfully used multiparametric MRI and CT imaging combined with clinical data to predict early recurrence and optimize treatment decisions: "Zhan, G., Wang, F., Wang, W., Li, Y., Chen, Q., Hu, H., & Chen, Y.-W. (2023). A transformer-based model for preoperative early recurrence prediction of hepatocellular carcinoma with multi-modality MRI. Computer Vision – ACCV 2022 Workshops, 185–194. https://doi.org/10.1007/978-3-031-27066-6_13". The novelty of HEROVision lies in its specific implementation of a specific Vision Transformer-based network and its integration with clinical decision-making tools, which could potentially enhance individualized treatment plans. The paper is therefore moderately innovative. Since the data are not publicly available, benchmarking by other groups is not possible which further dampens enthusiasm.]

Our Response:

Thank you for your thoughtful comment. We appreciate your recognition of HEROVision's specific implementation and its potential contribution to personalized treatment plans. While it is true that the data used in our study are not publicly available due to patient privacy considerations, we have focused on demonstrating the potential of HEROVision to improve patient outcomes and offer personalized treatment recommendations. We are committed to further validation and dissemination of our findings, and we hope that future efforts will include making data available for broader benchmarking.

3. *[The paper fails to compare its results with state-of-the-art (SOTA) Vision Transformer-based models that support multimodal data or publicly available datasets. Vision Transformers rely on the quality and diversity of the dataset because their performance is tied to the patterns they learn during training. Without testing on different publicly available datasets, it is difficult to establish the generalizability and robustness of the proposed approach. Moreover, the diversity of the training data is important for training Vision Transformers. Any bias in the training data, such as race or gender bias, can affect the model's predictions.]*

Our Response:

Thank you for your insightful comment.

1. Comparison with State-of-the-Art Models:

We compared our ViT model with other state-of-the-art (SOTA) models on both TA and SR datasets using the same experimental setup. This included two CNN models: ResNet50[1] and ConvNext[2], and two Transformer-based models: original ViT[3] and GC-ViT[4]. The results showed that our ViT model outperformed other SOTA models in terms of the 2-year C-index and AUC on both the TA dataset (2-year C-index: 0.72 vs. 0.68–0.71, $P = 0.073$ –0.633; 2-year AUC: 0.78 vs. 0.72–0.77, $P = 0.019$ –0.744) and SR dataset (2-year C-index: 0.72 vs. 0.63–0.68, $P = 0.064$ –0.381; 2-year AUC: 0.81 vs. 0.72–0.79, $P = 0.022$ –0.647), regardless of whether they were in the external testing cohort. Detailed results can be found in **Figures S12**,

S13 and Table S6. However, it is important to note that currently available publicly accessible datasets do not include both CEUS and MRI data for HCC patients prior to ablation and surgical treatment. To further assess our approach, we conducted ablation studies which showed that: (1) Varying the dataset partitioning ratios had minimal impact on model performance (**Figure S10 and Table S4**). (2) The fusion modality (CEUS + MRI) outperformed solitary modalities in both C-index and time-independent AUC (**Figure S11 and Table S5**).

2. Clinical Relevance:

Our model was specifically developed to address clinical challenges and fill gaps in current guidelines. The primary focus was to ensure that the model and its strategy are effective and feasible in a clinical setting. Our research demonstrated that HEROVision is accurate and effective in the largest rHCC cohort studied to date. While further prospective clinical trials on a larger scale are needed to validate the robustness and generalizability of the model, this will be the focus of our future research.

3. Robustness and Diversity of the Dataset:

To further demonstrate the robustness of our HEROVision model, we included data from two additional centers and conducted subgroup analyses on the 2ndSR and 2ndTA datasets, considering variables such as sex, age, tumor size, number of tumors, manufacturer, and centers. Although our dataset lacks racial diversity, it includes significant variability across other important factors. Our subgroup analyses indicated that differences between these subgroups did not significantly impact the performance of our model. Detailed results are presented in **Figures S14, S15 and Table S7**.

References:

- [1] K. He, X. Zhang, S. Ren, and J. Sun, "Deep residual learning for image recognition," in *Proceedings of the IEEE conference on computer vision and pattern recognition*, 2016, pp. 770-778.
- [2] Z. Liu, H. Mao, C.-Y. Wu, C. Feichtenhofer, T. Darrell, and S. Xie, "A convnet for the 2020s," in *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*, 2022, pp. 11976-11986.
- [3] A. Dosovitskiy, "An image is worth 16x16 words: Transformers for image recognition at scale," *arXiv preprint arXiv:2010.11929*, 2020.
- [4] A. Hatamizadeh, H. Yin, G. Heinrich, J. Kautz, and P. Molchanov, "Global context vision transformers," in *International Conference on Machine Learning*, 2023: PMLR, pp. 12633-12646.

Revised text:

(page 14-15/Methods section of the revised manuscript)

[Ablation Studies

We performed three ablation studies, investigating different training and internal validation cohort splitting ratios, various modality models, and different DL models. For the splitting ratios of the TA and SR datasets, based on chronological order, we retrained our ViT model under three settings, using 3/4, 4/5, and 5/6 of the training cohort, respectively, and then validated the model in each setting on the internal validation cohort. In exploring different modality models, we assessed the contributions of CEUS and MRI modalities to the final performance of our ViT model on both the training and internal validation cohorts. Lastly, for different DL models, we compared our ViT model with other DL models on both TA and SR datasets using the same experimental setup. This included two CNN models: ResNet^{50 49} and ConvNext⁵⁰, and two Transformer-based models: original ViT⁵¹ and GC-ViT⁵².]

(page 15/Methods section of the revised manuscript)

[Subgroup Analyses

To address potential variability and ensure the robustness of the findings, we conducted subgroup analyses in the 2ndTA and 2ndSR datasets, including sex (female and male), age (≤ 60 years and > 60 years), tumor size (≤ 3 cm and > 3 cm), the number of tumors (solitary and multiple), manufacturers (MyLab90 Xvision, Logiq E9, Siemens Acuson Sequoia, and Resona 7), and centers (Centers A, B, C, D, and E).]

(page 39/Figure S10 of the revised supplementary material)

[

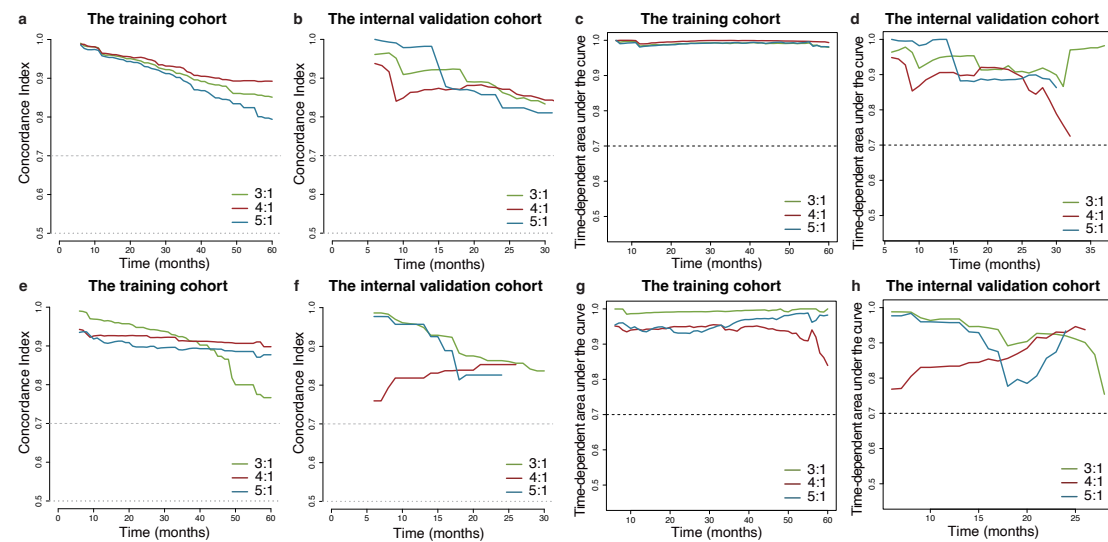


Figure S10. The Discriminative Performances of Different Training and Internal Validation Cohort Splitting Ratios

For TA dataset (a-d) and SR dataset (e-h), the C-index in the training and internal validation cohorts (a, b, e and f), and the time-dependent AUC in both cohorts (c, d, g and h) reflected the models' discriminatory performance at various points.

TA, thermal ablation; SR, surgical resection; C-index, concordance index; AUC, area under the curve.

(page 40/Figure S11 of the revised supplementary material)

[

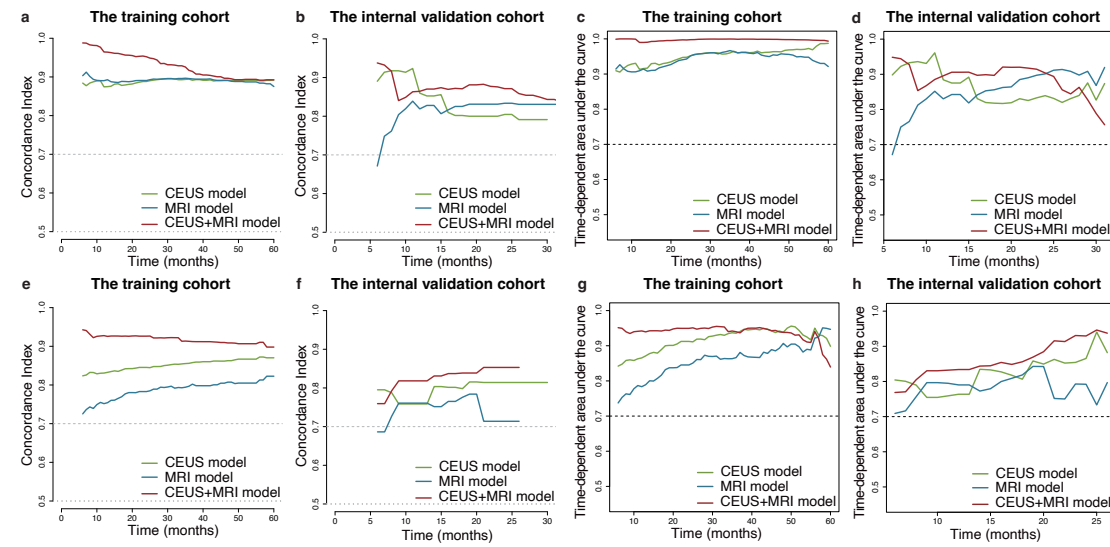


Figure S11. The Discriminative Performances of Different Modality Models in the Training and Internal Validation Cohorts

For TA dataset (a-d) and SR dataset (e-h), the C-index in the training and internal validation cohorts (a, b, e and f), and the time-dependent AUC in both cohorts (c, d, g and h) reflected the models' discriminatory performance at various points.

TA, thermal ablation; SR, surgical resection; C-index, concordance index; AUC, area under the curve; MRI, magnetic resonance imaging; CEUS, contrast-enhanced ultrasound.

(page 41/Figure S12 of the revised supplementary material)

[

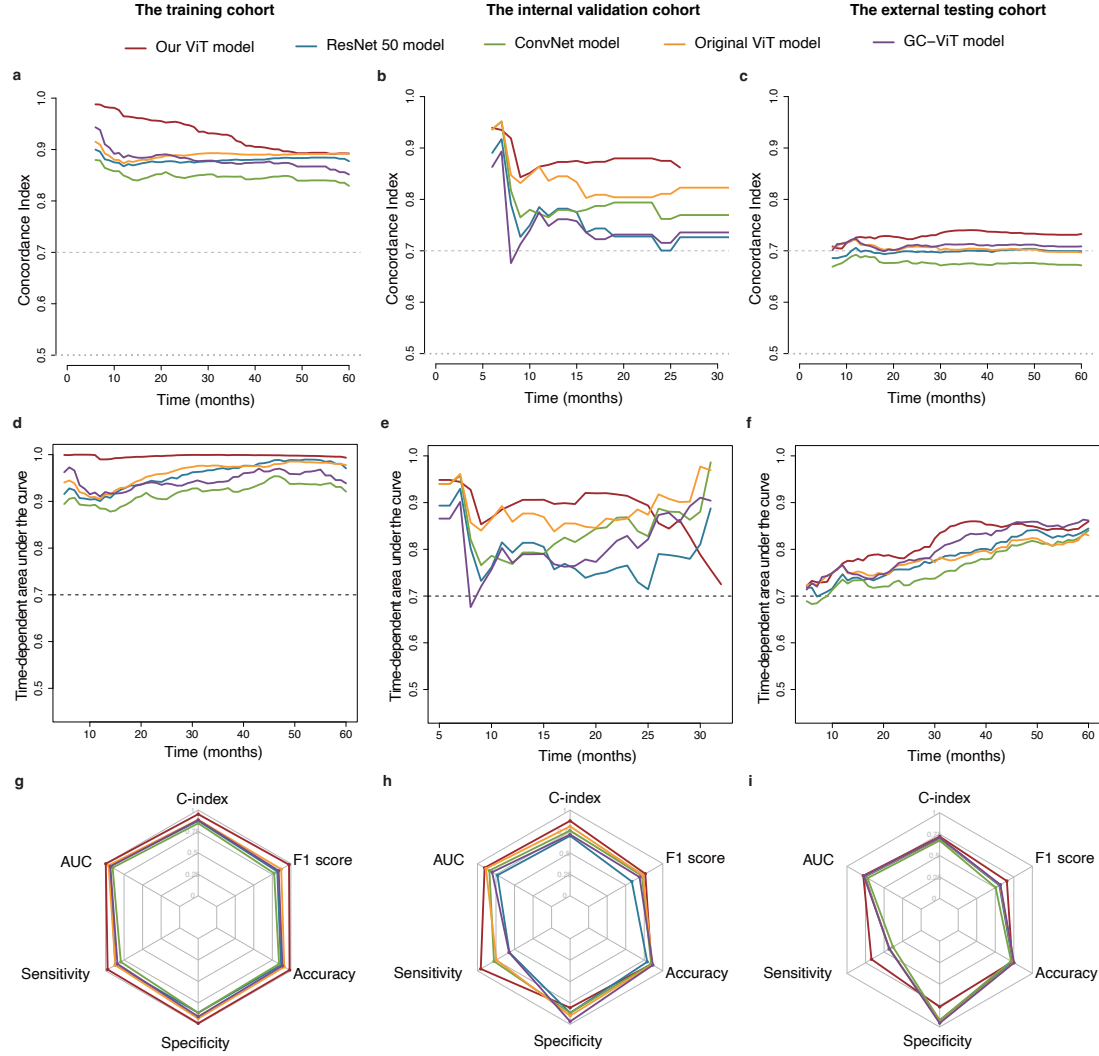


Figure S12. The Discriminative Performances of Different Deep Learning Models for the TA Dataset

The C-index (a-c) and time-dependent AUC (d-f) of different deep learning models demonstrated the models' discriminatory performance at various points in the training, internal validation, and external testing cohorts. (g-i) Radar charts illustrated the C-index, AUC, specificity, sensitivity, accuracy, and F1 score for 2-year PFS across these three cohorts. TA, thermal ablation; ViT, Vision Transformer; C-index, concordance index; AUC, area under the curve; PFS, progression-free survival.]

(page 42/Figure S13 of the revised supplementary material)

[

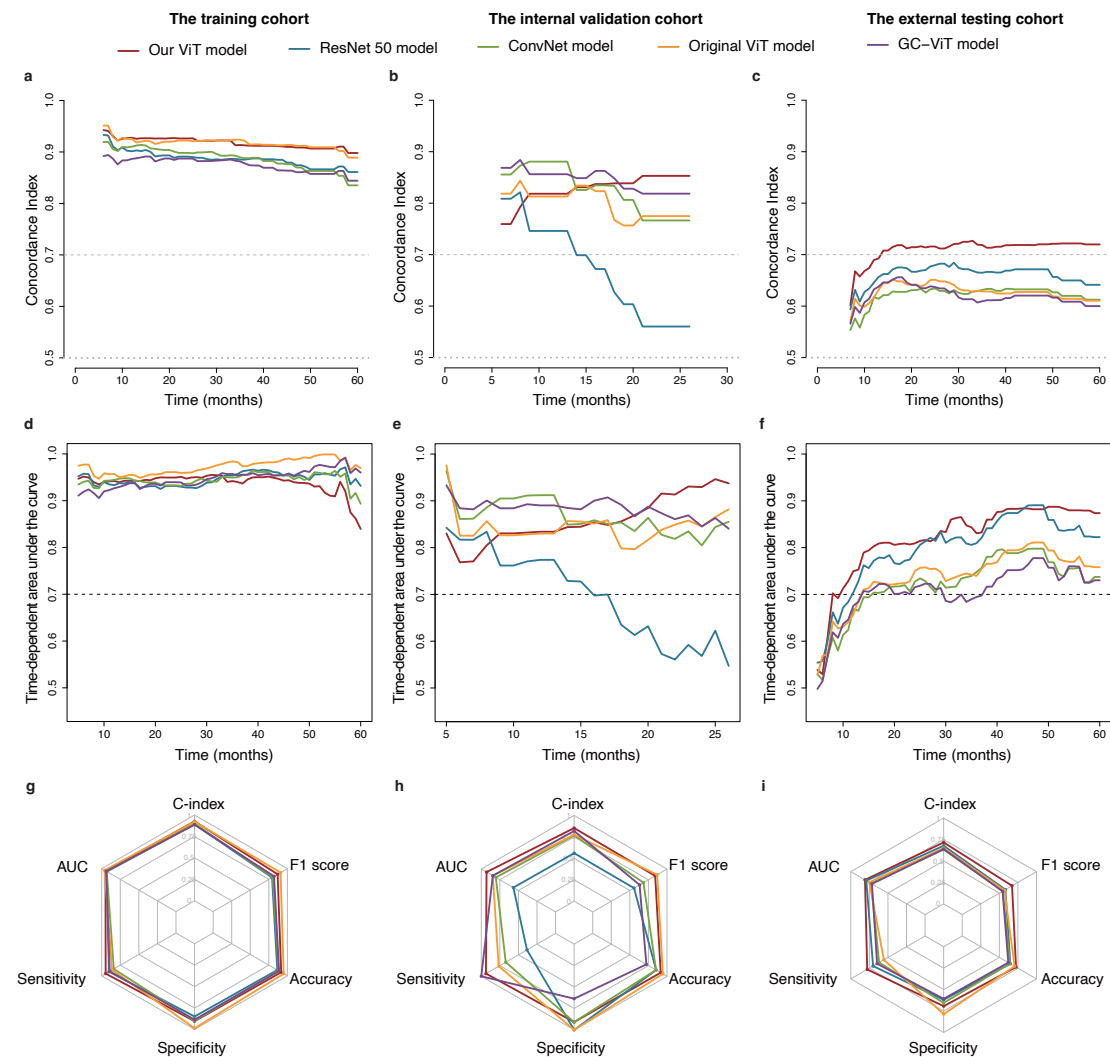


Figure S13. The Discriminative Performances of Different Deep Learning Models for the SR Dataset

The C-index (a-c) and time-dependent AUC (d-f) of different deep learning models demonstrated the models' discriminatory performance at various points in the training, internal validation, and external testing cohorts. (g-i) Radar charts illustrated the C-index, AUC, specificity, sensitivity, accuracy, and F1 score for 2-year PFS across these three cohorts. SR, surgical resection; ViT, Vision Transformer; C-index, concordance index; AUC, area under the curve; PFS, progression-free survival.]

(page 43-44/Figure S14 of the revised supplementary material)

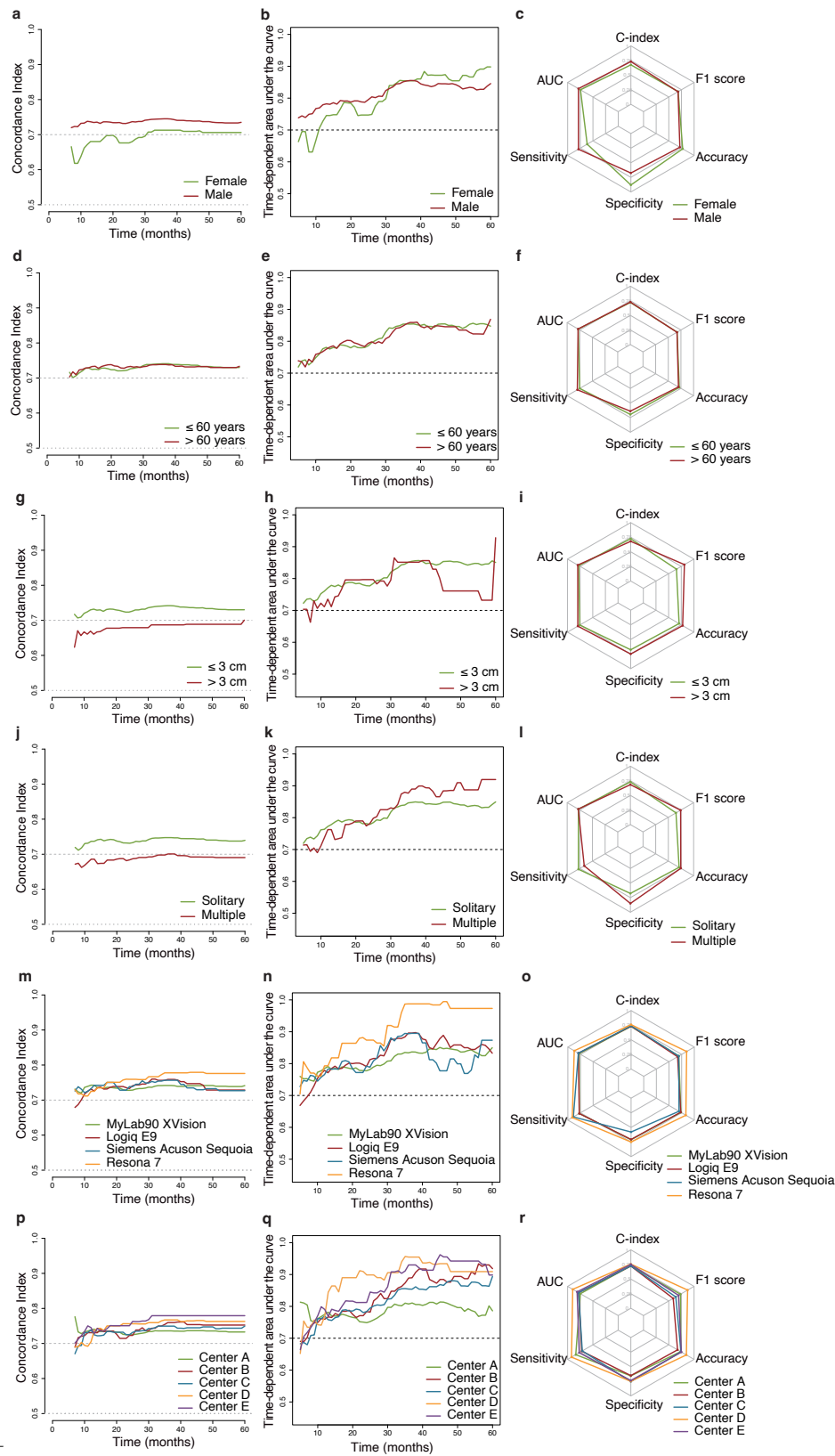


Figure S14. The Discriminative Performances of Different Subgroups for the 2ndTA Dataset
The C-index (a, d, g, j, m, and p) and time-dependent AUC (b, e, h, k, n, and q) for different
subgroups, including sex, age, tumor size, the number of tumors, manufacturers, and centers.

demonstrated the models' discriminatory performance at various points in the 2ndTA cohort. (c, f, i, l, o, and r) Radar charts illustrated the C-index, AUC, specificity, sensitivity, accuracy, and F1 score for 2-year PFS across these subgroups. TA, thermal ablation; ViT, Vision Transformer; C-index, concordance index; AUC, area under the curve; PFS, progression-free survival.]

(page 45-46/Figure S15 of the revised supplementary material)

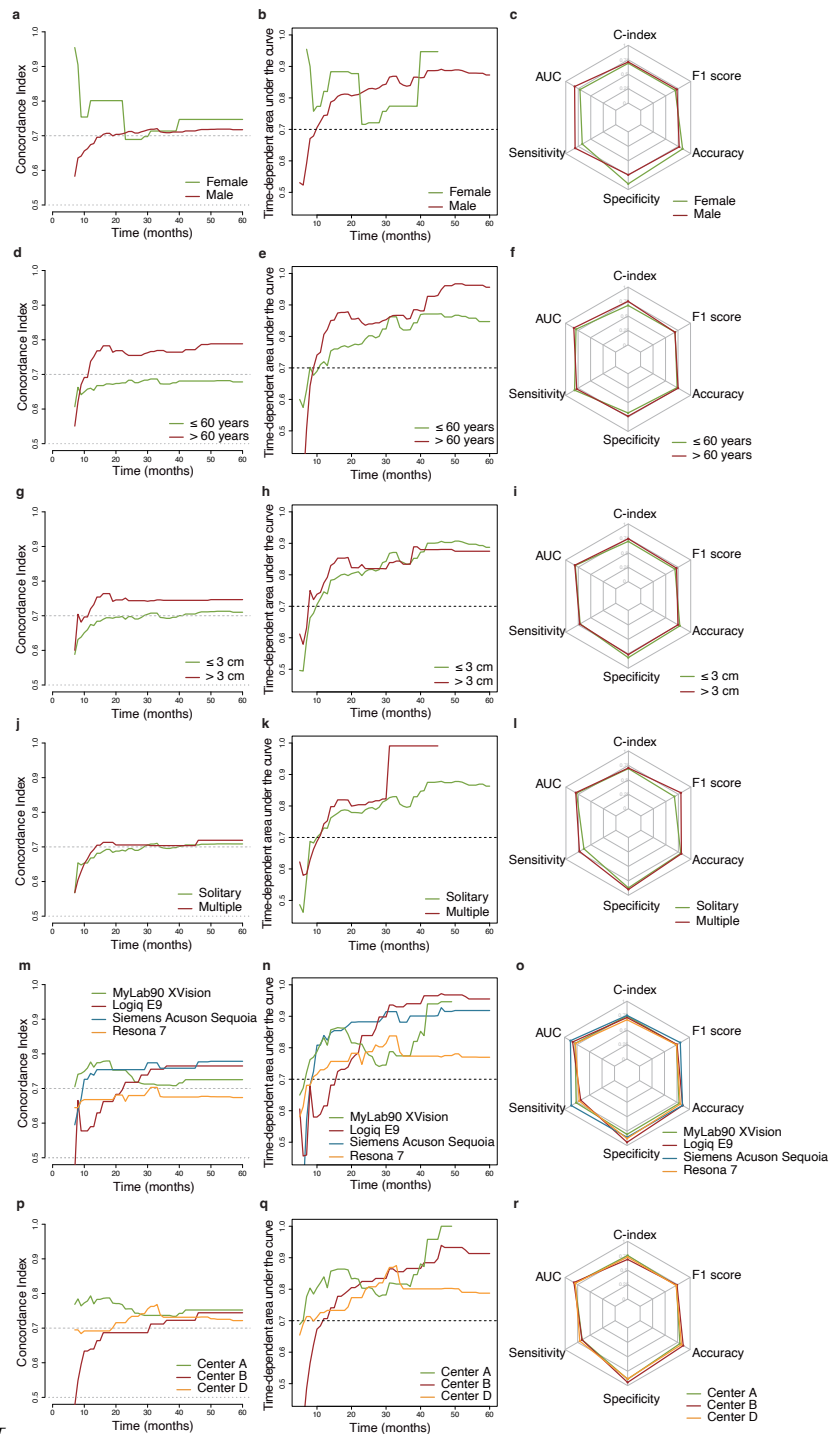


Figure S15. The Discriminative Performances of Different Subgroups for the 2ndSR Dataset

The C-index (a, d, g, j, m, and p) and time-dependent AUC (b, e, h, k, n, and q) for different subgroups, including sex, age, tumor size, the number of tumors, manufacturers, and centers, demonstrated the models' discriminatory performance at various points in the 2ndSR cohort. (c, f, i, l, o, and r) Radar charts illustrated the C-index, AUC, specificity, sensitivity, accuracy, and F1 score for 2-year PFS across these subgroups.

TA, thermal ablation; ViT, Vision Transformer; C-index, concordance index; AUC, area under the curve; PFS, progression-free survival.]

(page 21/Table S4 of the revised supplementary material)

[Table S4. Performance Comparison of Different Training and Internal Validation Cohort Splitting Ratios for 2-Year PFS

Cohorts	Ratios	TA dataset				SR dataset			
		2-year C-index	P Value	2-year AUC	P Value	2-year C-index	P Value	2-year AUC	P Value
The training cohort	3:1	0.94 (0.91 - 0.97)	Ref.	0.99 (0.97 - 1.00)	Ref.	0.95 (0.92 - 0.97)	Ref.	0.99 (0.98 - 1.00)	Ref.
	4:1	0.95 (0.93 - 0.98)	0.496	1.00 (0.99 - 1.00)	0.341	0.93 (0.90 - 0.95)	0.266	0.95 (0.92 - 0.98)	0.001
	5:1	0.93 (0.90 - 0.96)	0.634	0.99 (0.97 - 1.00)	0.913	0.90 (0.86 - 0.93)	0.016	0.93 (0.90 - 0.96)	<0.001
The internal validation cohort	3:1	0.86 (0.79 - 0.93)	Ref.	0.91 (0.82 - 1.00)	Ref.	0.86 (0.79 - 0.94)	Ref.	0.92 (0.83 - 1.00)	Ref.
	4:1	0.87 (0.80 - 0.94)	0.866	0.90 (0.81 - 1.00)	0.932	0.85 (0.76 - 0.94)	0.866	0.93 (0.82 - 1.00)	0.835
	5:1	0.82 (0.73 - 0.92)	0.502	0.89 (0.74 - 1.00)	0.663	0.83 (0.72 - 0.94)	0.589	0.93 (0.80 - 1.00)	0.772

TA, thermal ablation; SR, surgical resection; PFS, progression-free survival; AUC, area under the curve; Ref, reference.

(page 22/Table S5 of the revised supplementary material)

[Table S5. Performance Comparison of the Different Modality Models for 2-year PFS in the Training and Internal Validation Cohorts

Cohorts	Models	TA dataset				SR dataset			
		2-year C-index	P Value	2-year AUC	P Value	2-year C-index	P Value	2-year AUC	P Value
The training cohort	CEUS+MRI	0.95 (0.93 - 0.98)	Ref.	1.00 (0.99 - 1.00)	Ref.	0.93 (0.90 - 0.95)	Ref.	0.95 (0.92 - 0.98)	Ref.
	CEUS	0.89 (0.85 - 0.92)	0.003	0.95 (0.92 - 0.98)	<0.001	0.85 (0.81 - 0.88)	0.001	0.91 (0.88 - 0.94)	0.063
	MRI	0.89 (0.86 - 0.93)	0.004	0.94 (0.92 - 0.96)	<0.001	0.78 (0.74 - 0.83)	<0.001	0.85 (0.80 - 0.89)	<0.001
The internal validation cohort	CEUS+MRI	0.87 (0.80 - 0.94)	Ref.	0.90 (0.81 - 1.00)	Ref.	0.85 (0.76 - 0.94)	Ref.	0.93 (0.82 - 1.00)	Ref.
	CEUS	0.80 (0.72 - 0.89)	0.254	0.84 (0.72 - 0.96)	0.224	0.81 (0.71 - 0.91)	0.570	0.87 (0.74 - 0.99)	0.250
	MRI	0.83 (0.75 - 0.92)	0.504	0.90 (0.80 - 1.00)	0.956	0.71 (0.60 - 0.83)	0.063	0.79 (0.58 - 1.00)	0.028

TA, thermal ablation; SR, surgical resection; PFS, progression-free survival; AUC, area under the curve; MRI, magnetic resonance imaging; CEUS, contrast-enhanced ultrasound; Ref, reference.

(page 23/Table S6 of the revised supplementary material)

[Table S6. Performance Comparison of the Different Deep Learning Models for 2-year PFS in the Training, Internal Validation, and External Testing Cohorts]

<i>Datasets</i>	<i>Cohorts</i>	<i>Models</i>	<i>2-year C-index</i>	<i>P value</i>	<i>2-year AUC</i>	<i>P value</i>	<i>Sensitivity</i>	<i>Specificity</i>	<i>Accuracy</i>	<i>F1 score</i>
<i>TA dataset</i>	<i>The training cohort</i>	<i>Our ViT model</i>	0.95 (0.93 - 0.98)	<i>Ref.</i>	1.00 (0.99 - 1.00)	<i>Ref.</i>	0.97 (0.95 - 0.99)	0.99 (0.98 - 1.00)	0.98 (0.97 - 1.00)	0.98 (0.96 - 0.99)
		<i>ResNet 50 model</i>	0.87 (0.84 - 0.91)	0.001	0.94 (0.92 - 0.97)	<0.001	0.87 (0.83 - 0.90)	0.87 (0.83 - 0.90)	0.87 (0.83 - 0.90)	0.82 (0.77 - 0.86)
		<i>ConvNext model</i>	0.85 (0.81 - 0.89)	<0.001	0.90 (0.87 - 0.94)	<0.001	0.79 (0.75 - 0.84)	0.87 (0.83 - 0.90)	0.84 (0.80 - 0.88)	0.77 (0.73 - 0.82)
		<i>Original ViT model</i>	0.89 (0.85 - 0.92)	0.003	0.96 (0.94 - 0.98)	0.001	0.87 (0.83 - 0.91)	0.93 (0.91 - 0.96)	0.91 (0.88 - 0.94)	0.87 (0.83 - 0.91)
		<i>GC-ViT model</i>	0.88 (0.85 - 0.92)	0.002	0.94 (0.91 - 0.96)	<0.001	0.84 (0.80 - 0.88)	0.91 (0.88 - 0.94)	0.89 (0.85 - 0.92)	0.84 (0.79 - 0.88)
	<i>The internal validation cohort</i>	<i>Our ViT model</i>	0.87 (0.80 - 0.95)	<i>Ref.</i>	0.90 (0.81 - 1.00)	<i>Ref.</i>	0.96 (0.91 - 1.00)	0.81 (0.72 - 0.89)	0.85 (0.77 - 0.92)	0.76 (0.67 - 0.86)
		<i>ResNet 50 model</i>	0.70 (0.60 - 0.80)	0.006	0.73 (0.58 - 0.88)	0.004	0.57 (0.46 - 0.68)	0.87 (0.79 - 0.94)	0.79 (0.70 - 0.88)	0.58 (0.47 - 0.69)
		<i>ConvNext model</i>	0.76 (0.67 - 0.86)	0.063	0.84 (0.71 - 0.96)	0.223	0.78 (0.68 - 0.87)	0.87 (0.79 - 0.94)	0.84 (0.76 - 0.92)	0.72 (0.62 - 0.82)
		<i>Original ViT model</i>	0.81 (0.72 - 0.90)	0.266	0.89 (0.79 - 0.98)	0.706	0.75 (0.65 - 0.84)	0.90 (0.83 - 0.97)	0.86 (0.78 - 0.94)	0.73 (0.63 - 0.83)
		<i>GC-ViT model</i>	0.72 (0.62 - 0.81)	0.011	0.80 (0.67 - 0.93)	0.070	0.57 (0.46 - 0.68)	0.97 (0.93 - 1.00)	0.87 (0.79 - 0.94)	0.69 (0.58 - 0.79)
	<i>The external testing cohort</i>	<i>Our ViT model</i>	0.72 (0.69 - 0.76)	<i>Ref.</i>	0.78 (0.74 - 0.82)	<i>Ref.</i>	0.67 (0.63 - 0.71)	0.77 (0.73 - 0.80)	0.73 (0.69 - 0.76)	0.66 (0.62 - 0.69)
		<i>ResNet 50 model</i>	0.70 (0.66 - 0.73)	0.355	0.76 (0.71 - 0.80)	0.325	0.43 (0.39 - 0.47)	0.92 (0.90 - 0.95)	0.74 (0.70 - 0.77)	0.56 (0.52 - 0.59)
		<i>ConvNext model</i>	0.68 (0.64 - 0.71)	0.073	0.72 (0.68 - 0.77)	0.019	0.38 (0.34 - 0.42)	0.92 (0.90 - 0.94)	0.71 (0.68 - 0.75)	0.50 (0.46 - 0.54)
		<i>Original ViT model</i>	0.71 (0.67 - 0.74)	0.562	0.77 (0.73 - 0.81)	0.560	0.43 (0.39 - 0.47)	0.95 (0.93 - 0.97)	0.75 (0.71 - 0.78)	0.57 (0.53 - 0.60)
		<i>GC-ViT model</i>	0.71 (0.67 - 0.75)	0.633	0.77 (0.73 - 0.81)	0.744	0.43 (0.39 - 0.47)	0.96 (0.94 - 0.97)	0.75 (0.72 - 0.79)	0.57 (0.53 - 0.61)
<i>SR dataset</i>	<i>The training cohort</i>	<i>Our ViT model</i>	0.93 (0.90 - 0.95)	<i>Ref.</i>	0.95 (0.92 - 0.98)	<i>Ref.</i>	0.95 (0.92 - 0.97)	0.90 (0.87 - 0.94)	0.92 (0.89 - 0.95)	0.87 (0.83 - 0.91)
		<i>ResNet 50 model</i>	0.89 (0.86 - 0.93)	0.120	0.93 (0.90 - 0.97)	0.333	0.90 (0.87 - 0.94)	0.85 (0.81 - 0.89)	0.86 (0.83 - 0.90)	0.79 (0.75 - 0.84)
		<i>ConvNext model</i>	0.90 (0.87 - 0.93)	0.207	0.93 (0.90 - 0.97)	0.368	0.84 (0.80 - 0.88)	0.90 (0.87 - 0.93)	0.88 (0.85 - 0.92)	0.80 (0.76 - 0.85)
		<i>Original ViT model</i>	0.92 (0.89 - 0.95)	0.851	0.96 (0.93 - 0.99)	0.479	0.86 (0.82 - 0.90)	0.99 (0.97 - 1.00)	0.95 (0.93 - 0.97)	0.91 (0.88 - 0.94)
		<i>GC-ViT model</i>	0.89 (0.85 - 0.92)	0.088	0.94 (0.91 - 0.97)	0.581	0.89 (0.86 - 0.93)	0.89 (0.85 - 0.92)	0.89 (0.85 - 0.92)	0.82 (0.78 - 0.86)
	<i>The internal</i>	<i>Our ViT model</i>	0.85 (0.76 - 0.94)	<i>Ref.</i>	0.93 (0.82 - 1.00)	<i>Ref.</i>	0.94 (0.88 - 1.00)	0.91 (0.84 - 0.98)	0.92 (0.84 - 0.99)	0.84 (0.75 - 0.93)

<i>Datasets</i>	<i>Cohorts</i>	<i>Models</i>	<i>2-year C-index</i>	<i>P value</i>	<i>2-year AUC</i>	<i>P value</i>	<i>Sensitivity</i>	<i>Specificity</i>	<i>Accuracy</i>	<i>F1 score</i>
	<i>validation cohort</i>	<i>ResNet 50 model</i>	0.56 (0.43 - 0.69)	<0.001	0.57 (0.34 - 0.80)	<0.001	0.39 (0.26 - 0.51)	1.00 (1.00 - 1.00)	0.85 (0.76 - 0.94)	0.56 (0.43 - 0.68)
		<i>ConvNext model</i>	0.77 (0.66 - 0.87)	0.228	0.80 (0.63 - 0.98)	0.042	0.67 (0.55 - 0.79)	0.91 (0.84 - 0.98)	0.85 (0.76 - 0.94)	0.68 (0.56 - 0.80)
		<i>Original ViT model</i>	0.77 (0.67 - 0.88)	0.273	0.85 (0.67 - 1.00)	0.145	0.77 (0.66 - 0.87)	1.00 (1.00 - 1.00)	0.94 (0.89 - 1.00)	0.87 (0.78 - 0.95)
		<i>GC-ViT model</i>	0.82 (0.72 - 0.92)	0.612	0.85 (0.69 - 1.00)	0.145	1.00 (1.00 - 1.00)	0.64 (0.51 - 0.76)	0.72 (0.61 - 0.84)	0.63 (0.51 - 0.75)
		<i>Our ViT model</i>	0.72 (0.66 - 0.78)	Ref.	0.81 (0.74 - 0.88)	Ref.	0.78 (0.72 - 0.83)	0.69 (0.63 - 0.75)	0.72 (0.67 - 0.78)	0.67 (0.61 - 0.73)
	<i>The external testing cohort</i>	<i>ResNet 50 model</i>	0.68 (0.62 - 0.74)	0.381	0.79 (0.72 - 0.86)	0.647	0.70 (0.64 - 0.76)	0.62 (0.55 - 0.68)	0.65 (0.58 - 0.71)	0.58 (0.51 - 0.65)
		<i>ConvNext model</i>	0.63 (0.57 - 0.70)	0.064	0.73 (0.65 - 0.81)	0.043	0.62 (0.55 - 0.68)	0.65 (0.59 - 0.72)	0.64 (0.58 - 0.71)	0.55 (0.48 - 0.61)
		<i>Original ViT model</i>	0.65 (0.59 - 0.71)	0.144	0.75 (0.68 - 0.82)	0.138	0.56 (0.49 - 0.62)	0.79 (0.73 - 0.84)	0.71 (0.64 - 0.77)	0.57 (0.50 - 0.63)
		<i>GC-ViT model</i>	0.64 (0.58 - 0.70)	0.084	0.72 (0.63 - 0.80)	0.022	0.65 (0.58 - 0.71)	0.61 (0.54 - 0.67)	0.62 (0.56 - 0.69)	0.54 (0.48 - 0.61)

[TA, thermal ablation; SR, surgical resection; PFS, progression-free survival; AUC, area under the curve; Ref, reference.]

(page 25-26/Table S7 of the revised supplementary material)

[Table S7. Performance Comparison of HEROVision Across Different Subgroups for 2-Year PFS in the External Testing Cohorts]

<i>Datasets</i>	<i>Subgroups</i>	<i>Number</i>	<i>2-year C-index</i>	<i>P value</i>	<i>2-year AUC</i>	<i>P value</i>	<i>Sensitivity</i>	<i>Specificity</i>	<i>Accuracy</i>	<i>F1 score</i>
<i>Sex</i>										
	<i>Female</i>	<i>n=85</i>	0.68 (0.58 - 0.78)	Ref.	0.75 (0.63 - 0.86)	Ref.	0.61 (0.51 - 0.72)	0.88 (0.81 - 0.95)	0.77 (0.69 - 0.86)	0.69 (0.59 - 0.78)
	<i>Male</i>	<i>n=526</i>	0.73 (0.69 - 0.77)	0.307	0.79 (0.75 - 0.83)	0.426	0.79 (0.75 - 0.82)	0.68 (0.64 - 0.72)	0.72 (0.68 - 0.76)	0.68 (0.64 - 0.72)
<i>Age</i>										
<i>2ndTA dataset</i>	<i>≤60 years</i>	<i>n=366</i>	0.72 (0.68 - 0.77)	Ref.	0.78 (0.73 - 0.83)	Ref.	0.75 (0.71 - 0.80)	0.70 (0.65 - 0.74)	0.72 (0.67 - 0.76)	0.67 (0.63 - 0.72)
	<i>>60 years</i>	<i>n=245</i>	0.73 (0.67 - 0.78)	0.852	0.78 (0.72 - 0.85)	0.929	0.80 (0.75 - 0.85)	0.64 (0.58 - 0.70)	0.70 (0.64 - 0.76)	0.67 (0.61 - 0.73)
<i>Tumor size</i>										
	<i>≤3 cm</i>	<i>n=563</i>	0.72 (0.69 - 0.76)	Ref.	0.78 (0.74 - 0.82)	Ref.	0.76 (0.72 - 0.79)	0.68 (0.64 - 0.72)	0.71 (0.67 - 0.74)	0.66 (0.62 - 0.69)
	<i>>3 cm</i>	<i>n=48</i>	0.68 (0.55 - 0.81)	0.525	0.80 (0.64 - 0.95)	0.745	0.80 (0.68 - 0.91)	0.75 (0.63 - 0.87)	0.78 (0.66 - 0.90)	0.81 (0.70 - 0.92)
<i>The number of tumors</i>										

<i>Datasets</i>	<i>Subgroups</i>	<i>Number</i>	<i>2-year C-index</i>	<i>P value</i>	<i>2-year AUC</i>	<i>P value</i>	<i>Sensitivity</i>	<i>Specificity</i>	<i>Accuracy</i>	<i>F1 score</i>
<i>2ndSR dataset</i>	<i>Solitary</i>	<i>n=488</i>	<i>0.73 (0.69 - 0.77)</i>	<i>Ref.</i>	<i>0.78 (0.73 - 0.82)</i>	<i>Ref.</i>	<i>0.78 (0.74 - 0.82)</i>	<i>0.68 (0.64 - 0.72)</i>	<i>0.71 (0.67 - 0.75)</i>	<i>0.65 (0.61 - 0.69)</i>
	<i>Multiple</i>	<i>n=123</i>	<i>0.68 (0.60 - 0.77)</i>	<i>0.308</i>	<i>0.78 (0.69 - 0.87)</i>	<i>0.951</i>	<i>0.66 (0.58 - 0.75)</i>	<i>0.85 (0.79 - 0.91)</i>	<i>0.75 (0.67 - 0.82)</i>	<i>0.74 (0.67 - 0.82)</i>
	<i>Manufacturers</i>									
	<i>MyLab90 Xvision</i>	<i>n=334</i>	<i>0.73 (0.68 - 0.78)</i>	<i>Ref.</i>	<i>0.78 (0.72 - 0.83)</i>	<i>Ref.</i>	<i>0.77 (0.73 - 0.82)</i>	<i>0.71 (0.66 - 0.76)</i>	<i>0.74 (0.69 - 0.78)</i>	<i>0.69 (0.64 - 0.74)</i>
	<i>Logiq E9</i>	<i>n=156</i>	<i>0.73 (0.66 - 0.80)</i>	<i>0.907</i>	<i>0.79 (0.71 - 0.87)</i>	<i>0.740</i>	<i>0.77 (0.71 - 0.84)</i>	<i>0.71 (0.64 - 0.78)</i>	<i>0.73 (0.66 - 0.80)</i>	<i>0.67 (0.60 - 0.74)</i>
	<i>Siemens Acuson Sequoia</i>	<i>n=76</i>	<i>0.74 (0.64 - 0.84)</i>	<i>0.852</i>	<i>0.79 (0.69 - 0.90)</i>	<i>0.777</i>	<i>0.89 (0.82 - 0.96)</i>	<i>0.58 (0.47 - 0.69)</i>	<i>0.70 (0.59 - 0.80)</i>	<i>0.69 (0.59 - 0.80)</i>
	<i>Resona 7</i>	<i>n=45</i>	<i>0.76 (0.63 - 0.88)</i>	<i>0.646</i>	<i>0.87 (0.76 - 0.98)</i>	<i>0.077</i>	<i>0.91 (0.83 - 1.00)</i>	<i>0.75 (0.62 - 0.88)</i>	<i>0.83 (0.72 - 0.94)</i>	<i>0.85 (0.74 - 0.95)</i>
	<i>Centers</i>									
	<i>Center A</i>	<i>n=231</i>	<i>0.72 (0.67 - 0.78)</i>	<i>Ref.</i>	<i>0.75 (0.68 - 0.82)</i>	<i>Ref.</i>	<i>0.84 (0.79 - 0.89)</i>	<i>0.66 (0.60 - 0.72)</i>	<i>0.74 (0.68 - 0.79)</i>	<i>0.73 (0.67 - 0.79)</i>
	<i>Center B</i>	<i>n=102</i>	<i>0.72 (0.64 - 0.81)</i>	<i>0.972</i>	<i>0.78 (0.68 - 0.88)</i>	<i>0.483</i>	<i>0.72 (0.63 - 0.81)</i>	<i>0.65 (0.56 - 0.74)</i>	<i>0.67 (0.58 - 0.76)</i>	<i>0.59 (0.49 - 0.68)</i>
	<i>Center C</i>	<i>n=161</i>	<i>0.73 (0.66 - 0.79)</i>	<i>0.996</i>	<i>0.78 (0.70 - 0.86)</i>	<i>0.419</i>	<i>0.71 (0.64 - 0.78)</i>	<i>0.76 (0.69 - 0.82)</i>	<i>0.74 (0.67 - 0.81)</i>	<i>0.64 (0.56 - 0.71)</i>
	<i>Center D</i>	<i>n=46</i>	<i>0.76 (0.63 - 0.88)</i>	<i>0.654</i>	<i>0.90 (0.81 - 0.99)</i>	<i>0.005</i>	<i>0.92 (0.85 - 1.00)</i>	<i>0.75 (0.62 - 0.88)</i>	<i>0.85 (0.74 - 0.95)</i>	<i>0.87 (0.78 - 0.97)</i>
	<i>Center E</i>	<i>n=71</i>	<i>0.75 (0.65 - 0.85)</i>	<i>0.663</i>	<i>0.81 (0.71 - 0.92)</i>	<i>0.244</i>	<i>0.77 (0.67 - 0.87)</i>	<i>0.74 (0.63 - 0.84)</i>	<i>0.75 (0.65 - 0.85)</i>	<i>0.69 (0.59 - 0.80)</i>
	<i>Sex</i>									
	<i>Female</i>	<i>n=26</i>	<i>0.69 (0.51 - 0.87)</i>	<i>Ref.</i>	<i>0.72 (0.41 - 1.00)</i>	<i>Ref.</i>	<i>0.67 (0.49 - 0.85)</i>	<i>0.90 (0.78 - 1.00)</i>	<i>0.84 (0.70 - 0.98)</i>	<i>0.69 (0.51 - 0.87)</i>
	<i>Male</i>	<i>n=196</i>	<i>0.71 (0.65 - 0.78)</i>	<i>0.812</i>	<i>0.82 (0.76 - 0.89)</i>	<i>0.246</i>	<i>0.81 (0.76 - 0.87)</i>	<i>0.74 (0.68 - 0.80)</i>	<i>0.77 (0.71 - 0.83)</i>	<i>0.72 (0.66 - 0.79)</i>
	<i>Age</i>									
	<i>≤60 years</i>	<i>n=131</i>	<i>0.68 (0.60 - 0.76)</i>	<i>Ref.</i>	<i>0.79 (0.70 - 0.88)</i>	<i>Ref.</i>	<i>0.82 (0.76 - 0.89)</i>	<i>0.68 (0.60 - 0.76)</i>	<i>0.73 (0.66 - 0.81)</i>	<i>0.69 (0.61 - 0.77)</i>
	<i>>60 years</i>	<i>n=91</i>	<i>0.75 (0.67 - 0.84)</i>	<i>0.238</i>	<i>0.84 (0.73 - 0.94)</i>	<i>0.404</i>	<i>0.78 (0.69 - 0.86)</i>	<i>0.74 (0.65 - 0.83)</i>	<i>0.75 (0.66 - 0.84)</i>	<i>0.69 (0.59 - 0.78)</i>
	<i>Tumor size</i>									
	<i>≤3 cm</i>	<i>n=166</i>	<i>0.70 (0.63 - 0.77)</i>	<i>Ref.</i>	<i>0.81 (0.73 - 0.89)</i>	<i>Ref.</i>	<i>0.71 (0.64 - 0.78)</i>	<i>0.82 (0.76 - 0.88)</i>	<i>0.78 (0.72 - 0.85)</i>	<i>0.68 (0.61 - 0.76)</i>
	<i>>3 cm</i>	<i>n=56</i>	<i>0.74 (0.63 - 0.86)</i>	<i>0.503</i>	<i>0.82 (0.70 - 0.94)</i>	<i>0.876</i>	<i>0.72 (0.60 - 0.84)</i>	<i>0.76 (0.65 - 0.88)</i>	<i>0.75 (0.63 - 0.86)</i>	<i>0.72 (0.60 - 0.83)</i>
	<i>The number of tumors</i>									
	<i>Solitary</i>	<i>n=180</i>	<i>0.69 (0.63 - 0.76)</i>	<i>Ref.</i>	<i>0.79 (0.71 - 0.87)</i>	<i>Ref.</i>	<i>0.64 (0.57 - 0.71)</i>	<i>0.87 (0.82 - 0.92)</i>	<i>0.80 (0.74 - 0.86)</i>	<i>0.67 (0.60 - 0.74)</i>

<i>Datasets</i>	<i>Subgroups</i>	<i>Number</i>	<i>2-year C-index</i>	<i>P value</i>	<i>2-year AUC</i>	<i>P value</i>	<i>Sensitivity</i>	<i>Specificity</i>	<i>Accuracy</i>	<i>F1 score</i>
	<i>Multiple</i>	<i>n=42</i>	<i>0.71 (0.57 - 0.84)</i>	<i>0.892</i>	<i>0.80 (0.61 - 1.00)</i>	<i>0.814</i>	<i>0.73 (0.59 - 0.86)</i>	<i>0.90 (0.81 - 0.99)</i>	<i>0.81 (0.69 - 0.93)</i>	<i>0.80 (0.68 - 0.92)</i>
	<i>Manufacturers</i>									
	<i>MyLab90 Xvision</i>	<i>n=91</i>	<i>0.74 (0.65 - 0.83)</i>	<i>Ref.</i>	<i>0.79 (0.67 - 0.91)</i>	<i>Ref.</i>	<i>0.77 (0.68 - 0.85)</i>	<i>0.81 (0.73 - 0.89)</i>	<i>0.79 (0.71 - 0.88)</i>	<i>0.74 (0.65 - 0.83)</i>
	<i>Logiq E9</i>	<i>n=46</i>	<i>0.72 (0.59 - 0.85)</i>	<i>0.758</i>	<i>0.84 (0.70 - 0.98)</i>	<i>0.478</i>	<i>0.68 (0.54 - 0.81)</i>	<i>0.94 (0.88 - 1.00)</i>	<i>0.86 (0.76 - 0.96)</i>	<i>0.75 (0.63 - 0.88)</i>
	<i>Siemens Acuson Sequoia</i>	<i>n=39</i>	<i>0.75 (0.62 - 0.89)</i>	<i>0.888</i>	<i>0.88 (0.74 - 1.00)</i>	<i>0.167</i>	<i>0.87 (0.76 - 0.97)</i>	<i>0.86 (0.75 - 0.97)</i>	<i>0.86 (0.75 - 0.97)</i>	<i>0.82 (0.69 - 0.94)</i>
	<i>Resona 7</i>	<i>n=46</i>	<i>0.68 (0.54 - 0.81)</i>	<i>0.445</i>	<i>0.78 (0.63 - 0.94)</i>	<i>0.942</i>	<i>0.71 (0.58 - 0.84)</i>	<i>0.88 (0.79 - 0.97)</i>	<i>0.82 (0.70 - 0.93)</i>	<i>0.74 (0.61 - 0.87)</i>
	<i>Centers</i>									
	<i>Center A</i>	<i>n=120</i>	<i>0.76 (0.68 - 0.83)</i>	<i>Ref.</i>	<i>0.80 (0.70 - 0.90)</i>	<i>Ref.</i>	<i>0.66 (0.58 - 0.75)</i>	<i>0.89 (0.83 - 0.95)</i>	<i>0.80 (0.72 - 0.87)</i>	<i>0.73 (0.65 - 0.81)</i>
	<i>Center B</i>	<i>n=48</i>	<i>0.69 (0.56 - 0.82)</i>	<i>0.375</i>	<i>0.82 (0.68 - 0.97)</i>	<i>0.739</i>	<i>0.66 (0.53 - 0.80)</i>	<i>0.94 (0.88 - 1.00)</i>	<i>0.86 (0.76 - 0.96)</i>	<i>0.74 (0.61 - 0.86)</i>
	<i>Center D</i>	<i>n=54</i>	<i>0.73 (0.61 - 0.85)</i>	<i>0.690</i>	<i>0.80 (0.66 - 0.94)</i>	<i>0.939</i>	<i>0.71 (0.59 - 0.83)</i>	<i>0.88 (0.80 - 0.97)</i>	<i>0.83 (0.73 - 0.93)</i>	<i>0.72 (0.60 - 0.84)</i>

TA, thermal ablation; SR, surgical resection; PFS, progression-free survival; AUC, area under the curve; Ref, reference.]

4. *[The introduction does not adequately justify the use of Vision Transformers. It does not contrast their advantages over Convolutional Neural Networks (CNNs), despite existing literature highlighting CNNs' benefits in medical image-based diagnosis, such as in the study "Deep Learning Methods in Medical Image-Based Hepatocellular Carcinoma Diagnosis: A Systematic Review and Meta-Analysis" by Wei et al., 2023. Additionally, the Materials and Methods section lacks explanations for some of the dataset division and the rationale behind the specific cohort division ratios. This could impact the experimental setup's validity as well as the model's training process.]*

Our Response:

Thank you for your valuable comment.

1. Justification for Vision Transformers:

In the revised introduction, we have expanded the discussion to better justify the use of Vision Transformers (ViTs) by highlighting their advantages over convolutional neural networks (CNNs), particularly in handling complex medical imaging tasks. ViTs offer distinct benefits, including improved global attention mechanisms that allow them to capture long-range dependencies and relationships within images[1]. This is especially useful in our application, where multimodal data like CEUS and MRI are used.

2. Comparison with CNN Models:

We compared our ViT model with two CNN models (ResNet50[2] and ConvNext[3]) on both TA and SR datasets using the same experimental setup. The results showed that our ViT model outperformed other SOTA models in terms of the C-index and time-independent AUC on both the TA and SR datasets, regardless of whether they were in the internal validation cohort or the external testing cohort. Detailed results can be found in **Figures S12, S13** and **Table S6**.

3. Dataset Division and Rationale:

The division of our dataset follows a clinical application approach rather than a natural image AI model development strategy. We allocated iHCC patients into training and internal validation cohorts, with rHCC patients used for external validation. This division mirrors real-world clinical scenarios, avoiding arbitrary ratios.

4. Details on Dataset Division:

We divided the training and internal validation cohorts based on chronological order: iHCC patients who underwent thermal ablation before January 2020 were included in the training cohort, while the remaining iHCC patients were included in the internal validation cohort, approximately in a 4:1 ratio. Similarly, iHCC patients who underwent surgical resection before April 2020 were included in the training cohort, with the rest in the internal validation cohort, also approximately in a 4:1 ratio. This approach simulates a prospective clinical trial setup, which is a common practice in clinical research[4-6].

5. Impact of Dataset Division Ratio:

We also evaluated different training-to-validation ratios (3:1, 4:1, and 5:1) for both the

1stTA and 1stSR datasets. Detailed results are provided in **Figure S10** and **Table S4**. For the TA dataset, the 2-year C-index values in the internal validation cohort were 0.86, 0.87, and 0.82 for the 3:1, 4:1, and 5:1 ratios ($P = 0.502$ and 0.866), and the 2-year AUC values were 0.91, 0.90, and 0.89 ($P = 0.663$ and 0.932), respectively. For the SR dataset, the 2-year C-index values in the internal validation cohort were 0.86, 0.85, and 0.83 ($P = 0.589$ and 0.866), respectively, and the 2-year AUC values were 0.92, 0.93, and 0.93 ($P = 0.772$ and 0.835), respectively. The results indicate that our ViT model's performance remained consistent across these ratios, suggesting that the model's effectiveness is robust and not significantly affected by the dataset division ratio. This supports the validity of the experimental setup and the training process.

References:

- [1] G. Bertasius, H. Wang, and L. Torresani, "Is space-time attention all you need for video understanding?," in *ICML*, 2021, vol. 2, no. 3, p. 4.
- [2] K. He, X. Zhang, S. Ren, and J. Sun, "Deep residual learning for image recognition," in *Proceedings of the IEEE conference on computer vision and pattern recognition*, 2016, pp. 770-778.
- [3] Z. Liu, H. Mao, C.-Y. Wu, C. Feichtenhofer, T. Darrell, and S. Xie, "A convnet for the 2020s," in *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*, 2022, pp. 11976-11986.
- [4] H. Zhu, X. Zhang, X. Li, Y. Shi, H. Zhu, and Y. Sun, "Prediction of pathological nodal stage of locally advanced rectal cancer by collective features of multiple lymph nodes in magnetic resonance images before and after neoadjuvant chemoradiotherapy," (in eng), *Chin J Cancer Res*, vol. 31, no. 6, pp. 984-992, Dec 2019, doi: 10.21147/j.issn.1000-9604.2019.06.14.
- [5] J. Wang *et al.*, "Preparing CT imaging datasets for deep learning in lung nodule analysis: Insights from four well-known datasets," (in eng), *Heliyon*, vol. 9, no. 6, p. e17104, Jun 2023, doi: 10.1016/j.heliyon.2023.e17104.
- [6] T. M. Kim *et al.*, "Differentiating renal epithelioid angiomyolipoma from clear cell carcinoma: using a radiomics model combined with CT imaging characteristics," (in eng), *Abdom Radiol (NY)*, vol. 47, no. 8, pp. 2867-2880, Aug 2022, doi: 10.1007/s00261-022-03571-9.

Revised text:

(page 4/Introduction section of the revised manuscript)

[The Vision Transformer (ViT) is a groundbreaking architecture, representing the first fully Transformer-based model adapted for image tasks. Networks that leverage the Transformer's self-attention mechanism have demonstrated greater efficiency in training and inference,

particularly for capturing long-range dependencies and relationships within images compared to conventional convolutional neural networks (CNNs)¹⁸. Additionally, Transformers can significantly outperform CNNs in generalization on out-of-distribution samples¹⁹.]

5. [The study also does not provide detailed information on sex and gender considerations, failing to comply with the SAGER guidelines (as recommended in the review instructions).]

Our Response:

Thank you for your comment. In this study, we utilized retrospective HCC data sourced from five hospitals in China, collected through clinical electronic medical records. Sex was determined based on government-issued ID information. Currently, Chinese hospitals do not include self-identified gender in their medical record statistics. As a result, self-identified gender information was not collected from the patients. The sex distributions for the training, internal validation, and external testing cohorts are provided in the revised manuscript (**Table 1**). However, sex-based analysis was not conducted, as sex was not relevant to the development or validation of the model. Therefore, while we acknowledge the importance of sex and gender considerations, they were not applicable to the focus of this study. Detailed information is provided in the **Nature Portfolio Reporting Summary**.

Revised text:

(Nature Portfolio Reporting Summary)

[In this study, we utilized retrospective HCC data collected through clinical practice. Sex was determined based on the government-issued ID. The sex distributions for the training, internal validation, and external testing cohorts are provided in the paper. Sex-based analysis was not conducted, as sex was not relevant to the model's development or validation. Additionally, self-identified gender information was not collected from the patients.]

(page 25-28/Table 1 of the revised manuscript)

[Table 1. Baseline Characteristics of Patients with HCC in the Training, Internal Validation, and External Testing Cohorts

Characteristics	1stTA dataset		2ndTA dataset					1stSR dataset			2ndSR dataset	
	The training cohort	The internal validation cohort	The external testing cohort (Center A)	The external testing cohort (Center B)	The external testing cohort (Center C)	External testing cohort (Center D)	External testing cohort (Center E)	The training cohort	The internal validation cohort	The external testing cohort (Center A)	The external testing cohort (Center B)	External testing cohort (Center D)
	N=311	N=80	N=231	N=102	N=161	N=46	N=71	N=322	N=59	N=120	N=48	N=54
Sex												
Female	55 (17.7)	15 (18.8)	36 (15.6)	20 (19.6)	13 (8.1)	2 (4.3)	14 (19.7)	59 (18.3)	7 (11.9)	18 (15.0)	7 (14.6)	1 (1.9)
Male	256 (82.3)	65 (81.2)	195 (84.4)	82 (80.4)	148 (91.9)	44 (95.7)	57 (80.3)	263 (81.7)	52 (88.1)	102 (85.0)	41 (85.4)	53 (98.1)
Age (years)†	59.00 [52.00, 65.00]	60.50 [54.00, 67.00]	57.00 [51.00, 65.00]	62.50 [55.00, 68.00]	56.00 [46.00, 63.00]	53.00 [46.00, 62.00]	57.00 [52.00, 63.00]	58.00 [51.00, 65.00]	62.00 [55.00, 67.00]	58.50 [50.00, 66.00]	59.00 [50.00, 68.00]	56.50 [46.00, 61.00]
Liver cirrhosis												
Absent	8 (2.6)	6 (7.5)	8 (3.5)	29 (28.4)	40 (24.8)	2 (4.3)	12 (16.9)	116 (36.0)	25 (42.4)	16 (13.3)	5 (10.4)	11 (20.4)
Present	303 (97.4)	74 (92.5)	223 (96.5)	73 (71.6)	121 (75.2)	44 (95.7)	59 (83.1)	206 (64.0)	34 (57.6)	104 (86.7)	43 (89.6)	43 (79.6)
Etiology												
HBV	258 (83.0)	68 (85.0)	206 (89.2)	84 (82.4)	155 (96.3)	40 (87.0)	62 (87.3)	256 (79.5)	44 (74.6)	108 (90.0)	41 (85.4)	47 (87.0)
HCV	8 (2.6)	0 (0.0)	0 (0.0)	11 (10.8)	1 (0.6)	3 (6.5)	2 (2.8)	0 (0.0)	2 (3.4)	0 (0.0)	4 (8.3)	2 (3.7)
ALC	8 (2.6)	1 (1.2)	1 (0.4)	1 (1.0)	0 (0.0)	0 (0.0)	0 (0.0)	5 (1.6)	0 (0.0)	0 (0.0)	0 (0.0)	1 (1.9)
PBC	2 (0.6)	0 (0.0)	0 (0.0)	0 (0.0)	0 (0.0)	0 (0.0)	0 (0.0)	0 (0.0)	0 (0.0)	0 (0.0)	0 (0.0)	0 (0.0)
NAFLD	9 (2.9)	3 (3.8)	10 (4.3)	0 (0.0)	0 (0.0)	0 (0.0)	0 (0.0)	23 (7.1)	7 (11.9)	4 (3.3)	1 (2.1)	0 (0.0)
Undetermined	26 (8.4)	8 (10.0)	14 (6.1)	6 (5.9)	5 (3.1)	3 (6.5)	7 (9.9)	38 (11.8)	6 (10.2)	8 (6.7)	2 (4.2)	4 (7.4)

CTP grade												
<i>A level</i>	286 (92.0)	75 (93.8)	225 (97.4)	95 (93.1)	137 (85.1)	45 (97.8)	71 (100.0)	289 (89.8)	56 (94.9)	116 (96.7)	34 (70.8)	52 (96.3)
<i>B level</i>	25 (8.0)	5 (6.2)	6 (2.6)	7 (6.9)	24 (14.9)	1 (2.2)	0 (0.0)	33 (10.2)	3 (5.1)	4 (3.3)	14 (29.2)	2 (3.7)
Portal hypertension												
<i>Absent</i>	195 (62.7)	40 (50.0)	164 (71.0)	94 (92.2)	116 (72.0)	13 (28.3)	60 (84.5)	260 (80.7)	46 (78.0)	101 (84.2)	38 (79.2)	46 (85.2)
<i>Present</i>	116 (37.3)	40 (50.0)	67 (29.0)	8 (7.8)	45 (28.0)	33 (71.7)	11 (15.5)	62 (19.3)	13 (22.0)	19 (15.8)	10 (20.8)	8 (14.8)
Location												
<i>Left lobe</i>	58 (18.6)	16 (20.0)	53 (22.9)	26 (25.5)	44 (27.3)	14 (30.4)	22 (31.0)	89 (27.6)	9 (15.3)	41 (34.2)	20 (41.7)	21 (38.9)
<i>Right lobe</i>	244 (78.5)	61 (76.2)	171 (74.0)	74 (72.5)	113 (70.2)	29 (63.0)	49 (69.0)	218 (67.7)	46 (78.0)	73 (60.8)	25 (52.1)	30 (55.6)
<i>Whole lobe</i>	9 (2.9)	3 (3.8)	7 (3.0)	2 (2.0)	4 (2.5)	3 (6.5)	0 (0.0)	15 (4.7)	4 (6.8)	6 (5.0)	3 (6.2)	3 (5.6)
The number of tumors												
<i>1</i>	249 (80.1)	67 (83.8)	200 (86.6)	74 (72.5)	119 (73.9)	31 (67.4)	64 (90.1)	279 (86.6)	52 (88.1)	94 (78.3)	41 (85.4)	45 (83.3)
<i>2</i>	54 (17.4)	12 (15.0)	26 (11.3)	22 (21.6)	34 (21.1)	12 (26.1)	6 (8.5)	36 (11.2)	5 (8.5)	15 (12.5)	6 (12.5)	8 (14.8)
<i>3</i>	8 (2.6)	1 (1.2)	5 (2.2)	6 (5.9)	8 (5.0)	3 (6.5)	1 (1.4)	7 (2.2)	2 (3.4)	11 (9.2)	1 (2.1)	1 (1.9)
Tumor size												
<i>≤ 2 cm</i>	142 (45.7)	45 (56.2)	169 (73.2)	66 (64.7)	121 (75.2)	24 (52.2)	34 (47.9)	63 (19.6)	16 (27.1)	67 (55.8)	19 (39.6)	16 (29.6)
<i>> 2 cm, ≤ 3 cm</i>	119 (38.3)	26 (32.5)	52 (22.5)	25 (24.5)	30 (18.6)	17 (37.0)	25 (35.2)	82 (25.5)	14 (23.7)	24 (20.0)	14 (29.2)	26 (48.1)
<i>> 3 cm</i>	50 (16.1)	9 (11.2)	10 (4.3)	11 (10.8)	10 (6.2)	5 (10.9)	12 (16.9)	177 (55.0)	29 (49.2)	29 (24.2)	15 (31.2)	12 (22.2)
Ultrasonic echo												
<i>Hypoecho</i>	205 (65.9)	47 (58.8)	153 (66.2)	75 (73.5)	145 (90.1)	42 (91.3)	59 (83.1)	221 (68.6)	42 (71.2)	81 (67.5)	37 (77.1)	41 (75.9)
<i>Isoecho</i>	46 (14.8)	19 (23.8)	27 (11.7)	14 (13.7)	3 (1.9)	3 (6.5)	5 (7.0)	9 (2.8)	2 (3.4)	5 (4.2)	2 (4.2)	5 (9.3)
<i>Hyperecho</i>	60 (19.3)	14 (17.5)	51 (22.1)	13 (12.7)	13 (8.1)	1 (2.2)	7 (9.9)	92 (28.6)	15 (25.4)	34 (28.3)	9 (18.8)	8 (14.8)
T2WI signal												
<i>Hypointensity</i>	7 (2.3)	1 (1.2)	3 (1.3)	2 (2.0)	5 (3.1)	1 (2.2)	1 (1.4)	1 (0.3)	1 (1.7)	0 (0.0)	1 (2.1)	0 (0.0)
<i>Isointensity</i>	10 (3.2)	2 (2.5)	3 (1.3)	1 (1.0)	1 (0.6)	0 (0.0)	1 (1.4)	5 (1.6)	1 (1.7)	7 (5.8)	0 (0.0)	1 (1.9)

<i>Hyperintensity</i>	294 (94.5)	77 (96.2)	225 (97.4)	99 (97.1)	155 (96.3)	45 (97.8)	69 (97.2)	316 (98.1)	57 (96.6)	113 (94.2)	47 (97.9)	53 (98.1)
<i>DWI signal</i>												
<i>Hypointensity</i>	10 (3.2)	0 (0.0)	4 (1.7)	2 (2.0)	3 (1.9)	0 (0.0)	1 (1.4)	4 (1.2)	1 (1.7)	3 (2.5)	1 (2.1)	1 (1.9)
<i>Isointensity</i>	11 (3.5)	3 (3.8)	6 (2.6)	2 (2.0)	4 (2.5)	1 (2.2)	2 (2.8)	3 (0.9)	1 (1.7)	7 (5.8)	1 (2.1)	1 (1.9)
<i>Hyperintensity</i>	290 (93.2)	77 (96.2)	221 (95.7)	98 (96.1)	154 (95.7)	45 (97.8)	68 (95.8)	315 (97.8)	57 (96.6)	110 (91.7)	46 (95.8)	52 (96.3)
<i>Subcapsular</i>												
<i>No</i>	203 (65.3)	52 (65.0)	167 (72.3)	63 (61.8)	107 (66.5)	38 (82.6)	43 (60.6)	93 (28.9)	22 (37.3)	64 (53.3)	19 (39.6)	39 (72.2)
<i>Yes</i>	108 (34.7)	28 (35.0)	64 (27.7)	39 (38.2)	54 (33.5)	8 (17.4)	28 (39.4)	229 (71.1)	37 (62.7)	56 (46.7)	29 (60.4)	15 (27.8)
<i>Perivascular</i>												
<i>No</i>	253 (81.4)	67 (83.8)	190 (82.3)	84 (82.4)	137 (85.1)	44 (95.7)	58 (81.7)	242 (75.2)	47 (79.7)	103 (85.8)	40 (83.3)	45 (83.3)
<i>Yes</i>	58 (18.6)	13 (16.2)	41 (17.7)	18 (17.6)	24 (14.9)	2 (4.3)	13 (18.3)	80 (24.8)	12 (20.3)	17 (14.2)	8 (16.7)	9 (16.7)
<i>ALT (U/L)</i>												
<i>≤40</i>	265 (85.2)	64 (80.0)	199 (86.1)	85 (83.3)	138 (85.7)	31 (67.4)	65 (91.5)	224 (69.6)	42 (71.2)	78 (65.0)	38 (79.2)	42 (77.8)
<i>>40</i>	46 (14.8)	16 (20.0)	32 (13.9)	17 (16.7)	23 (14.3)	15 (32.6)	6 (8.5)	98 (30.4)	17 (28.8)	42 (35.0)	10 (20.8)	12 (22.2)
<i>AST (U/L)</i>												
<i>≤45</i>	268 (86.2)	69 (86.2)	207 (89.6)	89 (87.3)	154 (95.7)	37 (80.4)	67 (94.4)	248 (77.0)	48 (81.4)	87 (72.5)	34 (70.8)	46 (85.2)
<i>>45</i>	43 (13.8)	11 (13.8)	24 (10.4)	13 (12.7)	7 (4.3)	9 (19.6)	4 (5.6)	74 (23.0)	11 (18.6)	33 (27.5)	14 (29.2)	8 (14.8)
<i>INR</i>												
<i>≤1.15</i>	220 (70.7)	58 (72.5)	194 (84.0)	74 (72.5)	132 (82.0)	43 (93.5)	53 (74.6)	262 (81.4)	51 (86.4)	96 (80.0)	27 (56.2)	47 (87.0)
<i>>1.15</i>	91 (29.3)	22 (27.5)	37 (16.0)	28 (27.5)	29 (18.0)	3 (6.5)	18 (25.4)	60 (18.6)	8 (13.6)	24 (20.0)	21 (43.8)	7 (13.0)
<i>CEA (ng/mL)</i>												
<i>≤5</i>	266 (85.5)	65 (81.2)	203 (87.9)	87 (85.3)	142 (88.2)	43 (93.5)	63 (88.7)	268 (83.2)	53 (89.8)	113 (94.2)	43 (89.6)	49 (90.7)
<i>>5</i>	45 (14.5)	15 (18.8)	28 (12.1)	15 (14.7)	19 (11.8)	3 (6.5)	8 (11.3)	54 (16.8)	6 (10.2)	7 (5.8)	5 (10.4)	5 (9.3)
<i>AFP (ng/mL)</i>												
<i><200</i>	265 (85.2)	73 (91.2)	206 (89.2)	91 (89.2)	142 (88.2)	36 (78.3)	61 (85.9)	256 (79.5)	52 (88.1)	102 (85.0)	43 (89.6)	42 (77.8)

≥ 200	46 (14.8)	7 (8.8)	25 (10.8)	11 (10.8)	19 (11.8)	10 (21.7)	10 (14.1)	66 (20.5)	7 (11.9)	18 (15.0)	5 (10.4)	12 (22.2)
CA19-9 (U/ml)												
≤ 37	294 (94.5)	76 (95.0)	222 (96.1)	87 (85.3)	156 (96.9)	44 (95.7)	63 (88.7)	302 (93.8)	57 (96.6)	117 (97.5)	44 (91.7)	54 (100.0)
> 37	17 (5.5)	4 (5.0)	9 (3.9)	15 (14.7)	5 (3.1)	2 (4.3)	8 (11.3)	20 (6.2)	2 (3.4)	3 (2.5)	4 (8.3)	0 (0.0)
PNI*												
≥ 42.8	107 (34.4)	35 (43.8)	36 (15.6)	95 (93.1)	143 (88.8)	42 (91.3)	38 (53.5)	145 (45.0)	36 (61.0)	59 (49.2)	36 (75.0)	50 (92.6)
< 42.8	204 (65.6)	45 (56.2)	195 (84.4)	7 (6.9)	18 (11.2)	4 (8.7)	33 (46.5)	177 (55.0)	23 (39.0)	61 (50.8)	12 (25.0)	4 (7.4)

Unless indicated otherwise, data are the number of patients, with percentages in parentheses.

† Data are medians, with interquartile ranges in parentheses.

* The optimal cutoff values for the PNI were determined using the maximum Youden index.

TA, thermal ablation; SR, surgical resection; PFS, progression-free survival; CTP, Child-Turcotte-Pugh; ALC, alcoholic cirrhosis; PBC, primary biliary cholangitis; NAFLD, non-alcoholic fatty liver disease; AST, aspartate aminotransferase; ALT, alanine aminotransferase; AFP, alpha-fetoprotein; CA19-9, carbohydrate antigen 19-9; CEA, carcinoembryonic antigen; INR, international normalized ratio; PNI, prognostic nutritional index.]

6. [The computational complexity of the Vision Transformer model and the potential loss of diagnostic details due to image resizing and strip partitioning limit the model's clinical feasibility. The paper does not address the variability in imaging quality or interpretation (many imaging modalities are used), which are important factors that could affect the accuracy of individualized predictions.]

Our Response:

Thank you for your comment.

1. Quality Control of Imaging Data:

We applied strict quality control measures to both CEUS and MRI data. Only images that passed quality control, regardless of the imaging modality or manufacturer, were included in the study. This approach helps mitigate potential issues arising from variability in imaging quality.

2. Image Resizing and Strip Partitioning:

Image resizing was performed as a standard preprocessing step to standardize the input dimensions (128×128 pixels) due to the varying tumor sizes. Strip partitioning was introduced as a modification to the ViT model to enhance its capability for localized tumor analysis, aligning with medical logic for evaluating tumor regions.

3. Comparative Analysis and Subgroup Analysis:

In the external testing cohort, we conducted ablation studies and further compared the performance of single-modality CEUS or single-modality MRI versus combined CEUS and MRI. Detailed results are provided in **Figure S11** and **Table S4**. Additionally, subgroup analyses considering sex, age, tumor size, number of tumors, manufacturers, and centers were conducted. These analyses demonstrated that our ViT model effectively addresses variability across different subgroups, ensuring stable and accurate prognostic predictions (**Figures S14, S15** and **Table S7**).

Revised text:

(page 3/Appendix S1 of supplementary material)

[For initial hepatocellular carcinoma (iHCC)...The exclusion criteria were: ... (6) presence of unqualified image artifacts.]

[For recurrent hepatocellular carcinoma (rHCC)...The exclusion criteria were: ... (6) presence of unqualified image artifacts.]

(page 6-7/Appendix S5 section of the revised supplementary material)

[To minimize the impact of variability due to different hospitals, equipment, and technical parameters on subsequent analysis, we employed denoising and image augmentation techniques^{2, 3}. Denoising was performed using advanced filtering methods, such as Gaussian filters, to remove random noise and enhance image quality². Image augmentation techniques,

including rotation, scaling, translation, and contrast adjustment, were utilized to diversify the dataset, improve model robustness, and reduce the risk of overfitting³. Then, in order to objectively extract tumor subregions within the tumor and alleviate human bias, we expanded all drawn regions of interest masks outward by 5mm, allowing for consideration of both the internal region of the lesion and the surrounding liver parenchyma, providing more feature information for deep learning (DL) networks. Python was used to extract irregular bounding boxes of tumor subregions from various patterns. In order to improve the processing efficiency of DL models, we transform irregular boundaries into regular square boundaries and then crop them. Additional preprocessing included normalization of size and grayscale values to produce resized and standardized grayscale ROI images for model development. Each subregion image was adjusted to a standardized size of 128×128 pixels, with grayscale values normalized to ensure uniformity across the dataset, thereby improving model training.

Gaussian filters:

$$G(x,y)= 1/(2\pi\sigma^2) e^{-(x^2+y^2)/(2\sigma^2)}$$

Among them, x, y represents the position of the pixel; σ is the standard deviation of the pixels in the training cohort.

Normalization:

$$\text{Standardization } (x) = (x-\mu)/(\max(\sigma, 1.0/\sqrt{N}))$$

$$\text{Norm } (x) = (x-\min(x))/(\max(x)-\min(x))$$

Among them, x represents the images; μ is the average value of the pixels in the training cohort; σ is the standard deviation of the pixels in the training cohort; N is the number of pixels; $\min(x)$ represents the minimum pixel value and $\max(x)$ represents maximum pixel value.]

7. [Moreover, the comparisons with major guideline methods do not address inherent biases or limitations in the staging systems. This could undermine the validity of the results, as these biases can impact the model's performance. There is also a drop in performance metrics when the model is applied to the external testing cohort. This could indicate potential overfitting and limited generalizability. In addition, the statistical analysis lacks detailed justification for the selection of some specific tests and thresholds. The choice of propensity score matching (PSM) and the methods used to evaluate statistical significance are not sufficiently explained.]

Our Response:

Thank you for your comment.

1. Inherent Biases in Guideline Methods:

Existing HCC guidelines primarily rely on tumor size, number, vascular invasion, intra- and extrahepatic metastasis, liver function, and performance status for staging[1-6]. These methods are well-established through previous clinical studies and have been widely accepted by experts to guide clinical practice. Our study's cohorts align with these guideline criteria, and we have adhered strictly to these guidelines. However, identifying and addressing inherent

biases and limitations of these guideline-based methods was not the primary focus of our study. Instead, our goal was to demonstrate that an AI-based approach, leveraging clinical factors along with CEUS and MRI images, can outperform these guideline methods in predictive accuracy.

2. Performance Drop in External Validation:

The observed reduction in performance metrics when applying the model to the external testing cohort is expected. This is because the training and internal validation cohorts consist of iHCC patients, whereas the external testing cohort comprises rHCC patients. As mentioned in our previous response to **Reviewer 1**, this approach was necessitated by the limited availability of rHCC cases. Despite this challenge, our model still provides a novel means for stratifying prognosis and aiding individualized treatment decisions for rHCC patients, addressing gaps not covered by current guidelines.

3. Potential Overfitting and Generalizability:

While slight overfitting may have occurred in the external testing cohort, our HEROVision model consistently outperformed the reference methods provided by clinical guidelines. Additionally, HEROVision model demonstrated strong stability in subgroup analyses, indicating its high generalization capability. To further validate the model's generalizability, we collected additional data from two other centers for SR and TA cohorts, and the results confirmed HEROVision's robustness. However, prospective clinical trials are necessary to further validate the model's generalizability, which we plan to pursue in the next step of our research.

4. Statistical Analysis:

Categorical variables were compared between groups using either the chi-square or Fisher's exact test, while continuous data were assessed using the two-sample t-test or Wilcoxon rank-sum test, as appropriate. These statistical methods do not require preset thresholds. For metrics such as AUC, C-index, sensitivity, specificity, accuracy, and F1 score, thresholds were determined based on the training cohort. We have added further details to the **Statistical Analysis section** in the revised manuscript.

5. Propensity Score Matching (PSM):

PSM is a popular method of using the propensity score in the medical literature[7]. PSM is a widely used method in rHCC researches[8-11], to ensure that baseline characteristics between groups are highly similar, thereby minimizing confounding factors in individualized treatment decisions[7, 12, 13]. Inferences about treatment effect made using PSM are valid only if, in the matched sample. In our study, we employed PSM with a caliper set at 0.02, considering variables such as sex, age, tumor location, tumor size, number of tumors, portal hypertension, subcapsular lesions, perivascular lesions, ALT, AST, INR, CEA, AFP, CA19-9, and PNI as independent variables. Detailed descriptions of the PSM process have been added to the revised supplementary materials (**Appendix S7**).

References:

- [1] M. B. Amin *et al.*, *AJCC cancer staging manual*. Springer, 2017.
- [2] A. G. Singal *et al.*, "AASLD Practice Guidance on prevention, diagnosis, and treatment of hepatocellular carcinoma," (in eng), *Hepatology*, vol. 78, no. 6, pp. 1922-1965, Dec 1 2023, doi: 10.1097/hep.0000000000000466.
- [3] G. O. o. N. H. Commission, "Standard for diagnosis and treatment of primary liver cancer (2022 edition)," *临床肝胆病杂志*, vol. 38, no. 2, pp. 288-303, 2022.
- [4] T. Yau, V. Y. Tang, T. J. Yao, S. T. Fan, C. M. Lo, and R. T. Poon, "Development of Hong Kong Liver Cancer staging system with treatment stratification for patients with hepatocellular carcinoma," (in eng), *Gastroenterology*, vol. 146, no. 7, pp. 1691-700.e3, Jun 2014, doi: 10.1053/j.gastro.2014.02.032.
- [5] F. Farinati *et al.*, "Development and Validation of a New Prognostic System for Patients with Hepatocellular Carcinoma," (in eng), *PLoS Med*, vol. 13, no. 4, p. e1002006, Apr 2016, doi: 10.1371/journal.pmed.1002006.
- [6] "2018 Korean Liver Cancer Association-National Cancer Center Korea Practice Guidelines for the Management of Hepatocellular Carcinoma," (in eng), *Korean J Radiol*, vol. 20, no. 7, pp. 1042-1113, Jul 2019, doi: 10.3348/kjr.2019.0140.
- [7] P. C. Austin, "Balance diagnostics for comparing the distribution of baseline covariates between treatment groups in propensity - score matched samples," *Statistics in medicine*, vol. 28, no. 25, pp. 3083-3107, 2009.
- [8] J. H. Zhong *et al.*, "Repeat hepatic resection versus radiofrequency ablation for recurrent hepatocellular carcinoma: retrospective multicentre study," (in eng), *Br J Surg*, vol. 109, no. 1, pp. 71-78, Dec 17 2021, doi: 10.1093/bjs/znab340.
- [9] K. D. Song *et al.*, "Repeated Hepatic Resection versus Radiofrequency Ablation for Recurrent Hepatocellular Carcinoma after Hepatic Resection: A Propensity Score Matching Study," (in eng), *Radiology*, vol. 275, no. 2, pp. 599-608, May 2015, doi: 10.1148/radiol.14141568.
- [10] N. Machairas *et al.*, "Meta-Analysis of Repeat Hepatectomy versus Radiofrequency Ablation for Recurrence of Hepatocellular Carcinoma," (in eng), *Cancers (Basel)*, vol. 14, no. 21, Nov 2 2022, doi: 10.3390/cancers14215398.
- [11] Y. Feng *et al.*, "Radiofrequency ablation versus repeat resection for recurrent hepatocellular carcinoma (≤ 5 cm) after initial curative resection," (in eng), *Eur Radiol*, vol. 30, no. 11, pp. 6357-6368, Nov 2020, doi: 10.1007/s00330-020-06990-8.
- [12] M. S. Ali *et al.*, "Reporting of covariate selection and balance assessment in propensity score analysis is suboptimal: a systematic review," (in eng), *J Clin Epidemiol*, vol. 68, no. 2, pp. 112-21, Feb 2015, doi: 10.1016/j.jclinepi.2014.08.011.

- [13] P. R. Rosenbaum and D. B. Rubin, "The central role of the propensity score in observational studies for causal effects," *Biometrika*, vol. 70, no. 1, pp. 41-55, 1983.

Revised text:

(page 15-16/Methods section of the revised manuscript)

[Categorical variables were compared between two groups using either the chi-square or Fisher's exact test. Continuous data were evaluated using either the two-sample t test or Wilcoxon rank-sum test, as appropriate. Clinical data are expressed as mean $\pm$ standard deviation or median (25 quantile, 75 quantile). The performance of models was assessed using the C-index and time-dependent AUC, and comparisons were made using the DeLong test. Survival curves were estimated using Kaplan–Meier analysis with the log-rank test. Univariate and multivariate analyses using Cox proportional hazards regression models were conducted to identify clinical factors associated with PFS. The optimal cutoff values for the prognostic nutritional index (PNI) were determined using the maximum Youden index. The analyses provided P-values, hazard ratios (HR), and 95% CI. The threshold for statistical significance was set at $P < 0.05$. Balance in baseline characteristics between 2ndTA and 2ndSR datasets was evaluated by the SMD.]

(page 15/ Methods section of the revised manuscript)

[PSM is a widely used method in medical research to balance covariates between treatment groups⁵³. However, the validity of inferences regarding treatment effects based on PSM depends on whether the matched groups have comparable distributions of the measured baseline covariates.]

(page 10-11/ Appendix S7 of the revised supplementary material)

[To reduce potential confounding effects arising from variations in baseline characteristics, the efficacy of the models in guiding cross-stratification was assessed through propensity score matching (PSM) for 2-year PFS^{6, 7, 8}. We utilized the "MatchIt" package in R software. The propensity score was estimated via logistic regression utilizing a 1:1 ratio (patients in the 2ndTA dataset vs. patients in the 2ndSR dataset) for optimal pair matching employing the nearest neighbor method. A caliper is the maximum permitted difference between matched subjects in PSM⁹. Without a caliper, substantial bias can occur; using the recommended caliper significantly reduces bias. A smaller caliper further decreases bias and results in closer matches, although some subjects may be excluded from matching. Therefore, this study chose a caliper value of 0.02, as it is a commonly accepted threshold in the literature for ensuring adequate matching while minimizing bias⁸. The independent variables were selected based on pretreatment clinical factors that could potentially influence survival and previous literature reports^{10, 11, 12, 13}, including sex, age, tumor location, tumor size, number of tumors, portal

hypertension, subcapsular lesions, perivascular lesions, ALT, AST, INR, CEA, AFP, CA19-9, and PNI. Standardized mean differences (SMDs) were computed for all confounders to assess covariate balance between groups. SMDs are interpreted as follows: values below 0.10 indicate very small differences, values between 0.10 and 0.30 suggest small differences, values between 0.30 and 0.50 reflect moderate differences, and values above 0.50 represent large differences.]

8. [The rationale behind the selection of specific variables and the chosen caliper value of 0.02 for PSM should be explicitly justified. Furthermore, while the section effectively details the re-stratification process using the HEROVision models, a more thorough discussion on the clinical implications of these re-stratifications is necessary. Specifically, elaborating on how the re-categorization from high-risk to low-risk influences patient management and treatment outcomes would provide valuable insights into the practical applications of the findings. Also, the title of this section should be in all capital letters to be consistent with previous headers.]

Our Response:

Thank you for your comment.

1. Selection of Variables and Caliper Value for PSM:

The specific variables selected for PSM were based on pre-treatment clinical factors known to potentially influence survival outcomes, as well as insights from previous literature[1-4]. These variables include sex, age, tumor location, tumor size, number of tumors, portal hypertension, subcapsular lesions, perivascular lesions, ALT, AST, INR, CEA, AFP, CA19-9, and PNI.

A caliper is the maximum permitted difference between matched subjects in PSM[5]. Without a caliper, substantial bias can occur; using the recommended caliper significantly reduces bias. A smaller caliper further decreases bias and results in closer matches, although some subjects may be excluded from matching. Therefore, this study chose a caliper value of 0.02, as it is a commonly accepted threshold in the literature for ensuring adequate matching while minimizing bias[6]. Further details regarding the selection of variables and the caliper value have been added to the **Methods section** and **Appendix S7** of the revised manuscript.

2. Clinical Implications of Re-Stratification:

Currently, there are no established guidelines or definitive classification systems for the management of rHCC [7-10], with most consensus recommendations still following the staging and treatment guidelines for iHCC. The HEROVision model offers a novel approach to individualized preferences in optimizing treatment for rHCC within the Milan criteria. By accurately re-stratifying patients into high-risk and low-risk categories based on their specific clinical and imaging profiles, HEROVision facilitates more personalized treatment decisions with significant clinical implications. A more detailed discussion on these clinical implications has been added to the **Figure 6** and **Discussion section** of the revised manuscript.

3. Title Formatting:

The title of this section has been updated to be in all capital letters to maintain consistency with previous headers in the revised manuscript.

References:

- [1] J. H. Zhong *et al.*, "Repeat hepatic resection versus radiofrequency ablation for recurrent hepatocellular carcinoma: retrospective multicentre study," (in eng), *Br J Surg*, vol. 109, no. 1, pp. 71-78, Dec 17 2021, doi: 10.1093/bjs/znab340.
- [2] K. D. Song *et al.*, "Repeated Hepatic Resection versus Radiofrequency Ablation for Recurrent Hepatocellular Carcinoma after Hepatic Resection: A Propensity Score Matching Study," (in eng), *Radiology*, vol. 275, no. 2, pp. 599-608, May 2015, doi: 10.1148/radiol.14141568.
- [3] N. Machairas *et al.*, "Meta-Analysis of Repeat Hepatectomy versus Radiofrequency Ablation for Recurrence of Hepatocellular Carcinoma," (in eng), *Cancers (Basel)*, vol. 14, no. 21, Nov 2 2022, doi: 10.3390/cancers14215398.
- [4] Y. Feng *et al.*, "Radiofrequency ablation versus repeat resection for recurrent hepatocellular carcinoma (≤ 5 cm) after initial curative resection," (in eng), *Eur Radiol*, vol. 30, no. 11, pp. 6357-6368, Nov 2020, doi: 10.1007/s00330-020-06990-8.
- [5] M. Lunt, "Selecting an appropriate caliper can be essential for achieving good balance with propensity score matching," *American journal of epidemiology*, vol. 179, no. 2, pp. 226-235, 2014.
- [6] M. S. Ali *et al.*, "Reporting of covariate selection and balance assessment in propensity score analysis is suboptimal: a systematic review," (in eng), *J Clin Epidemiol*, vol. 68, no. 2, pp. 112-21, Feb 2015, doi: 10.1016/j.jclinepi.2014.08.011.
- [7] J. A. Marrero *et al.*, "Diagnosis, Staging, and Management of Hepatocellular Carcinoma: 2018 Practice Guidance by the American Association for the Study of Liver Diseases," (in eng), *Hepatology*, vol. 68, no. 2, pp. 723-750, Aug 2018, doi: 10.1002/hep.29913.
- [8] "EASL Clinical Practice Guidelines: Management of hepatocellular carcinoma," (in eng), *J Hepatol*, vol. 69, no. 1, pp. 182-236, Jul 2018, doi: 10.1016/j.jhep.2018.03.019.
- [9] A. Vogel *et al.*, "Hepatocellular carcinoma: ESMO Clinical Practice Guidelines for diagnosis, treatment and follow-up," (in eng), *Ann Oncol*, vol. 29, no. Suppl 4, pp. iv238-iv255, Oct 1 2018, doi: 10.1093/annonc/mdy308.
- [10] J. K. Heimbach *et al.*, "AASLD guidelines for the treatment of hepatocellular carcinoma," (in eng), *Hepatology*, vol. 67, no. 1, pp. 358-380, Jan 2018, doi: 10.1002/hep.29086.

Revised text:

(page 10-11/Appendix S7 of the revised supplementary material)

[To reduce potential confounding effects arising from variations in baseline characteristics, the efficacy of the models in guiding cross-stratification was assessed through propensity score matching (PSM) for 2-year PFS^{6, 7, 8}. We utilized the "MatchIt" package in R software. The propensity score was estimated via logistic regression utilizing a 1:1 ratio (patients in the 2ndTA dataset vs. patients in the 2ndSR dataset) for optimal pair matching employing the nearest neighbor method. A caliper is the maximum permitted difference between matched subjects in PSM⁹. Without a caliper, substantial bias can occur; using the recommended caliper significantly reduces bias. A smaller caliper further decreases bias and results in closer matches, although some subjects may be excluded from matching. Therefore, this study chose a caliper value of 0.02, as it is a commonly accepted threshold in the literature for ensuring adequate matching while minimizing bias⁸. The independent variables were selected based on pretreatment clinical factors that could potentially influence survival and previous literature reports^{10, 11, 12, 13}, including sex, age, tumor location, tumor size, number of tumors, portal hypertension, subcapsular lesions, perivascular lesions, ALT, AST, INR, CEA, AFP, CA19-9, and PNI. Standardized mean differences (SMDs) were computed for all confounders to assess covariate balance between groups. SMDs are interpreted as follows: values below 0.10 indicate very small differences, values between 0.10 and 0.30 suggest small differences, values between 0.30 and 0.50 reflect moderate differences, and values above 0.50 represent large differences.]

(page 10-11/Discussion section of the revised manuscript)

[Currently, there are no established guidelines or definitive classification systems for the management of rHCC^{7, 8, 9, 10}, with most consensus recommendations still following the staging and treatment guidelines for iHCC. The HEROVision model offers a novel approach to individualized preferences in optimizing treatment for rHCC within the Milan criteria. By accurately re-stratifying patients into high-risk and low-risk categories based on their specific clinical and imaging profiles, HEROVision facilitates more personalized treatment decisions with significant clinical implications (Figure 6). For patients initially categorized as high-risk but downgraded to low-risk after re-stratification, a switch from TA to RHR could potentially improve prognosis and extend survival time, despite the higher average costs associated with surgery^{23, 24}. Conversely, for patients who switch from RHR to TA after being downgraded to low-risk, they might benefit from a less invasive procedure with lower complication rates, reduced economic burden, and better preservation of liver function^{25, 26}. For patients initially categorized as low-risk but upgraded to high-risk following re-stratification, a change in treatment strategy—from TA to RHR, or vice versa—might be unnecessary. In these cases, the original treatment approach may have been more effective in managing early-stage rHCC, and altering the treatment plan could potentially worsen outcomes. For those patients who remain high-risk after re-stratification, more aggressive sequential treatment strategies along with

implementing closer monitoring may be warranted. Ultimately, the integration of HEROVision into clinical practice has the potential to enhance treatment outcomes, improve hospital efficiency, and contribute to more personalized patient management. By tailoring treatment strategies to the individual risk profiles of rHCC patients, HEROVision could play a critical role in advancing the standard of care and optimizing resource allocation in clinical settings.]

(page 31/Figure 6 of the revised manuscript)

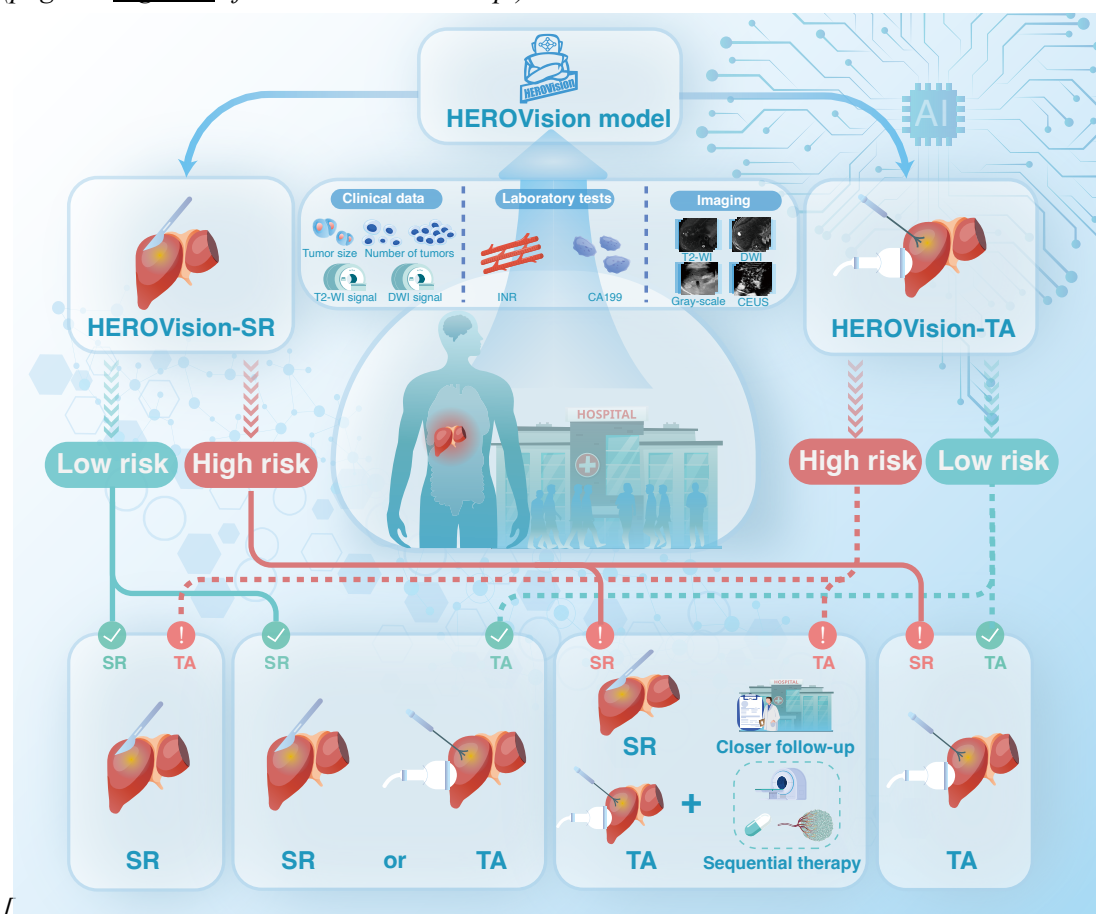


Figure 6. Envisioning the future of rHCC with the clinical integration of the HEROVision system.

First, patients with early-stage rHCC would undergo a comprehensive evaluation, including medical history collection, laboratory assessments, CEUS, and MRI. Subsequently, the HEROVision model processes the relevant data to provide risk stratification for rHCC. Following this, the HEROVision model offers personalized treatment recommendations. rHCC, recurrent hepatocellular carcinoma; CEUS: contrast-enhanced ultrasound; MRI: magnetic resonance imaging; TA, thermal ablation; SR, surgical resection; T2WI, T2-weighted imaging; DWI, diffusion-weighted imaging.]

9. [Remarks on code availability:

The provided GitHub repository is well-documented. The README file offers an overview of

the model's purpose, prerequisites, and instructions for installing necessary libraries and running the application. The dataset handling script (dataset.py) details the preprocessing steps, ensuring reproducibility. The model architecture (Model_hovertrans.py) is documented, including hyperparameters and training specifics, which are crucial for replicating the results. The training scripts (train_hover_SR.py and train_hover_TA.py) provide instructions and use established methods such as the Adam optimizer and Cox Loss, with callbacks for early stopping and model checkpointing. Overall, the repository's organization and detailed documentation make it user-friendly and support the reproducibility of the paper's findings.]

Our Response:

Thank you for your positive feedback on the code repository. We are pleased that you found the documentation and organization user-friendly and supportive of reproducibility. We appreciate your recognition of the efforts made to ensure that the code and accompanying documentation are clear and comprehensive, allowing others to replicate the findings of our study.

- **Response to Reviewer 3**

1. *[I co-reviewed this manuscript with one of the reviewers who provided the listed reports. This is part of the Nature Communications initiative to facilitate training in peer review and to provide appropriate recognition for Early Career Researchers who co-review manuscripts.]*

Our Response:

Thank you once again for your valuable feedback and for your contribution to the review process.

● **Response to Reviewer 4**

1. *[By narrowing the in/exclusion criteria so much, it is not surprising that the clinical classification models (guideline systems) are not performing well. By your inclusion restrictions you focus on an indication niche, which was left liberally open for individual treatment decision making based on knowing the specific patient and the possibilities of the specific medical center. It basically does not make sense to use the clinical scores and classification inside this narrow group at all.]*

Our Response:

Thank you for your comment. We understand and agree with your perspective.

1. Focus on rHCC within Milan Criteria

While the clinical guidelines are designed for all HCC patients, our study specifically focuses on early-stage HCC patients, particularly those meeting the Milan criteria. This group includes both initial HCC (iHCC) and recurrent HCC (rHCC) patients with solitary tumors under 5 cm in diameter, or 2-3 tumors, each with a maximum diameter of 3 cm. According to the Barcelona Clinic Liver Cancer (BCLC) guidelines, only patients at the very early (BCLC-0) or early (BCLC-A) stages of HCC are eligible for curative treatment such as surgical resection or ablation[1, 2]. Therefore, our inclusion/exclusion criteria aim to focus on these early-stage rHCC patients for whom treatment decisions pose a clinical challenge. We believe developing better methods for this niche patient group is critically important.

2. Significance of Comparing to Clinical Guidelines

While we acknowledge that clinical guidelines are designed to encompass a broader HCC population, they are the only available tools for guiding treatment decisions for early-stage rHCC patients in clinical practice. Therefore, comparing our HEROVision model with these guideline-based classification systems is necessary to demonstrate its added value in this specific cohort. Consequently, comparing our HEROVision model against these existing methods is highly relevant. This comparison demonstrates that our model is better suited to early-stage rHCC patients, offering higher prognostic accuracy.

3. Filling the Gap in Existing Guidelines

Our comparison is not intended to discredit existing guideline methods but rather to illustrate that our AI-based model, HEROVision, offers superior performance for early-stage rHCC patients. As noted in the literature, Jiang et al.[3] developed a model that combines serum neutrophil count, tumor size, and arterial phase hyperenhancement proportion to predict recurrence in advanced-stage HCC patients better than existing staging systems. Our model, therefore, aims to fill this critical gap by offering a tool that can provide more precise prognostic information. We believe readers will recognize that this comparison sheds light on the limitations of current guidelines within this narrow patient cohort. We have clarified this point in the **Discussion section** of the revised manuscript, emphasizing that our model may

complement existing guidelines for early-stage rHCC, providing clinicians with enhanced tools to optimize curative treatments. We further acknowledge that for broader HCC populations, clinical guidelines may outperform our model. This highlights the significance of our research, as it addresses a critical gap in the existing clinical guidelines.

References:

- [1] A. G. Singal *et al.*, "AASLD Practice Guidance on prevention, diagnosis, and treatment of hepatocellular carcinoma," (in eng), *Hepatology*, vol. 78, no. 6, pp. 1922-1965, Dec 1 2023, doi: 10.1097/hep.0000000000000466.
- [2] G. O. o. N. H. Commission, "Standard for diagnosis and treatment of primary liver cancer (2022 edition)," *临床肝胆病杂志*, vol. 38, no. 2, pp. 288-303, 2022.

2. [The actual goal of your study and the algorithm is not making a decision regarding which local therapy to use or if it should be used at all but to give an outlook on the prognosis of the patient, namely the OS/PFS. So the decision derived from this information would be to treat or not to treat locally – with the alternative being systemic treatment – or TACE/TARE (which we know to perform inferior to local ablation and surgery in suitable cases). Therefore the title is misleading. You do not investigate on how to improve HCC treatment. In order to give aid in the treatment decision, you would have to include patients in whom the therapy decision was wrong or suboptimal. But how do you know in your cohorts that the treatment they received was inferior to any other treatment option? You cant know that and neither your AI algorithm. So the whole paper should be changed from the title on into “prognostication tool in HCC to be treated by means of local therapy”.]

Our Response:

Thank you for your insightful comments.

The study focuses on rHCC patients who meet the Milan criteria (solitary tumors <5 cm or 2–3 tumors with a maximum diameter <3 cm). According to the BCLC and CNLC staging systems, these patients are categorized as early-stage, for which the recommended treatments are surgical resection or ablation[1, 2]. We recognize that we cannot definitively determine if other treatments such as TACE, TARE, or systemic therapy would result in better PFS or OS compared to surgery and ablation. However, since both surgical resection and thermal ablation are considered curative treatments[3], our study aimed to support optimal decision-making between these options, aligning with clinical guidelines and real-world practice.

We agree that for patients with tumors larger than 5 cm or those with multiple tumors, the benefits of TACE, TARE, or systemic therapies may outweigh those of surgical resection or thermal ablation. However, these patients are not the target population of our HEROVision

model, which is designed specifically for early-stage HCCs where curative treatment options are most optimal.

We emphasize that in China, the follow-up of postoperative HCC patients follows a standardized protocol, which enables the early detection of rHCC in most patients. Therefore, our HEROVision model has practical clinical significance for this specific population.

In the future, we plan to develop models targeting more complex, intermediate-to-advanced HCCs, where TACE, TARE, or systemic therapies might be more beneficial. However, this falls outside the scope of the current study.

As you mentioned, “local therapy” generally encompasses surgical resection, ablation, intra-arterial therapies (e.g., TACE or radioembolization), and radiotherapy (e.g., stereotactic body radiotherapy, SBRT)[4, 5]. Our study specifically addresses surgery and thermal ablation for early-stage rHCC patients. Given this focus, we believe our title, “Vision Transformer-Based Biomarker can Optimize Curative Treatment for Patients with Recurrent Hepatocellular Carcinoma,” is appropriate and accurately reflects the content of the study. Therefore, we suggest retaining the original title.

References:

- [1] A. G. Singal *et al.*, "AASLD Practice Guidance on prevention, diagnosis, and treatment of hepatocellular carcinoma," (in eng), *Hepatology*, vol. 78, no. 6, pp. 1922-1965, Dec 1 2023, doi: 10.1097/hep.0000000000000466.
- [2] G. O. o. N. H. Commission, "Standard for diagnosis and treatment of primary liver cancer (2022 edition)," *临床肝胆病杂志*, vol. 38, no. 2, pp. 288-303, 2022.
- [3] M. Reig *et al.*, "BCLC strategy for prognosis prediction and treatment recommendation: The 2022 update," (in eng), *J Hepatol*, vol. 76, no. 3, pp. 681-693, Mar 2022, doi: 10.1016/j.jhep.2021.11.018.
- [4] V. Arndt, M. Tim, S. Gonzalo, S. Riad, and S. Anna, "Hepatocellular carcinoma," *Lancet*, vol. 400, no. 10360, 2022, doi: 10.1016/s0140-6736(22)01200-4.
- [5] S. Amit G, K. Fasiha, and L. Josep M, "Global trends in hepatocellular carcinoma epidemiology: implications for screening, prevention and therapy," *Nat Rev Clin Oncol*, vol. 20, no. 12, 2023, doi: 10.1038/s41571-023-00825-3.

3. [To include just T2w, DWI and CEUS sounds surprising. How did you decide to use these specific parts of the diagnostic work-up? What do you expect the AI to read on these images? We know that T2 signal and diffusion restriction loosely correlates with the grading of the tumors. What if you provided the algorithm with the grading of the tumors instead of MRI? Or to give just SI-values/radiomics from these images instead of letting the algorithm read this on

its own? Or if you run the algorithm against sets of (para)clinical parameters, which are known to determine the prognosis and outcome of patients?

As said previously, the clinical guideline based categories are not designed to give this fine granulated info on prognosis.]

Our Response:

Thank you for your comment.

1. Rationale for Selecting Imaging Modalities:

The selection of T2WI, DWI, and CEUS modalities was based on the guidelines and clinical practices for HCC surveillance[1]. The American Association for the Study of Liver Diseases (AASLD), European Association for the Study of the Liver (EASL), and Asian Pacific Association for the Study of the Liver (APASL) recommend ultrasound as the primary screening method for HCC[2-4]. However, MRI is often used as a follow-up diagnostic tool due to its higher specificity (98%) compared to ultrasound[5]. Among the functional sequences of MRI, T2WI and DWI are the most commonly used. In China, the majority of HCC patients undergo these imaging tests during follow-up, making these modalities suitable for constructing our model.

2. Approach to Tumor Grading and Diagnosis:

Our HEROVision model does not involve tumor grading. The study focuses on early-stage rHCC patients, with the diagnosis based on tumor size, number, liver function, and other factors. This diagnosis is determined through multidisciplinary discussions involving hepatobiliary surgery and interventional radiology, rather than solely by ultrasound, MRI, or an AI model.

3. Model's Functionality and Imaging Analysis:

Our HEROVision model automatically extracts features from images of these rHCC patients, using Cox regression to associate them with prognosis, followed by parameter training and modeling—essentially employing deep learning techniques. This process does not involve the use of SI values. Once the model is trained with imaging and prognosis data, it can predict prognosis information based on the images. The model then performs risk stratification based on the prognosis, which is a different approach from the traditional clinical staging systems for HCC.

4. Use of Clinical Guidelines for Comparison:

As previously mentioned, the staging systems and clinical guidelines used in our study are intended for prognostic stratification and treatment recommendations for iHCC patients. Currently, the management and treatment of rHCC patients are still aligned with those of iHCC in real-world clinical practice. Although these clinical guidelines are not specifically designed for early-stage HCC patients, due to the lack of guidelines for this particular group (early-stage rHCC), we compared our results with the existing six staging systems and guidelines.

References:

- [1] S. Amit G, K. Fasiha, and L. Josep M, "Global trends in hepatocellular carcinoma epidemiology: implications for screening, prevention and therapy," *Nat Rev Clin Oncol*, vol. 20, no. 12, 2023, doi: 10.1038/s41571-023-00825-3.
- [2] A. G. Singal *et al.*, "AASLD Practice Guidance on prevention, diagnosis, and treatment of hepatocellular carcinoma," (in eng), *Hepatology*, vol. 78, no. 6, pp. 1922-1965, Dec 1 2023, doi: 10.1097/hep.0000000000000466.
- [3] "EASL Clinical Practice Guidelines: Management of hepatocellular carcinoma," (in eng), *J Hepatol*, vol. 69, no. 1, pp. 182-236, Jul 2018, doi: 10.1016/j.jhep.2018.03.019.
- [4] M. Omata *et al.*, "Asia-Pacific clinical practice guidelines on the management of hepatocellular carcinoma: a 2017 update," (in eng), *Hepatol Int*, vol. 11, no. 4, pp. 317-370, Jul 2017, doi: 10.1007/s12072-017-9799-9.
- [5] P. Gupta, R. Soundararajan, A. Patel, M. P. Kumar, V. Sharma, and N. Kalra, "Abbreviated MRI for hepatocellular carcinoma screening: A systematic review and meta-analysis," (in eng), *J Hepatol*, vol. 75, no. 1, pp. 108-119, Jul 2021, doi: 10.1016/j.jhep.2021.01.041.

4. [Aim: "This approach aimed to optimize individual treatment selection between RHR and TA for patients with early-stage rHCC in real-world scenarios."]

Well, but the algorithm was not product of the study at hand, right? So what was the actual aim of the present study? Please phrase a specific aim or hypothesis.]

Our Response:

Thank you for your comment. We have revised the statement to clearly define the actual aim of the study. The revised aim is: "This approach aimed to develop and validate a ViT-based model designed for optimizing individual treatment selection between RHR and TA for patients with early-stage rHCC in real-world scenarios." This has been updated in the revised manuscript.

Revised text:

(page 5/Introduction section of the revised manuscript)

[This approach aimed to develop and validate a ViT-based model designed for optimizing individual treatment selection between RHR and TA for patients with early-stage rHCC in real-world scenarios.]

5. [Methods: Did you perform any tumor segmentation on the MRI images or the US images to

let the algorithm know which is the tumor and which the surrounding liver? Did the algorithm use any anatomic information, which is the most important input a surgeon or radiologist need to make a decision regarding local treatment? For me the plausibility is lacking if these important informations were not used by the algorithm.]

Our Response:

Thank you for your insightful comments.

Our HEROVision model does not require tumor segmentation to differentiate between the tumor and surrounding liver tissue. Instead, it focuses on analyzing imaging features from both the tumor and its surrounding tissue, including but not limited to anatomical features, texture characteristics, and high-dimensional hidden features. Specifically, the model automatically extracted 768 features from MRI images and 640 features from US images, which were then integrated into the model to achieve the current prognostic stratification results.

While we did not perform explicit tumor segmentation, we did conduct ROI selection, which is detailed in the **Appendix S5** of the supplementary materials. We believe that including surrounding tissue within the ROI is crucial, as the AI model likely extracts features related to tumor aggressiveness from this area, which is significantly associated with prognosis[1-3].

Our approach allows the extraction of both intra-tumoral and peri-tumoral features, contributing to a more effective prognostic stratification. This is one of the key advantages of our HEROVision model.

References:

- [1] F. Wang *et al.*, "Multitask deep learning for prediction of microvascular invasion and recurrence-free survival in hepatocellular carcinoma based on MRI images," (in eng), *Liver Int*, vol. 44, no. 6, pp. 1351-1362, Jun 2024, doi: 10.1111/liv.15870.
- [2] W. Wang *et al.*, "A Transformer-Based microvascular invasion classifier enhances prognostic stratification in HCC following radiofrequency ablation," (in eng), *Liver Int*, vol. 44, no. 4, pp. 894-906, Apr 2024, doi: 10.1111/liv.15846.
- [3] X. Xu *et al.*, "Radiomic analysis of contrast-enhanced CT predicts microvascular invasion and outcome in hepatocellular carcinoma," (in eng), *J Hepatol*, vol. 70, no. 6, pp. 1133-1144, Jun 2019, doi: 10.1016/j.jhep.2019.02.023.

Revised text:

(page 6-7/Appendix S5 of the revised supplementary material)

[...Three-dimensional regions of interest (ROIs) for HCC lesions were manually delineated slice-by-slice on both axial T2WI and DWI. Two-dimensional ROIs were manually contoured onto the grayscale and CEUS images using the same approach.

To minimize the impact of variability due to different hospitals, equipment, and technical parameters on subsequent analysis, we employed denoising and image augmentation techniques^{2, 3}. Denoising was performed using advanced filtering methods, such as Gaussian filters, to remove random noise and enhance image quality². Image augmentation techniques, including rotation, scaling, translation, and contrast adjustment, were utilized to diversify the dataset, improve model robustness, and reduce the risk of overfitting³. Then, in order to objectively extract tumor subregions within the tumor and alleviate human bias, we expanded all drawn regions of interest masks outward by 5mm, allowing for consideration of both the internal region of the lesion and the surrounding liver parenchyma, providing more feature information for deep learning (DL) networks. Python was used to extract irregular bounding boxes of tumor subregions from various patterns. In order to improve the processing efficiency of DL models, we transform irregular boundaries into regular square boundaries and then crop them...]

6. [Fig 1: Legend should explain all abbreviations. These abbreviations should then be used in all occasions where they appear inside the table (e.g. not hepatectomy but SR).

Inclusion: clarify by writing “... of PRETHERAPEUTIC contrast enhanced US and liver MRI within a 1-month interval.”

Exclusion: Please clarify: “unqualified artefacts” – shouldn’t this read: “unqualifying artefacts”?

In the external testing cohort, you include patients with resection of rHCC. How can you end up with a TA testing cohort?]

Our Response:

Thank you for your comments.

1. Abbreviations in Figure 1

We have updated the legend in **Figure 1** to explain all abbreviations. All instances of abbreviations in the table are now consistent with those used in the legend.

2. Inclusion Criteria Clarification

We have revised the inclusion criteria to specify: “... of pretherapeutic contrast-enhanced ultrasound (CEUS) and magnetic resonance imaging (MRI) data within a 1-month interval.”

3. Term Clarification for Artifacts

We have considered your suggestion and would like to clarify that the term “unqualified artifacts” was used intentionally. In this context, “unqualified artifacts” refers to artifacts that

do not meet the necessary quality standards or criteria for analysis. The term “unqualifying artifacts” might imply that the artifacts themselves do not disqualify the image, which is not the intended meaning. Therefore, we believe that “unqualified artifacts” more accurately describes artifacts of insufficient quality that are excluded from consideration.

4. Testing Cohort Explanation

As previously mentioned, the HEROVision model consists of two sub-models: HEROVision-SR and HEROVision-TA. These models were trained using data from iHCC patients who underwent surgical resection and thermal ablation, respectively. The external testing cohorts include rHCC patients who received SR and TA, respectively. The inclusion of both types of patients in the testing cohorts aligns with the model’s design to evaluate performance across different treatment modalities.

Revised text:

(page 12/Discussion section of the revised manuscript)

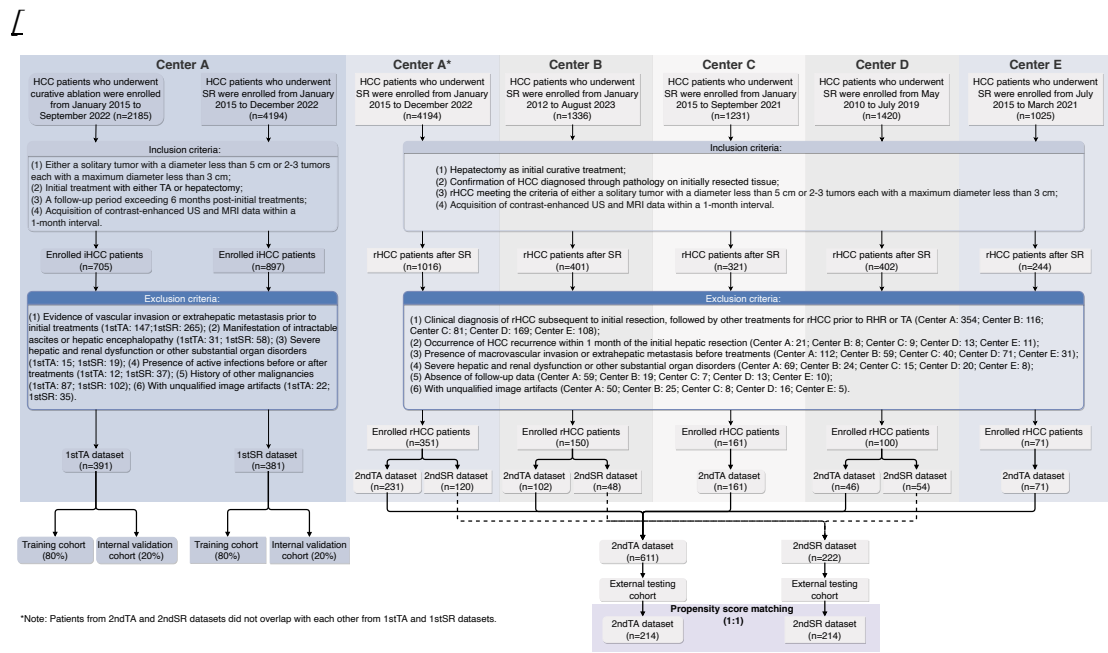


Figure 1. Flowchart of Patient Inclusion and Exclusion

iHCC, initial hepatocellular carcinoma; rHCC, recurrent hepatocellular carcinoma; TA, thermal ablation; SR, surgical resection.]

(page 3/Appendix S1 of supplementary material)

[For initial hepatocellular carcinoma (iHCC)...The exclusion criteria were: ... (4) acquisition of pretherapeutic contrast-enhanced ultrasound (CEUS) and magnetic resonance imaging (MRI) data within a 1-month interval...

For recurrent hepatocellular carcinoma (rHCC)...The exclusion criteria were: ... (4) acquisition of pretherapeutic CEUS and MRI data within a 1-month interval...]

Response to the Comments of All Reviewers

Revision Report

Manuscript Title: Vision Transformer-Based Biomarker can Optimize Curative-Intent Treatment for Patients with Recurrent Hepatocellular Carcinoma

Manuscript Number: NCOMMS-24-36298A

Date of this Revision: January 10, 2025

Thank you very much for kindly giving us an opportunity to revise our manuscript. We have revised the manuscript accordingly and gave point-by-point response to all the reviewers. Great thanks to our reviewers. They really helped us a lot to improve this work. We hope the revision has met the standard of *Nature Communications*. Please feel free to contact us if any further concerns should be addressed.

Table of Contents:

●	REVIEWER COMMENTS	2
●	Response to Reviewer 1	5
●	Response to Reviewer 2	6
●	Response to Reviewer 3	14
●	Response to Reviewer 5	15

● **REVIEWER COMMENTS**

Reviewer #1 (Remarks to the Author):

Thank you for addressing my main methodological concerns. The revisions have satisfactorily clarified the issues.

Reviewer #1 (Remarks on code availability):

I have broadly reviewed the provided code for structure. Although I did not run it, it appears well-documented and aligns with the methodology described in the manuscript.

Reviewer #2 (Remarks to the Author):

EDITORIAL NOTE: Reviewer #1 assessed your Response to Reviewer #2. Reviewer #1 indicated that, while most of Reviewer #2's comments were addressed, the following points from Reviewer #2's previous report were still pending:

- The newly added Figure 6 is confusing and overly complex; the Reviewer believes it may have been generated by a vision-language model (please note that we cannot publish AI-generated images). It remains unclear what is the role that HEROVision would play in supporting physicians and the points across diagnosis and therapy where the model would be applied.
- Reviewer #2's concern about the variability in imaging quality and interpretation across different modalities was not properly addressed in the revision. This variability is critical for the accuracy of the model's predictions.

Reviewer #3 (Remarks to the Author):

I co-reviewed this manuscript with one of the reviewers who provided the listed reports. This is part of the Nature Communications initiative to facilitate training in peer review and to provide appropriate recognition for Early Career Researchers who co-review manuscripts.

Reviewer #3 (Remarks on code availability):

The provided GitHub repository is well-documented, with the README file offering an overview of the model's purpose, prerequisites, and instructions for installing necessary libraries and running the application. The dataset handling script (dataset.py) details the

preprocessing steps, ensuring reproducibility. The model architecture (Model_hovertrans.py) is thoroughly documented, including hyperparameters and training specifics, which are crucial for replicating the results. The training scripts (train_hover_SR.py and train_hover_TA.py) provide clear instructions and utilize established methods such as the Adam optimizer and Cox Loss, with callbacks for early stopping and model checkpointing. Overall, the repository is well-organized, making it user-friendly and supporting the reproducibility of the paper's findings.

Reviewer #5 (Remarks to the Author): Clinical expert in hepatocellular carcinoma radiology; replaces Reviewer #4

I serve as a new reviewer for the second round of review of this article.

I have been able to appreciate the additional work done by the authors in response to the first round of comments.

The methodology is comprehensive and employs state-of-the-art deep learning techniques. The use of multimodal imaging data and propensity score matching enhances the study's validity. The decision to validate the HEROVision model on external datasets is a significant strength, providing evidence of generalizability. Nonetheless, some concerns include:

- Training on initial hepatocellular carcinoma (iHCC) data for application to rHCC may introduce biases, despite biological similarities between the two.
- The geographic and demographic variability of datasets is noted but not deeply analyzed for potential biases
- While the model architecture is described, details on hyperparameter optimization and model interpretability are limited.
- The explainability of the models is limited. Did the authors attempt to understand why the model classifies patients into low- or high-risk categories? Providing insights into this aspect would enhance interpretability and clinical relevance.
- As noted by a previous reviewer, the study focuses exclusively on patients who are candidates for both surgical resection and thermal ablation, excluding those eligible for other treatment modalities. This limitation should be addressed or acknowledged in the discussion.
- The most stimulating part of the study is the comparison between the two models, which may suggest which treatment is better. However, the models do not provide clear guidance in cases where both classify patients into the same risk group. This limitation should be explored further.
- The authors should focus more on discrepant cases, where one model reclassifies a patient into a different risk group. Detailed characteristics of these patients and their tumors should be provided to better understand why they were initially allocated to the “wrong” group and how

reclassification impacts outcomes.

- Current staging systems, such as the BCLC, are not designed for recurrent disease. Therefore, it is not surprising that they perform less effectively than the proposed models. This limitation should be explicitly discussed or mentioned in the manuscript.
- Minor comment: The term "curative-intent" is preferable to "curative" for accuracy and consistency in describing the therapeutic goals.
- Additionally, CEUS is rarely used as a surveillance tool in clinical practice, whereas CT is more commonly employed. This discrepancy should be addressed in the manuscript, as the lack of discussion on CT may limit the applicability of the study's findings.

Overall, the study provides valuable insights into the use of advanced deep learning models for optimizing rHCC treatment. However, revisions are necessary to address concerns. With these improvements, the manuscript would be a strong candidate for publication.

- **Response to Reviewer 1**

1. *[Thank you for addressing my main methodological concerns. The revisions have satisfactorily clarified the issues.]*

Our Response:

Thank you for your positive feedback and for recognizing our efforts in addressing your methodological concerns. We are very glad that the revisions have clarified the issues to your satisfaction.

2. *[I have broadly reviewed the provided code for structure. Although I did not run it, it appears well-documented and aligns with the methodology described in the manuscript.]*

Our Response:

Thank you for taking the time to review the code. We appreciate your recognition of its documentation quality and alignment with the described methodology.

● **Response to Reviewer 2**

1. *[The newly added Figure 6 is confusing and overly complex; the Reviewer believes it may have been generated by a vision-language model (please note that we cannot publish AI-generated images). It remains unclear what is the role that HEROVision would play in supporting physicians and the points across diagnosis and therapy where the model would be applied.]*

Our Response:

Thank you for your valuable feedback. We appreciate your observations regarding Figure 6 and its potential clarity issues. Below, we address your concerns in detail:

1) Clarification on Figure Creation

We confirm that all figures in our manuscript, including Figure 6, were manually created by our research team and not generated by any vision-language model or AI-based tools. We value the integrity of academic publication standards and have ensured that all visualizations are original and specifically tailored to our study.

2) Simplification and Enhancement of Figure 6

To improve clarity and reduce complexity, we have revised Figure 6. The updated figure (which has been placed in supplementary **Figure S18**) provides a more streamlined depiction of HEROVision's functionalities and its integration into clinical workflows for recurrent HCC (rHCC).

3) Role of HEROVision in Supporting Physicians

(1) Predicting Progression-Free Survival (PFS):

HEROVision serves as an effective tool for predicting PFS in early-stage HCC patients who are planning to undergo surgical resection or thermal ablation, including both initial HCC (iHCC) and rHCC.

(2) Supplementing Guidelines in the Absence of rHCC-Specific Guidelines:

Given the lack of clinical guidelines and staging systems tailored to rHCC, HEROVision bridges this gap by offering reliable preoperative risk stratification.

(3) Personalized Treatment Recommendations:

For rHCC patients within the Milan criteria, HEROVision assists in personalized treatment decisions, particularly between surgical resection and thermal ablation. For example, the model identifies high-risk patients initially scheduled for ablation who could benefit from being reclassified into the low-risk category following a switch to surgical resection, resulting in significantly improving survival outcomes.

These roles and clinical implications of HEROVision have been further elaborated in the revised discussion section to provide a comprehensive understanding of its significance and application.

Revised text:

(page 55/Figure S18 of the revised supplementary material)

[

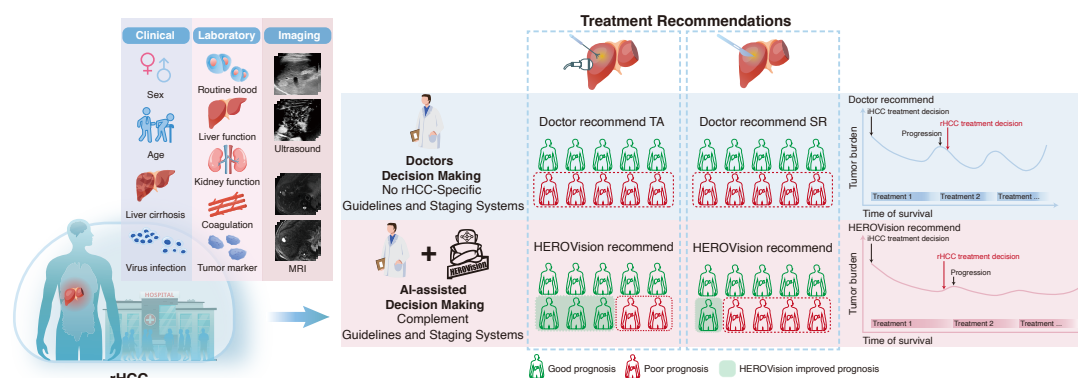


Figure S18. Envisioning the Future of rHCC Management with Clinical Integration of the HEROVision System

First, patients with early-stage rHCC undergo a comprehensive evaluation, including medical history collection, laboratory assessments, CEUS, and MRI. Subsequently, HEROVision processes the relevant data to provide prognostic stratification for patients with rHCC. Following this, HEROVision assists doctors in making personalized treatment decisions. Furthermore, after HEROVision predicts tumor progression, doctors can optimize curative-intend treatment before tumor resistance and uncontrolled growth.

iHCC, initial hepatocellular carcinoma; rHCC, recurrent hepatocellular carcinoma; CEUS, contrast-enhanced ultrasound; MRI, magnetic resonance imaging; TA, thermal ablation; SR, surgical resection.]

(page 10-11/Discussion section of the revised manuscript)

[Currently, there are no established guidelines or definitive classification systems for the management of rHCC^{7, 8, 9, 10}, with most consensus recommendations still following the staging and treatment guidelines for iHCC. The HEROVision model offers a novel approach to individualized preferences in optimizing treatment for rHCC within the Milan criteria. By accurately re-stratifying patients into good and poor prognosis groups based on their specific clinical and imaging profiles, HEROVision facilitates more personalized treatment decisions with significant clinical implications (Figure S18). For TA, 35.9% of patients initially categorized as high-risk could be downgraded to low-risk after re-stratification, and switching from TA to RHR could probably improve their prognosis and extend survival time, despite the higher average cost associated with surgery^{23, 24}. Conversely, for RHR, although only 6.6% of high-risk patients who could benefit from switching RHR to TA, they would have a higher probability of better prognosis, less invasive procedure with lower complication rates, reduced economic burden, and better preservation of liver function^{25, 26}. Obviously, the above personalized treatment changes are of great clinical significance. For patients initially

categorized as low-risk but upgraded to high-risk following re-stratification, any change in treatment strategy would be unnecessary and might make outcomes worse. For patients who remain low-risk after re-stratification, both treatment options could be effective, and the decision should be made collaboratively by doctors and patients. For those who remained high-risk after re-stratification, more aggressive sequential treatment strategies (such as combining with transarterial chemoembolization [TACE] or systemic therapies) along with implementing closer monitoring might be warranted. Ultimately, the integration of HEROVision into clinical practice has the potential to enhance treatment outcomes, improve hospital efficiency, and contribute to more personalized patient management. By tailoring treatment strategies to the individual risk profiles of rHCC patients, HEROVision could play a critical role in advancing the standard of care and optimizing resource allocation in clinical settings.]

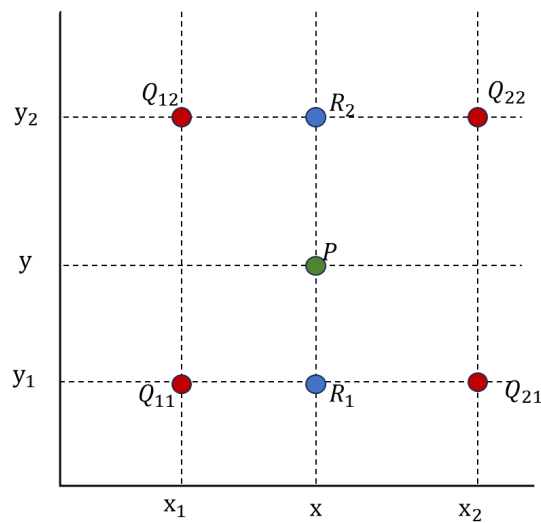
2. [Reviewer #2's concern about the variability in imaging quality and interpretation across different modalities was not properly addressed in the revision. This variability is critical for the accuracy of the model's predictions.]

Our Response:

Thank you for your insightful comment regarding the variability in imaging quality and its potential impact on the accuracy of the model's predictions.

1) Standardization of Image Processing:

This study included only HCC tumors with a size of less than 5 cm, regardless of whether they were iHCC or rHCC. To standardize input dimensions, we cropped ROIs and resized them to 128×128 pixels using bilinear interpolation[1]. This method preserves essential image details while avoiding significant quality loss.



$$f(x, y_1) \approx \frac{x_2 - x}{x_2 - x_1} f(Q_{11}) + \frac{x - x_1}{x_2 - x_1} f(Q_{21}) \quad (1-1)$$

$$f(x, y_2) \approx \frac{x_2 - x}{x_2 - x_1} f(Q_{12}) + \frac{x - x_1}{x_2 - x_1} f(Q_{22}) \quad (1-2)$$

$$f(x, y) \approx \frac{y_2 - y}{y_2 - y_1} f(x, y_1) + \frac{y - y_1}{y_2 - y_1} f(x, y_2) \quad (1-3)$$

Bilinear interpolation calculates the pixel value of a target point (P) based on the values of four neighboring points (Q_{11} , Q_{12} , Q_{21} , Q_{22}). First, linear interpolation is performed along the x-axis to calculate intermediate values (R_1 , R_2), and then along the y-axis to derive the final pixel value for P. This process preserves essential image features while standardizing dimensions for model compatibility.

2) Maintaining Image Detail and Quality:

The selection of 128×128 pixels as the standard size was based on maintaining a balance between computational efficiency and image quality. For ViT-based models, the image was divided into 16×16 patch blocks, resulting in a grid of 8×8 patches. This approach minimized the loss of heterogeneity between pixels, a critical factor for capturing subtle early liver cancer features, and facilitated robust comparison across models.

3) Analysis of Tumor Size Variability:

Resizing ratios vary depending on tumor size:

- (1) For smaller tumors, the resizing ratio is higher.
- (2) For larger tumors, the resizing ratio is lower.

To ensure resizing does not adversely affect image quality or model performance, we conducted subgroup analyses based on tumor size (≤ 3 cm and > 3 cm) in the 2ndSR and 2ndTA datasets. Results, as shown in **Figures S14, S15** and **Table S7**, revealed no significant differences in model performance across subgroups. This finding indicated that resizing did not introduce bias or degrade image quality in a way that affected the model's accuracy.

4) Implications for Imaging Variability Across Modalities:

By standardizing image preprocessing and employing resizing techniques that preserve key image features, we mitigated the variability in imaging quality inherent to different modalities. Our approach ensures that the model processes images consistently, irrespective of their original resolution or modality. To further address variability in imaging modalities, we conducted ablation studies and further compared the performance of single-modality CEUS or single-modality MRI versus combined CEUS and MR in both C-index and time-independent AUC. Detailed results are provided in **Figure S11** and **Table S5**. These analyses demonstrated stable and accurate prognostic predictions, confirming the model's robustness against variability.

We have incorporated these details into the revised manuscript to clarify the robustness of our preprocessing pipeline and its ability to handle variability in imaging quality. Thank you for highlighting this critical aspect. We hope those results can remove your concerns this time.

References:

[1] E. J. Kirkland, *Advanced computing in electron microscopy*. Springer, 1998.

Revised text:

(page 8-9/ Appendix S5 section of the revised supplementary material)

[Resizing with bilinear interpolation:

$$f(x,y \ 1) \approx (x \ 2-x)/(x \ 2-x \ 1) f(Q \ 11) + (x-x \ 1)/(x \ 2-x \ 1) f(Q \ 21)$$

$$f(x,y \ 2) \approx (x \ 2-x)/(x \ 2-x \ 1) f(Q \ 12) + (x-x \ 1)/(x \ 2-x \ 1) f(Q \ 22)$$

$$f(x,y) \approx (y \ 2-y)/(y \ 2-y \ 1) f(x,y \ 1) + (y-y \ 1)/(y \ 2-y \ 1) f(x,y \ 2)$$

Bilinear interpolation calculates the pixel value of a target point (P) based on the values of four neighboring points (Q_{11} , Q_{12} , Q_{21} , Q_{22}). First, linear interpolation is performed along the x-axis to calculate intermediate values, and then along the y-axis to derive the final pixel value for P.]

(page 46/Figure S11 of the revised supplementary material)

[

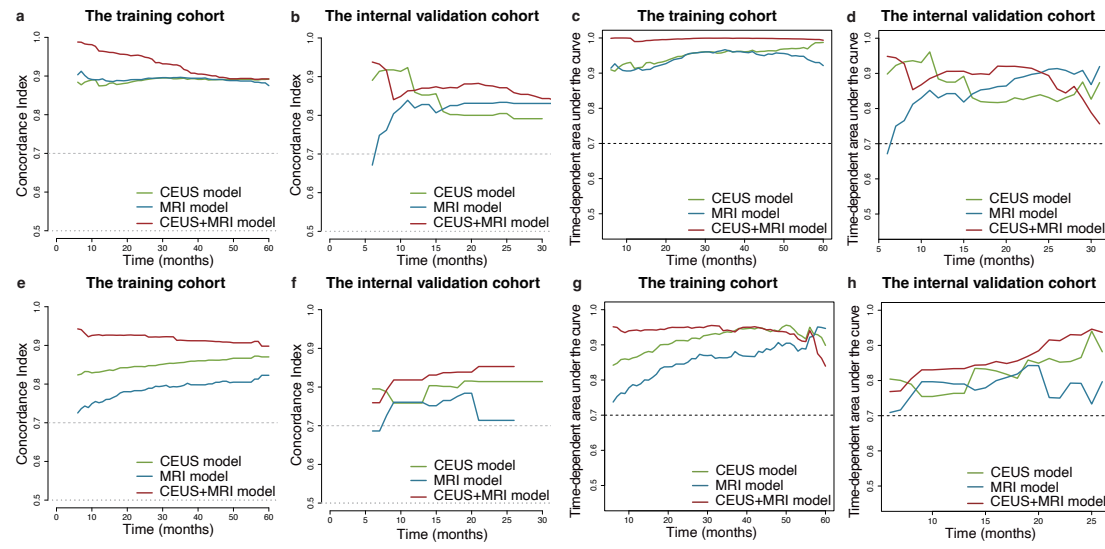


Figure S11. The Discriminative Performances of Different Modality Models in the Training and Internal Validation Cohorts

For TA dataset (a-d) and SR dataset (e-h), the C-index in the training and internal validation cohorts (a, b, e and f), and the time-dependent AUC in both cohorts (c, d, g and h) reflected the models' discriminatory performance at various points.

TA, thermal ablation; SR, surgical resection; C-index, concordance index; AUC, area under the curve; MRI, magnetic resonance imaging; CEUS, contrast-enhanced ultrasound.]

(page 49-50/Figure S14 of the revised supplementary material)

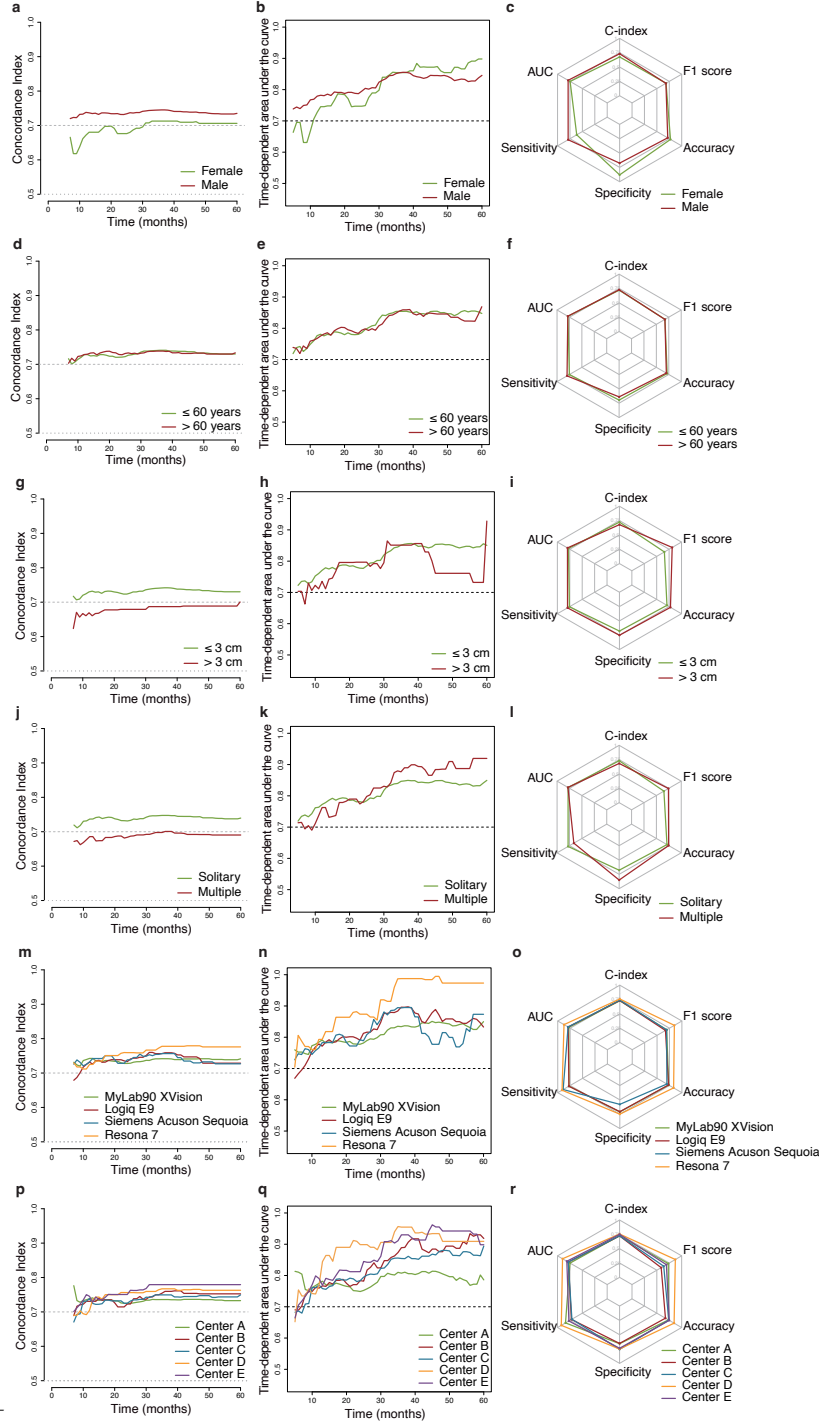


Figure S14. The Discriminative Performances of Different Subgroups for the 2ndTA Dataset
The C-index (a, d, g, j, m, and p) and time-dependent AUC (b, e, h, k, n, and q) for different subgroups, including sex, age, tumor size, the number of tumors, manufacturers, and centers, demonstrated the models' discriminatory performance at various points in the 2ndTA cohort. (c, f, i, l, o, and r) Radar charts illustrated the C-index, AUC, specificity, sensitivity, accuracy, and F1 score for 2-year PFS across these subgroups.
TA, thermal ablation; ViT, Vision Transformer; C-index, concordance index; AUC, area under the curve; PFS, progression-free survival.]

(page 51-52/Figure S15 of the revised supplementary material)

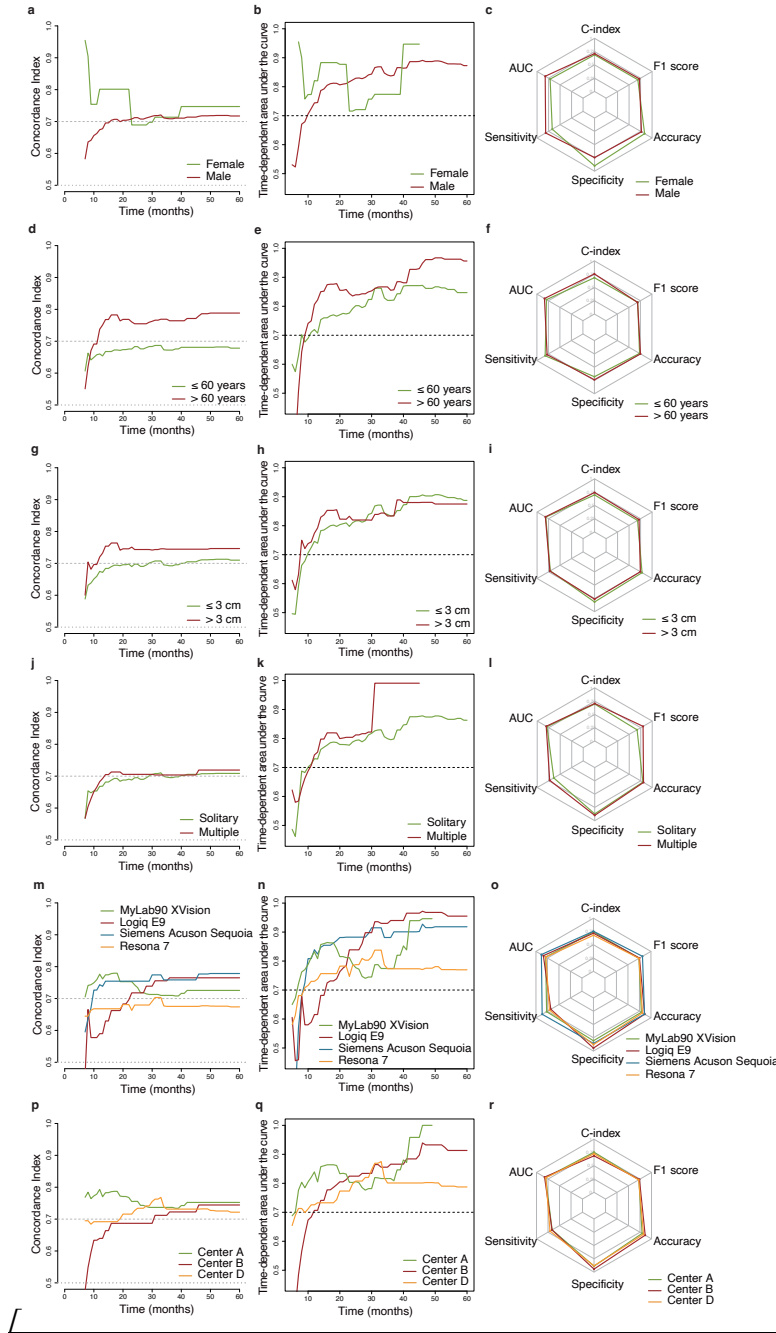


Figure S15. The Discriminative Performances of Different Subgroups for the 2ndSR Dataset
The C-index (a, d, g, j, m, and p) and time-dependent AUC (b, e, h, k, n, and q) for different subgroups, including sex, age, tumor size, the number of tumors, manufacturers, and centers, demonstrated the models' discriminatory performance at various points in the 2ndSR cohort. (c, f, i, l, o, and r) Radar charts illustrated the C-index, AUC, specificity, sensitivity, accuracy, and F1 score for 2-year PFS across these subgroups.
TA, thermal ablation; ViT, Vision Transformer; C-index, concordance index; AUC, area under the curve; PFS, progression-free survival.]

(page 24/Table S5 of the revised supplementary material)

[Table S5. Performance Comparison of the Different Modality Models for 2-year PFS in the Training and Internal Validation Cohorts]

Cohorts	Models	TA dataset				SR dataset			
		2-year C-index	P Value	2-year AUC	P Value	2-year C-index	P Value	2-year AUC	P Value
The training cohort	CEUS+MRI	0.95 (0.93 - 0.98)	Ref.	1.00 (0.99 - 1.00)	Ref.	0.93 (0.90 - 0.95)	Ref.	0.95 (0.92 - 0.98)	Ref.
	CEUS	0.89 (0.85 - 0.92)	0.003	0.95 (0.92 - 0.98)	<0.001	0.85 (0.81 - 0.88)	0.001	0.91 (0.88 - 0.94)	0.063
	MRI	0.89 (0.86 - 0.93)	0.004	0.94 (0.92 - 0.96)	<0.001	0.78 (0.74 - 0.83)	<0.001	0.85 (0.80 - 0.89)	<0.001
The internal validation cohort	CEUS+MRI	0.87 (0.80 - 0.94)	Ref.	0.90 (0.81 - 1.00)	Ref.	0.85 (0.76 - 0.94)	Ref.	0.93 (0.82 - 1.00)	Ref.
	CEUS	0.80 (0.72 - 0.89)	0.254	0.84 (0.72 - 0.96)	0.224	0.81 (0.71 - 0.91)	0.570	0.87 (0.74 - 0.99)	0.250
	MRI	0.83 (0.75 - 0.92)	0.504	0.90 (0.80 - 1.00)	0.956	0.71 (0.60 - 0.83)	0.063	0.79 (0.58 - 1.00)	0.028

1

(page 27-28/Table S7 of the revised supplementary material)

[Table S7. Performance Comparison of HEROVision Across Different Subgroups for 2-Year PFS in the External Testing Cohorts]

Datasets	Subgroups	Number	2-year C-index	P value	2-year AUC	P value	Sensitivity	Specificity	Accuracy	F1 score
2ndTA dataset	Tumor size									
	≤3 cm	n=563	0.72 (0.69 - 0.76)	Ref.	0.78 (0.74 - 0.82)	Ref.	0.76 (0.72 - 0.79)	0.68 (0.64 - 0.72)	0.71 (0.67 - 0.74)	0.66 (0.62 - 0.69)
	>3 cm	n=48	0.68 (0.55 - 0.81)	0.525	0.80 (0.64 - 0.95)	0.745	0.80 (0.68 - 0.91)	0.75 (0.63 - 0.87)	0.78 (0.66 - 0.90)	0.81 (0.70 - 0.92)
2ndSR dataset	Tumor size									
	≤3 cm	n=166	0.70 (0.63 - 0.77)	Ref.	0.81 (0.73 - 0.89)	Ref.	0.71 (0.64 - 0.78)	0.82 (0.76 - 0.88)	0.78 (0.72 - 0.85)	0.68 (0.61 - 0.76)
	>3 cm	n=56	0.74 (0.63 - 0.86)	0.503	0.82 (0.70 - 0.94)	0.876	0.72 (0.60 - 0.84)	0.76 (0.65 - 0.88)	0.75 (0.63 - 0.86)	0.72 (0.60 - 0.83)

[TA, thermal ablation; SR, surgical resection; PFS, progression-free survival; AUC, area under the curve; Ref, reference.]

- **Response to Reviewer 3**

1. [I co-reviewed this manuscript with one of the reviewers who provided the listed reports. This is part of the Nature Communications initiative to facilitate training in peer review and to provide appropriate recognition for Early Career Researchers who co-review manuscripts. The provided GitHub repository is well-documented, with the README file offering an overview of the model's purpose, prerequisites, and instructions for installing necessary libraries and running the application. The dataset handling script (dataset.py) details the preprocessing steps, ensuring reproducibility. The model architecture (Model_hovertrans.py) is thoroughly documented, including hyperparameters and training specifics, which are crucial for replicating the results. The training scripts (train_hover_SR.py and train_hover_TA.py) provide clear instructions and utilize established methods such as the Adam optimizer and Cox Loss, with callbacks for early stopping and model checkpointing. Overall, the repository is well-organized, making it user-friendly and supporting the reproducibility of the paper's findings.]

Our Response:

Thank you for your detailed review and for highlighting the strengths of our GitHub repository. We are delighted that you found the documentation and organization of the repository to be thorough and user-friendly. Ensuring reproducibility and clarity was a primary goal, and we greatly appreciate your recognition of these efforts. If there are any additional suggestions for improvement, we would be happy to incorporate them.

● **Response to Reviewer 5**

1. *[Training on initial hepatocellular carcinoma (iHCC) data for application to rHCC may introduce biases, despite biological similarities between the two.]*

Our Response:

Thank you for raising this important point. We acknowledge the potential biases that may arise from using initial HCC (iHCC) for model training and appreciate the opportunity to explain our rationale and address the implications in detail.

1) Rationale for Using iHCC Data in Model Development:

The decision to train the model using non-treated iHCC data and validate it on recurrent HCC (rHCC) data was driven by practical constraints in data availability. Assembling a sufficiently large and high-quality dataset exclusively from rHCC patients who underwent repeated surgical resection or thermal ablation, with complete follow-up and no additional treatments between the initial and subsequent interventions, proved to be highly challenging, even when collaborating with major tertiary hospitals across northern, central, southern, and western China. Given these constraints, numerous studies have demonstrated significant biological similarities between iHCC and rHCC in terms of tumor behavior and liver environment[1-4]. Based on this evidence, we adopted the strategy of using iHCC data for model training, and our external testing results (e.g., a 2-year AUC of 0.78 in the TA dataset and 0.81 in the SR dataset) further support the feasibility of this approach.

2) Impact of Dataset Differences on Model Performance:

We recognize that training on iHCC data introduces potential biases due to inherent differences between iHCC and rHCC. Despite this, our results demonstrate acceptable performance gaps between internal validation (e.g., 2-year AUC of 0.90–0.92) and external testing (e.g., 2-year AUC of 0.78–0.81), with differences of 0.11–0.12. These variations are within a reasonable range considering the challenges of rHCC-specific data collection and highlight the model's robustness under these constraints.

3) Novelty and Significance of Our Approach:

Despite the limitations, our study represents a pioneering effort to develop an AI model specifically addressing prognostic prediction in rHCC patients, a population currently underserved by existing clinical guidelines and staging systems. Our model outperformed six widely used clinical guidelines and staging systems, demonstrating superior prognostic accuracy and offering a significant step forward in addressing this clinical need. Although this may not represent the ideal approach, it is a practical and innovative solution to a longstanding challenge, paving the way for future improvements in this domain.

4) Future Directions and Generalizability:

To further assess the model's generalizability, we included data from two additional centers involving repeated surgical resection and thermal ablation. These analyses confirmed the

model's robustness and applicability across various clinical settings. However, we acknowledge the need for further independent validation, particularly with prospective datasets, to fully establish the model's efficacy and generalizability. This limitation is recognized as a key area for future research.

To enhance the generalizability of our findings, we included data from five centers, encompassing patients who underwent repeated surgical resection and thermal ablation. These results confirmed the model's robustness across diverse clinical scenarios. However, we recognize the need for further prospective validation to fully establish the model's efficacy. Future efforts will focus on acquiring larger, multicenter datasets specifically from rHCC patients and refining the model to further improve its performance and reliability. This limitation and our future plans have been explicitly acknowledged in the **revised discussion section**.

In conclusion, while we acknowledge the potential biases and limitations of our approach, our study offers a feasible and impactful strategy for addressing the lack of AI models tailored to rHCC. We are committed to further validating and refining the model through prospective studies to ensure its practical utility in clinical decision-making.

References:

- [1] T. Yoh *et al.*, "Surgery for Recurrent Hepatocellular Carcinoma: Achieving Long-term Survival," (in eng), *Ann Surg*, vol. 273, no. 4, pp. 792-799, Apr 1 2021, doi: 10.1097/sla.0000000000003358.
- [2] H. Imamura *et al.*, "Risk factors contributing to early and late phase intrahepatic recurrence of hepatocellular carcinoma after hepatectomy," (in eng), *J Hepatol*, vol. 38, no. 2, pp. 200-7, Feb 2003, doi: 10.1016/s0168-8278(02)00360-4.
- [3] J. C. Wu *et al.*, "Risk factors for early and late recurrence in hepatitis B-related hepatocellular carcinoma," (in eng), *J Hepatol*, vol. 51, no. 5, pp. 890-7, Nov 2009, doi: 10.1016/j.jhep.2009.07.009.
- [4] X. Ding *et al.*, "Genomic and Epigenomic Features of Primary and Recurrent Hepatocellular Carcinomas," (in eng), *Gastroenterology*, May 16 2020, doi: 10.1053/j.gastro.2019.09.056.

Revised text:

(page 11/Discussion section of the revised manuscript)

[In our study, we developed HEROVision using iHCC data and validated it with rHCC data for several reasons. On one hand, the strategy to use iHCC for model development and rHCC for validation was primarily driven by practical constraints. Despite our collaboration with hospitals across northern, central, southern, and western China, assembling a large rHCC dataset for training was challenging due to difficulties in collecting a sufficient number of rHCC patients with comprehensive follow-up data and no additional treatments between

interventions. On the other hand, from a pathological and tumor microenvironment (TME) perspective, the biological behavior of rHCC closely mirrors that of iHCC. Time to recurrence has been widely recognized as a biological surrogate marker: early recurrence, typically within 2 years post-surgery, is believed to share the same clonal origin as the iHCC, while late recurrence, occurring after more than 2 years, is thought to arise from a de novo origin, driven by underlying liver damage that persists despite tumor resection^{27, 28, 29}. Furthermore, Ding et al.³⁰ compared genetic features and found that 52% of rHCCs share the clonal lineage of the initial tumor. Both conditions are characterized by a similar liver environment due to underlying cirrhosis. This biological similarity supports our decision to use iHCC data for model development, offering a practical solution to the challenge of obtaining sufficient rHCC data. Although this approach may not be perfect, it effectively addresses data scarcity and enables meaningful model development. HEROVision demonstrated effectiveness in rHCC patients, as evidenced by our study: the 2-year AUC was 0.90 for the internal validation cohort and 0.78 for the external testing cohort in the TA dataset, and 0.92 and 0.81, respectively, in the SR dataset.]

(page 12/Limitation section of the revised manuscript)

[... Second, because the number of patients with rHCC underwent RHR is limited in clinical practice, we were forced to train HEROVision by employing iHCC data. Although the results proved that our strategy was correct, the performance of HEROVision is likely to be enhanced, if it was sufficiently trained by real rHCC data ...]

2. [The geographic and demographic variability of datasets is noted but not deeply analyzed for potential biases.]

Our Response:

Thank you for your thoughtful comments. We acknowledge that geographic and demographic variability in our datasets may introduce potential biases, possibly impacting the model's generalizability.

To address this concern, we conducted comprehensive subgroup analyses to evaluate HEROVision's consistent performance across the 2ndSR and 2ndTA datasets. These analyses considered key variables, including sex, age, tumor size, number of tumors, manufacturer, and center. Our findings demonstrated no statistically significant differences in the 2-year C-index across subgroups in the 2ndTA dataset (0.68–0.76, $P = 0.307$ – 0.996) and the 2ndSR dataset (0.68–0.76, $P = 0.238$ – 0.892). In the 2ndTA dataset, the 2-year AUC for Center D showed a significant difference compared to Center A (0.90 versus 0.75, $P = 0.005$). However, there were no significant statistical differences in other subgroups within the 2ndTA dataset (0.75–0.87, $P = 0.077$ – 0.951) or across all subgroups in the 2ndSR dataset (0.72–0.88, $P = 0.167$ – 0.942).

These results suggest that variations in geographic and demographic factors minimally affected the model's performance. Detailed subgroup analysis results are provided in **Figures S14, S15,** and **Table S7.**

While our dataset includes variability in several critical aspects, it lacks racial diversity and international representation. Future research will aim to incorporate diverse populations to further minimize potential biases and enhance the model's robustness.

This limitation has been explicitly acknowledged and addressed in the discussion section of the revised manuscript.

Revised text:

(page 10/Discussion section of the revised manuscript)

[Geographic and demographic variability in our retrospective datasets may introduce potential information biases, possibly impacting the model's generalizability. To address this, we conducted comprehensive subgroup analyses to evaluate HEROVision's consistent performance across the 2ndSR and 2ndTA datasets. In addition to performing well on datasets from different geographical regions of China, HEROVision's performance was also assessed across subjects with varying demographics, including age and sex, as well as tumor characteristics such as tumor size and the number of tumors, and different manufacturers. The model's performance remained consistent across these subgroups, confirming its robustness and generalizability. However, our dataset lacks racial diversity and international data, which will be incorporated in future research to further minimize potential biases.]

(page 49-50/Figure S14 of the revised supplementary material)

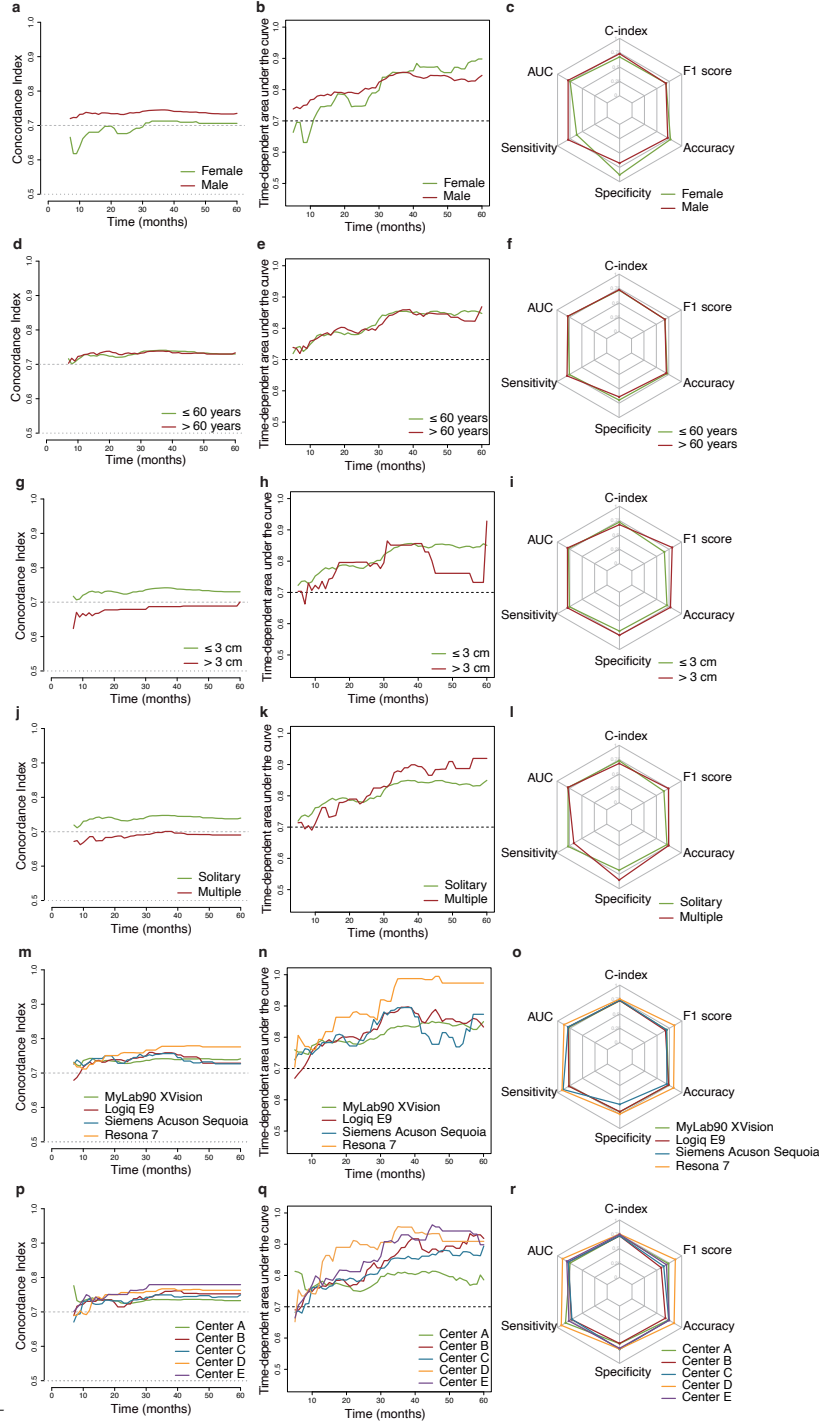


Figure S14. The Discriminative Performances of Different Subgroups for the 2ndTA Dataset
The C-index (a, d, g, j, m, and p) and time-dependent AUC (b, e, h, k, n, and q) for different subgroups, including sex, age, tumor size, the number of tumors, manufacturers, and centers, demonstrated the models' discriminatory performance at various points in the 2ndTA cohort. (c, f, i, l, o, and r) Radar charts illustrated the C-index, AUC, specificity, sensitivity, accuracy, and F1 score for 2-year PFS across these subgroups.
TA, thermal ablation; ViT, Vision Transformer; C-index, concordance index; AUC, area under the curve; PFS, progression-free survival.]

(page 51-52/Figure S15 of the revised supplementary material)

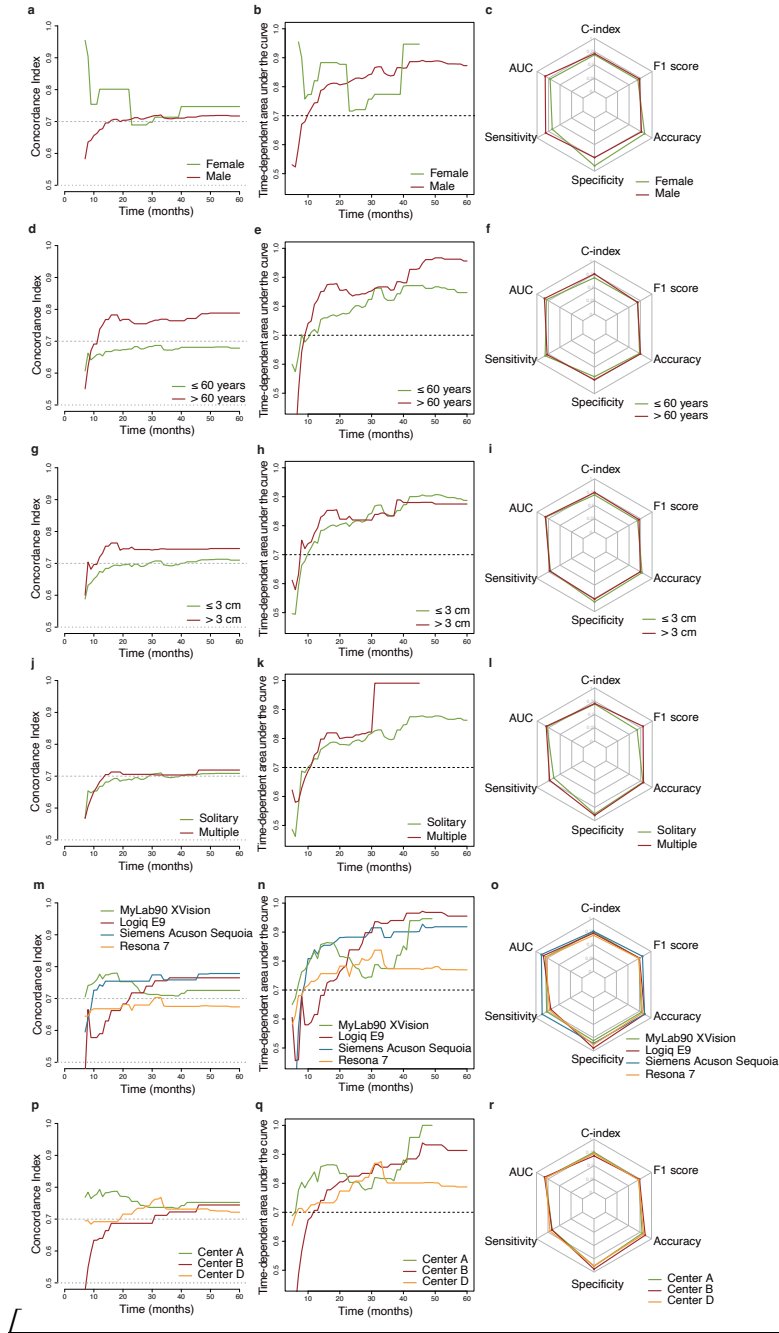


Figure S15. The Discriminative Performances of Different Subgroups for the 2ndSR Dataset
The C-index (a, d, g, j, m, and p) and time-dependent AUC (b, e, h, k, n, and q) for different subgroups, including sex, age, tumor size, the number of tumors, manufacturers, and centers, demonstrated the models' discriminatory performance at various points in the 2ndSR cohort. (c, f, i, l, o, and r) Radar charts illustrated the C-index, AUC, specificity, sensitivity, accuracy, and F1 score for 2-year PFS across these subgroups.
TA, thermal ablation; ViT, Vision Transformer; C-index, concordance index; AUC, area under the curve; PFS, progression-free survival.]

(page 27-28/Table S7 of the revised supplementary material)

[Table S7. Performance Comparison of HEROVision Across Different Subgroups for 2-Year PFS in the External Testing Cohorts

Datasets	Subgroups	Number	2-year C-index	P value	2-year AUC	P value	Sensitivity	Specificity	Accuracy	F1 score
Sex										
	Female	n=85	0.68 (0.58 - 0.78)	Ref.	0.75 (0.63 - 0.86)	Ref.	0.61 (0.51 - 0.72)	0.88 (0.81 - 0.95)	0.77 (0.69 - 0.86)	0.69 (0.59 - 0.78)
	Male	n=526	0.73 (0.69 - 0.77)	0.307	0.79 (0.75 - 0.83)	0.426	0.79 (0.75 - 0.82)	0.68 (0.64 - 0.72)	0.72 (0.68 - 0.76)	0.68 (0.64 - 0.72)
Age										
	≤60 years	n=366	0.72 (0.68 - 0.77)	Ref.	0.78 (0.73 - 0.83)	Ref.	0.75 (0.71 - 0.80)	0.70 (0.65 - 0.74)	0.72 (0.67 - 0.76)	0.67 (0.63 - 0.72)
	>60 years	n=245	0.73 (0.67 - 0.78)	0.852	0.78 (0.72 - 0.85)	0.929	0.80 (0.75 - 0.85)	0.64 (0.58 - 0.70)	0.70 (0.64 - 0.76)	0.67 (0.61 - 0.73)
Tumor size										
	≤3 cm	n=563	0.72 (0.69 - 0.76)	Ref.	0.78 (0.74 - 0.82)	Ref.	0.76 (0.72 - 0.79)	0.68 (0.64 - 0.72)	0.71 (0.67 - 0.74)	0.66 (0.62 - 0.69)
	>3 cm	n=48	0.68 (0.55 - 0.81)	0.525	0.80 (0.64 - 0.95)	0.745	0.80 (0.68 - 0.91)	0.75 (0.63 - 0.87)	0.78 (0.66 - 0.90)	0.81 (0.70 - 0.92)
2ndTA dataset	The number of tumors									
	Solitary	n=488	0.73 (0.69 - 0.77)	Ref.	0.78 (0.73 - 0.82)	Ref.	0.78 (0.74 - 0.82)	0.68 (0.64 - 0.72)	0.71 (0.67 - 0.75)	0.65 (0.61 - 0.69)
	Multiple	n=123	0.68 (0.60 - 0.77)	0.308	0.78 (0.69 - 0.87)	0.951	0.66 (0.58 - 0.75)	0.85 (0.79 - 0.91)	0.75 (0.67 - 0.82)	0.74 (0.67 - 0.82)
Manufacturers										
	MyLab90 Xvision	n=334	0.73 (0.68 - 0.78)	Ref.	0.78 (0.72 - 0.83)	Ref.	0.77 (0.73 - 0.82)	0.71 (0.66 - 0.76)	0.74 (0.69 - 0.78)	0.69 (0.64 - 0.74)
	Logiq E9	n=156	0.73 (0.66 - 0.80)	0.907	0.79 (0.71 - 0.87)	0.740	0.77 (0.71 - 0.84)	0.71 (0.64 - 0.78)	0.73 (0.66 - 0.80)	0.67 (0.60 - 0.74)
	Siemens Acuson Sequoia	n=76	0.74 (0.64 - 0.84)	0.852	0.79 (0.69 - 0.90)	0.777	0.89 (0.82 - 0.96)	0.58 (0.47 - 0.69)	0.70 (0.59 - 0.80)	0.69 (0.59 - 0.80)
	Resona 7	n=45	0.76 (0.63 - 0.88)	0.646	0.87 (0.76 - 0.98)	0.077	0.91 (0.83 - 1.00)	0.75 (0.62 - 0.88)	0.83 (0.72 - 0.94)	0.85 (0.74 - 0.95)
Centers										
	Center A	n=231	0.72 (0.67 - 0.78)	Ref.	0.75 (0.68 - 0.82)	Ref.	0.84 (0.79 - 0.89)	0.66 (0.60 - 0.72)	0.74 (0.68 - 0.79)	0.73 (0.67 - 0.79)
	Center B	n=102	0.72 (0.64 - 0.81)	0.972	0.78 (0.68 - 0.88)	0.483	0.72 (0.63 - 0.81)	0.65 (0.56 - 0.74)	0.67 (0.58 - 0.76)	0.59 (0.49 - 0.68)

<i>Datasets</i>	<i>Subgroups</i>	<i>Number</i>	<i>2-year C-index</i>	<i>P value</i>	<i>2-year AUC</i>	<i>P value</i>	<i>Sensitivity</i>	<i>Specificity</i>	<i>Accuracy</i>	<i>F1 score</i>
<i>2ndSR dataset</i>	<i>Center C</i>	<i>n=161</i>	<i>0.73 (0.66 - 0.79)</i>	<i>0.996</i>	<i>0.78 (0.70 - 0.86)</i>	<i>0.419</i>	<i>0.71 (0.64 - 0.78)</i>	<i>0.76 (0.69 - 0.82)</i>	<i>0.74 (0.67 - 0.81)</i>	<i>0.64 (0.56 - 0.71)</i>
	<i>Center D</i>	<i>n=46</i>	<i>0.76 (0.63 - 0.88)</i>	<i>0.654</i>	<i>0.90 (0.81 - 0.99)</i>	<i>0.005</i>	<i>0.92 (0.85 - 1.00)</i>	<i>0.75 (0.62 - 0.88)</i>	<i>0.85 (0.74 - 0.95)</i>	<i>0.87 (0.78 - 0.97)</i>
	<i>Center E</i>	<i>n=71</i>	<i>0.75 (0.65 - 0.85)</i>	<i>0.663</i>	<i>0.81 (0.71 - 0.92)</i>	<i>0.244</i>	<i>0.77 (0.67 - 0.87)</i>	<i>0.74 (0.63 - 0.84)</i>	<i>0.75 (0.65 - 0.85)</i>	<i>0.69 (0.59 - 0.80)</i>
	<i>Sex</i>									
	<i>Female</i>	<i>n=26</i>	<i>0.69 (0.51 - 0.87)</i>	<i>Ref.</i>	<i>0.72 (0.41 - 1.00)</i>	<i>Ref.</i>	<i>0.67 (0.49 - 0.85)</i>	<i>0.90 (0.78 - 1.00)</i>	<i>0.84 (0.70 - 0.98)</i>	<i>0.69 (0.51 - 0.87)</i>
	<i>Male</i>	<i>n=196</i>	<i>0.71 (0.65 - 0.78)</i>	<i>0.812</i>	<i>0.82 (0.76 - 0.89)</i>	<i>0.246</i>	<i>0.81 (0.76 - 0.87)</i>	<i>0.74 (0.68 - 0.80)</i>	<i>0.77 (0.71 - 0.83)</i>	<i>0.72 (0.66 - 0.79)</i>
	<i>Age</i>									
	<i>≤60 years</i>	<i>n=131</i>	<i>0.68 (0.60 - 0.76)</i>	<i>Ref.</i>	<i>0.79 (0.70 - 0.88)</i>	<i>Ref.</i>	<i>0.82 (0.76 - 0.89)</i>	<i>0.68 (0.60 - 0.76)</i>	<i>0.73 (0.66 - 0.81)</i>	<i>0.69 (0.61 - 0.77)</i>
	<i>>60 years</i>	<i>n=91</i>	<i>0.75 (0.67 - 0.84)</i>	<i>0.238</i>	<i>0.84 (0.73 - 0.94)</i>	<i>0.404</i>	<i>0.78 (0.69 - 0.86)</i>	<i>0.74 (0.65 - 0.83)</i>	<i>0.75 (0.66 - 0.84)</i>	<i>0.69 (0.59 - 0.78)</i>
	<i>Tumor size</i>									
	<i>≤3 cm</i>	<i>n=166</i>	<i>0.70 (0.63 - 0.77)</i>	<i>Ref.</i>	<i>0.81 (0.73 - 0.89)</i>	<i>Ref.</i>	<i>0.71 (0.64 - 0.78)</i>	<i>0.82 (0.76 - 0.88)</i>	<i>0.78 (0.72 - 0.85)</i>	<i>0.68 (0.61 - 0.76)</i>
	<i>>3 cm</i>	<i>n=56</i>	<i>0.74 (0.63 - 0.86)</i>	<i>0.503</i>	<i>0.82 (0.70 - 0.94)</i>	<i>0.876</i>	<i>0.72 (0.60 - 0.84)</i>	<i>0.76 (0.65 - 0.88)</i>	<i>0.75 (0.63 - 0.86)</i>	<i>0.72 (0.60 - 0.83)</i>
	<i>The number of tumors</i>									
	<i>Solitary</i>	<i>n=180</i>	<i>0.69 (0.63 - 0.76)</i>	<i>Ref.</i>	<i>0.79 (0.71 - 0.87)</i>	<i>Ref.</i>	<i>0.64 (0.57 - 0.71)</i>	<i>0.87 (0.82 - 0.92)</i>	<i>0.80 (0.74 - 0.86)</i>	<i>0.67 (0.60 - 0.74)</i>
	<i>Multiple</i>	<i>n=42</i>	<i>0.71 (0.57 - 0.84)</i>	<i>0.892</i>	<i>0.80 (0.61 - 1.00)</i>	<i>0.814</i>	<i>0.73 (0.59 - 0.86)</i>	<i>0.90 (0.81 - 0.99)</i>	<i>0.81 (0.69 - 0.93)</i>	<i>0.80 (0.68 - 0.92)</i>
	<i>Manufacturers</i>									
	<i>MyLab90 Xvision</i>	<i>n=91</i>	<i>0.74 (0.65 - 0.83)</i>	<i>Ref.</i>	<i>0.79 (0.67 - 0.91)</i>	<i>Ref.</i>	<i>0.77 (0.68 - 0.85)</i>	<i>0.81 (0.73 - 0.89)</i>	<i>0.79 (0.71 - 0.88)</i>	<i>0.74 (0.65 - 0.83)</i>
	<i>Logiq E9</i>	<i>n=46</i>	<i>0.72 (0.59 - 0.85)</i>	<i>0.758</i>	<i>0.84 (0.70 - 0.98)</i>	<i>0.478</i>	<i>0.68 (0.54 - 0.81)</i>	<i>0.94 (0.88 - 1.00)</i>	<i>0.86 (0.76 - 0.96)</i>	<i>0.75 (0.63 - 0.88)</i>
	<i>Siemens Acuson Sequoia</i>	<i>n=39</i>	<i>0.75 (0.62 - 0.89)</i>	<i>0.888</i>	<i>0.88 (0.74 - 1.00)</i>	<i>0.167</i>	<i>0.87 (0.76 - 0.97)</i>	<i>0.86 (0.75 - 0.97)</i>	<i>0.86 (0.75 - 0.97)</i>	<i>0.82 (0.69 - 0.94)</i>
	<i>Resona 7</i>	<i>n=46</i>	<i>0.68 (0.54 - 0.81)</i>	<i>0.445</i>	<i>0.78 (0.63 - 0.94)</i>	<i>0.942</i>	<i>0.71 (0.58 - 0.84)</i>	<i>0.88 (0.79 - 0.97)</i>	<i>0.82 (0.70 - 0.93)</i>	<i>0.74 (0.61 - 0.87)</i>
	<i>Centers</i>									
	<i>Center A</i>	<i>n=120</i>	<i>0.76 (0.68 - 0.83)</i>	<i>Ref.</i>	<i>0.80 (0.70 - 0.90)</i>	<i>Ref.</i>	<i>0.66 (0.58 - 0.75)</i>	<i>0.89 (0.83 - 0.95)</i>	<i>0.80 (0.72 - 0.87)</i>	<i>0.73 (0.65 - 0.81)</i>

<i>Datasets</i>	<i>Subgroups</i>	<i>Number</i>	<i>2-year C-index</i>	<i>P value</i>	<i>2-year AUC</i>	<i>P value</i>	<i>Sensitivity</i>	<i>Specificity</i>	<i>Accuracy</i>	<i>F1 score</i>
	<i>Center B</i>	<i>n=48</i>	<i>0.69 (0.56 - 0.82)</i>	<i>0.375</i>	<i>0.82 (0.68 - 0.97)</i>	<i>0.739</i>	<i>0.66 (0.53 - 0.80)</i>	<i>0.94 (0.88 - 1.00)</i>	<i>0.86 (0.76 - 0.96)</i>	<i>0.74 (0.61 - 0.86)</i>
	<i>Center D</i>	<i>n=54</i>	<i>0.73 (0.61 - 0.85)</i>	<i>0.690</i>	<i>0.80 (0.66 - 0.94)</i>	<i>0.939</i>	<i>0.71 (0.59 - 0.83)</i>	<i>0.88 (0.80 - 0.97)</i>	<i>0.83 (0.73 - 0.93)</i>	<i>0.72 (0.60 - 0.84)</i>

TA, thermal ablation; SR, surgical resection; PFS, progression-free survival; AUC, area under the curve; Ref, reference.]

3. [While the model architecture is described, details on hyperparameter optimization and model interpretability are limited.]

Our Response:

Thank you for your insightful comments.

1) Hyperparameter Optimization:

Based on our previous study[1], we selected similar network architectures and depths to ensure robustness. The range of hyperparameter values for grid search was determined based on the overall data size. Specifically, a grid search was conducted to identify the optimal learning rate and batch size, key parameters influencing the model's performance. To enhance convergence, we employed the StepLR learning rate scheduler, which dynamically adjusts the learning rate during training. This approach allowed the use of a larger learning rate initially for rapid convergence and a smaller learning rate during later stages for fine-tuning, thus optimizing both efficiency and precision.

2) Model Interpretability:

We acknowledge that interpretability remains a common challenge for deep learning models, including the one developed in this study. This limitation arises from the inherent complexity of neural networks, which operate as "black boxes" by nature[2-4]. However, we have made efforts to enhance interpretability through clinically relevant methods. For example, we visualized the model's outputs using a nomogram, which aligns with clinical decision-making practices and provides an intuitive representation of the model's predictions.

3) Future Directions for Interpretability:

While the scope of this study focuses on prognostic predictions for rHCC patients, we recognize the importance of enhancing model interpretability. In future work, we aim to explore multi-omics data integration (e.g., genomics and transcriptomics) to identify potential biomarkers and improve interpretability. These advancements could provide insights into the underlying mechanisms of the model's predictions and align more closely with clinical applications.

We have supplemented the discussion section of the revised manuscript and the methods section of the revised supplementary material to include these points, highlighting both the efforts made and the potential for future enhancements in hyperparameter optimization and model interpretability.

References:

- [1] K. Zhang *et al.*, "ViT-based quantification of intratumoral heterogeneity for predicting the early recurrence in HCC following multiple ablation," (in eng), *Liver Int*, Nov 11 2024, doi: 10.1111/liv.16051.
- [2] C. Rudin, "Stop Explaining Black Box Machine Learning Models for High Stakes Decisions and Use Interpretable Models Instead," (in eng), *Nat Mach Intell*, vol. 1, no.

- 5, pp. 206-215, May 2019, doi: 10.1038/s42256-019-0048-x.
- [3] A. J. London, "Artificial Intelligence and Black-Box Medical Decisions: Accuracy versus Explainability," (in eng), *Hastings Cent Rep*, vol. 49, no. 1, pp. 15-21, Jan 2019, doi: 10.1002/hast.973.
- [4] A. Holzinger, G. Langs, H. Denk, K. Zatloukal, and H. Müller, "Causability and explainability of artificial intelligence in medicine," (in eng), *Wiley Interdiscip Rev Data Min Knowl Discov*, vol. 9, no. 4, p. e1312, Jul-Aug 2019, doi: 10.1002/widm.1312.

Revised text:

(page 11-12/ Appendix S6 section of the revised supplementary material)

[Based on our previous study⁶, we selected similar network architectures and depths to ensure robustness. The range of hyperparameter values for grid search was determined based on the overall data size. Specifically, a grid search was conducted to identify the optimal learning rate and batch size, key parameters influencing the model's performance. To enhance convergence, we employed the StepLR learning rate scheduler, which dynamically adjusts the learning rate during training. This approach allowed the use of a larger learning rate initially for rapid convergence and a smaller learning rate during later stages for fine-tuning, thus optimizing both efficiency and precision.]

(page 12/Discussion section of the revised manuscript)

[The lack of interpretability in DL presents challenges, particularly in supervising systems based on complex neural networks³⁴. While HEROVision model generates a risk score to classify patients into high- or low-risk groups, the computational process remains a "black box" due to the inherent complexity of neural networks. Gradient-weighted Class Activation Mapping (Grad-CAM), a visualization technique, identifies critical regions in the input image by calculating the gradient of the target class with respect to a specific convolutional layer³⁵. However, the patch-based structure of ViT models often results in heatmaps with scattered distributions across the image, which may lack correlation with clinically relevant features and may not always provide a meaningful understanding of the decision-making^{36, 37}. Consequently, these heatmaps may fail to provide concentrated or clinically meaningful visualizations. To address this, we used nomograms to provide a clear and clinically applicable visualization of how individual variables contribute to risk predictions. In the future, integrating molecular data such as genomic markers with DL features holds great potential for improving interpretability, and we plan to explore this avenue in subsequent studies.]

4. [The explainability of the models is limited. Did the authors attempt to understand why the model classifies patients into low- or high-risk categories? Providing insights into this aspect

would enhance interpretability and clinical relevance.]

Our Response:

Thank you for your insightful comment. We acknowledge the importance of model explainability in enhancing clinical relevance and understanding, and we address your concerns as follows:

1) Current Challenges in Explainability:

Our study employs a deep learning algorithm to analyze imaging data (e.g., ultrasound and MRI) from HCC patients, using prognosis as the target label. This approach extracts high-throughput prognostic features from images and combines them with clinical factors to construct a comprehensive model. The final output is a risk score that stratifies patients into high- or low-risk categories. However, the computational process underlying deep learning remains a “black box” due to the inherent complexity of neural networks[1-4]. This limitation is a widespread challenge in the field and underscores the urgent need for advancements in explainability.

2) Heatmap Visualization and Its Limitations:

In contemporary research, the most commonly used method to interpret deep learning models is gradient-based visualization techniques such as Grad-CAM (Gradient-weighted Class Activation Mapping)[5]. Grad-CAM identifies critical regions in the input image by calculating the gradient of the target class with respect to a specific convolutional layer. However, the patch-based structure of ViT models often results in heatmaps with scattered distributions across the image[6, 7]. Moreover, this approach often highlights regions that may not directly correlate with clinically relevant features, resulting in more invalid areas and diluting the interpretability of the visualization. Furthermore, the highlighted regions may not always provide a meaningful understanding of the decision-making[8]. Consequently, these heatmaps often fail to provide concentrated or clinically meaningful visualizations. This limitation has been noted in previous research, and similar issues are observed in our model, which is why we chose not to include heatmaps in this study.

3) Nomograms as a Practical Interpretative Tool:

To enhance interpretability, we opted for the use of nomograms (**Figure S2**), which allow for a clear visualization of how individual variables contribute to the overall risk prediction. This approach provides a straightforward and clinically applicable explanation of the model’s outputs, serving as a bridge between complex computational predictions and practical decision-making.

4) Potential for Molecular and Multi-Omics Integration:

Emerging research suggests that combining molecular information (e.g., genomic and transcriptomic data) with deep learning outputs may offer a pathway to greater interpretability[9-11]. For example, linking specific imaging features identified by the model to underlying molecular markers could enhance our understanding of the biological basis of

risk classification. While this was beyond the scope of the current study, we consider this a promising avenue for future research and plan to explore these integrative approaches in subsequent studies.

By addressing these challenges and exploring potential solutions, we aim to bridge the gap between complex computational methods and practical clinical applications, ensuring that our model continues to evolve in both accuracy and relevance.

References:

- [1] C. Rudin, "Stop Explaining Black Box Machine Learning Models for High Stakes Decisions and Use Interpretable Models Instead," (in eng), *Nat Mach Intell*, vol. 1, no. 5, pp. 206-215, May 2019, doi: 10.1038/s42256-019-0048-x.
- [2] A. J. London, "Artificial Intelligence and Black-Box Medical Decisions: Accuracy versus Explainability," (in eng), *Hastings Cent Rep*, vol. 49, no. 1, pp. 15-21, Jan 2019, doi: 10.1002/hast.973.
- [3] A. Holzinger, G. Langs, H. Denk, K. Zatloukal, and H. Müller, "Causability and explainability of artificial intelligence in medicine," (in eng), *Wiley Interdiscip Rev Data Min Knowl Discov*, vol. 9, no. 4, p. e1312, Jul-Aug 2019, doi: 10.1002/widm.1312.
- [4] V. Hassija *et al.*, "Interpreting black-box models: a review on explainable artificial intelligence," *Cognitive Computation*, vol. 16, no. 1, pp. 45-74, 2024.
- [5] R. R. Selvaraju, M. Cogswell, A. Das, R. Vedantam, D. Parikh, and D. Batra, "Grad-cam: Visual explanations from deep networks via gradient-based localization," in *Proceedings of the IEEE international conference on computer vision*, 2017, pp. 618-626.
- [6] Z. Qiu, H. Rivaz, and Y. Xiao, "Is visual explanation with Grad-CAM more reliable for deeper neural networks? A case study with automatic pneumothorax diagnosis," in *International Workshop on Machine Learning in Medical Imaging*, 2023: Springer, pp. 224-233.
- [7] S. Leem and H. Seo, "Attention Guided CAM: Visual Explanations of Vision Transformer Guided by Self-Attention," in *Proceedings of the AAAI Conference on Artificial Intelligence*, 2024, vol. 38, no. 4, pp. 2956-2964.
- [8] G. Shamai *et al.*, "Deep learning-based image analysis predicts PD-L1 status from H&E-stained histopathology images in breast cancer," (in eng), *Nat Commun*, vol. 13, no. 1, p. 6753, Nov 8 2022, doi: 10.1038/s41467-022-34275-9.
- [9] T. Y. Xia *et al.*, "Predicting Microvascular Invasion in Hepatocellular Carcinoma Using CT-based Radiomics Model," (in eng), *Radiology*, vol. 307, no. 4, p. e222729, May 2023, doi: 10.1148/radiol.222729.
- [10] S. Wang *et al.*, "Mining whole-lung information by artificial intelligence for predicting EGFR genotype and targeted therapy response in lung cancer: a multicohort study," (in eng), *Lancet Digit Health*, vol. 4, no. 5, pp. e309-e319, May 2022, doi: 10.1016/s2589-7500(22)00024-3.
- [11] X. Zhang *et al.*, "Exploring non-invasive precision treatment in non-small cell lung cancer patients through deep learning radiomics across imaging features and molecular phenotypes," (in eng), *Biomark Res*, vol. 12, no. 1, p. 12, Jan 25 2024, doi:

Revised text:

(page 12/Discussion section of the revised manuscript)

[The lack of interpretability in DL presents challenges, particularly in supervising systems based on complex neural networks³⁴. While HEROVision model generates a risk score to classify patients into high- or low-risk groups, the computational process remains a “black box” due to the inherent complexity of neural networks. Gradient-weighted Class Activation Mapping (Grad-CAM), a visualization technique, identifies critical regions in the input image by calculating the gradient of the target class with respect to a specific convolutional layer³⁵. However, the patch-based structure of ViT models often results in heatmaps with scattered distributions across the image, which may lack correlation with clinically relevant features and may not always provide a meaningful understanding of the decision-making^{36, 37}. Consequently, these heatmaps may fail to provide concentrated or clinically meaningful visualizations. To address this, we used nomograms to provide a clear and clinically applicable visualization of how individual variables contribute to risk predictions. In the future, integrating molecular data such as genomic markers with DL features holds great potential for improving interpretability, and we plan to explore this avenue in subsequent studies.]

(page 37/Figures section of the revised supplementary material)

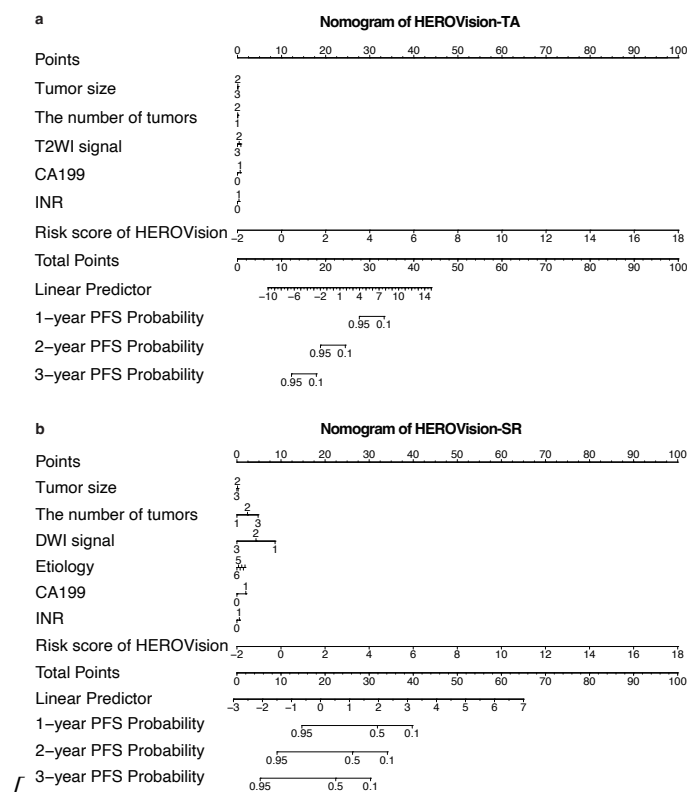


Figure S2. Constructed Nomograms to Estimate Individualized PFS for TA and SR
a Nomogram for predicting the PFS probability after receiving TA. b Nomogram for predicting
the PFS probability after receiving SR.
TA, thermal ablation; SR, surgical resection; PFS, progression-free survival.]

5. [As noted by a previous reviewer, the study focuses exclusively on patients who are candidates
for both surgical resection and thermal ablation, excluding those eligible for other treatment
modalities. This limitation should be addressed or acknowledged in the discussion.]

Our Response:

Thank you for pointing this out. We agree with your comment and have acknowledged this limitation in the revised discussion section.

Revised text:

(page 12-13/Discussion section of the revised manuscript)

[... Third, HEROVision was developed specifically for patients experiencing early-stage
recurrence after SR. As such, it may not be applicable to patients with advanced-stage
recurrence or those who received non-surgical or non-ablative treatments, such as TACE or
systemic therapies, between initial resection and subsequent recurrences ...]

6. [The most stimulating part of the study is the comparison between the two models, which
may suggest which treatment is better. However, the models do not provide clear guidance in
cases where both classify patients into the same risk group. This limitation should be explored
further.]

Our Response:

Thank you for your insightful comments. We agree that exploring strategies for patients who are classified into the same risk group by both models is essential for enhancing clinical applicability.

For patients who remain low-risk after re-stratification, our findings suggest that both treatment options can be effective, allowing the decision to be made collaboratively by doctors and patients. This shared decision-making process should consider the patient's preferences, clinical characteristics, and quality-of-life concerns, ensuring that the chosen treatment aligns with the patient's overall goals.

For patients who remain high-risk after re-stratification, more aggressive sequential treatment strategies—such as combining surgical resection or thermal ablation with transarterial chemoembolization (TACE) or systemic therapies—along with implementing

closer monitoring, may be warranted. These strategies could potentially improve outcomes, although further studies are needed to validate their efficacy, which we plan to explore in future research.

We have included this discussion in the revised manuscript to acknowledge this limitation and provide a direction for future investigation.

Revised text:

(page 10-11/Discussion section of the revised manuscript)

[... For TA, 35.9% of patients initially categorized as high-risk could be downgraded to low-risk after re-stratification, and switching from TA to RHR could probably improve their prognosis and extend survival time, despite the higher average cost associated with surgery^{23, 24}. Conversely, for RHR, although only 6.6% of high-risk patients who could benefit from switching RHR to TA, they would have a higher probability of better prognosis, less invasive procedure with lower complication rates, reduced economic burden, and better preservation of liver function^{25, 26}. Obviously, the above personalized treatment changes are of great clinical significance. For patients initially categorized as low-risk but upgraded to high-risk following re-stratification, any change in treatment strategy would be unnecessary and might make outcomes worse. For patients who remain low-risk after re-stratification, both treatment options could be effective, and the decision should be made collaboratively by doctors and patients. For those who remained high-risk after re-stratification, more aggressive sequential treatment strategies (such as combining with transarterial chemoembolization [TACE] or systemic therapies) along with implementing closer monitoring might be warranted ...]

(page 13/Discussion section of the revised manuscript)

[... Forth, for patients who remain high-risk after re-stratification, HEROVision does not recommend a specific next-step treatment. However, more aggressive sequential treatment and closer monitoring strategies may potentially improve outcomes. Further studies are needed to validate their efficacy ...]

7. [The authors should focus more on discrepant cases, where one model reclassifies a patient into a different risk group. Detailed characteristics of these patients and their tumors should be provided to better understand why they were initially allocated to the “wrong” group and how reclassification impacts outcomes.]

Our Response:

Thank you for your valuable suggestion. It is extremely challenging to identify and understand the biological mechanisms by which different patients with rHCC have different outcomes with surgical resection and thermal ablation. Current studies, both clinical and basic,

are still far from reaching clear conclusions. Our hypothesis is that a well-developed AI model can identify which patients are placed in the “wrong” group by identifying subtle differences in imaging features and correlating them with differences in treatment outcomes. Our results supported our hypothesis, but also suggested that the non-imaging characteristics of patients are likely far from sufficient for us to understand why some patients were initially allocated to the “wrong” group.

Nevertheless, to address your concern, we have supplemented the revised manuscript and supplementary materials with the individual clinical variables and risk scores of patients in the matched 2ndTA and 2ndSR groups (**Figure S16**). Specifically, we provided baseline characteristics (**Table S9**) for the following subgroups:

1) Matched 2ndTA Group:

(a) Patients downgraded from high-risk to low-risk (n = 23)

(b) Patients remaining high-risk after re-stratification (n = 41)

Results indicated no significant differences between these subgroups in baseline characteristics ($P = 0.101\text{--}1.000$), except for AFP levels ($P < 0.001$).

2) Matched 2ndSR Group:

(a) Patients downgraded from high-risk to low-risk (n = 6)

(b) Patients remaining high-risk after re-stratification (n = 85)

Similarly, no significant differences were observed between these subgroups ($P = 0.055\text{--}1.000$), except for DWI signal intensity ($P = 0.001$).

We believe this detailed analysis of discrepant cases further demonstrated that routine clinical baseline variables and simple imaging features contribute very limitedly to the prognostic stratification of rHCC patients. However, the risk score provided by the HEROVision model can play a crucial role.

For rHCC patients within the Milan criteria, there are no established guidelines or staging systems with detailed characteristics to guide treatment selection. Decisions are primarily based on clinician and patient preferences, and clinical baseline variables often play a limited role in this process. Our results underscore the potential of the HEROVision model to extract and identify potential prognostic features from clinical and imaging data, especially from imaging data, enabling effective risk stratification. This capability highlights its value in optimizing treatment decisions beyond conventional clinical parameters.

Revised text:

(page 8/Results section of the revised manuscript)

[Optimize Treatment Selection in Real World rHCC

... Figure S16 presents the individual clinical variables and risk scores of these patients ...

... Baseline characteristics (Table S9) between the subgroup downgraded from high-risk to low-risk (n=23) and the subgroup remaining high-risk after re-stratification (n=41) showed no significant differences ($P = 0.101\text{--}1.000$), except for AFP levels ($P < 0.001$) ...

...Baseline characteristics (Table S9) between the subgroup downgraded from high-risk to low-risk (n=6) and the subgroup remaining high-risk after re-stratification (n=85) revealed no significant differences ($P = 0.055$ – 1.000), except for DWI signal ($P = 0.001$) ...]

(page 32-35/Tables section of the revised supplementary material)

[Table S9. Baseline Characteristics of Patients Initially Categorized as High-Risk but Downgraded to Low-Risk vs. Those Remaining High-Risk After Re-Stratification in Matched 2ndTA and 2ndSR Groups]

<i>Characteristics</i>	<i>Matched 2ndTA Group</i>				<i>Matched 2ndSR Group</i>			
	<i>Overall (N=64)</i>	<i>Downgraded from high-risk to low-risk subgroup (N=23)</i>	<i>Remaining high-risk subgroup (N=41)</i>	<i>P value</i>	<i>Overall (N=91)</i>	<i>Downgraded from high-risk to low-risk subgroup (N=6)</i>	<i>Remaining high-risk subgroup (N=85)</i>	<i>P value</i>
<i>Sex</i>				0.279				0.666
Female	8 (12.5)	1 (4.3)	7 (17.1)		13 (14.3)	0 (0.0)	13 (15.3)	
Male	56 (87.5)	22 (95.7)	34 (82.9)		78 (85.7)	6 (100.0)	72 (84.7)	
<i>Age (years)†</i>	61.00 [50.00, 66.50]	61.00 [54.50, 67.00]	61.00 [50.00, 66.00]	0.533	58.00 [46.50, 65.50]	58.00 [47.25, 61.25]	58.00 [47.00, 66.00]	0.533
<i>Liver cirrhosis</i>				0.411				1.000
Absent	7 (10.9)	4 (17.4)	3 (7.3)		7 (7.7)	0 (0.0)	7 (8.2)	
Present	57 (89.1)	19 (82.6)	38 (92.7)		84 (92.3)	6 (100.0)	78 (91.8)	
<i>Etiology</i>				0.651				0.703
HBV	56 (87.5)	20 (87.0)	36 (87.8)		82 (90.1)	6 (100.0)	76 (89.4)	
HCV	2 (3.1)	0 (0.0)	2 (4.9)		0 (0.0)	0 (0.0)	0 (0.0)	
ALC	0 (0.0)	0 (0.0)	0 (0.0)		0 (0.0)	0 (0.0)	0 (0.0)	
PBC	0 (0.0)	0 (0.0)	0 (0.0)		0 (0.0)	0 (0.0)	0 (0.0)	
NAFLD	2 (3.1)	1 (4.3)	1 (2.4)		3 (3.3)	0 (0.0)	3 (3.5)	
Undetermined	4 (6.2)	2 (8.7)	2 (4.9)		6 (6.6)	0 (0.0)	6 (7.1)	
<i>CTP grade</i>				0.558				0.716

Characteristics	Matched 2ndTA Group				Matched 2ndSR Group			
	Overall (N=64)	Downgraded from high-risk to low-risk subgroup (N=23)	Remaining high-risk subgroup (N=41)	P value	Overall (N=91)	Downgraded from high-risk to low-risk subgroup (N=6)	Remaining high-risk subgroup (N=85)	P value
A level	58 (90.6)	22 (95.7)	36 (87.8)	0.561	79 (86.8)	6 (100.0)	73 (85.9)	1.000
B level	6 (9.4)	1 (4.3)	5 (12.2)		12 (13.2)	0 (0.0)	12 (14.1)	
Portal hypertension								
Absent	43 (67.2)	17 (73.9)	26 (63.4)	0.336	78 (85.7)	5 (83.3)	73 (85.9)	0.134
Present	21 (32.8)	6 (26.1)	15 (36.6)		13 (14.3)	1 (16.7)	12 (14.1)	
Location								
Left lobe	26 (40.6)	11 (47.8)	15 (36.6)	0.101	34 (37.4)	0 (0.0)	34 (40.0)	0.055
Right lobe	35 (54.7)	12 (52.2)	23 (56.1)		50 (54.9)	5 (83.3)	45 (52.9)	
Whole lobe	3 (4.7)	0 (0.0)	3 (7.3)		7 (7.7)	1 (16.7)	6 (7.1)	
The number of tumors				0.524				0.236
1	50 (78.1)	21 (91.3)	29 (70.7)		65 (71.4)	2 (33.3)	63 (74.1)	
2	8 (12.5)	2 (8.7)	6 (14.6)		15 (16.5)	3 (50.0)	12 (14.1)	
3	6 (9.4)	0 (0.0)	6 (14.6)	0.911	11 (12.1)	1 (16.7)	10 (11.8)	0.216
Tumor size								
≤ 2 cm	28 (43.8)	8 (34.8)	20 (48.8)		48 (52.7)	5 (83.3)	43 (50.6)	
> 2 cm, ≤ 3 cm	18 (28.1)	7 (30.4)	11 (26.8)	0.911	19 (20.9)	1 (16.7)	18 (21.2)	0.216
> 3 cm	18 (28.1)	8 (34.8)	10 (24.4)		24 (26.4)	0 (0.0)	24 (28.2)	
Ultrasonic echo								

<i>Characteristics</i>	<i>Matched 2ndTA Group</i>				<i>Matched 2ndSR Group</i>			
	<i>Overall (N=64)</i>	<i>Downgraded from high-risk to low-risk subgroup (N=23)</i>	<i>Remaining high-risk subgroup (N=41)</i>	<i>P value</i>	<i>Overall (N=91)</i>	<i>Downgraded from high-risk to low-risk subgroup (N=6)</i>	<i>Remaining high-risk subgroup (N=85)</i>	<i>P value</i>
<i>Hypoecho</i>	49 (76.6)	18 (78.3)	31 (75.6)	0.743	65 (71.4)	5 (83.3)	60 (70.6)	0.752
<i>Isoecho</i>	8 (12.5)	3 (13.0)	5 (12.2)		5 (5.5)	1 (16.7)	4 (4.7)	
<i>Hyperecho</i>	7 (10.9)	2 (8.7)	5 (12.2)		21 (23.1)	0 (0.0)	21 (24.7)	
<i>T2WI signal</i>								
<i>Hypointensity</i>	2 (3.1)	0 (0.0)	2 (4.9)	0.476	0 (0.0)	0 (0.0)	0 (0.0)	0.001
<i>Isointensity</i>	0 (0.0)	0 (0.0)	0 (0.0)		5 (5.5)	1 (16.7)	4 (4.7)	
<i>Hyperintensity</i>	62 (96.9)	23 (100.0)	39 (95.1)		86 (94.5)	5 (83.3)	81 (95.3)	
<i>DWI signal</i>								
<i>Hypointensity</i>	3 (4.7)	0 (0.0)	3 (7.3)	0.768	3 (3.3)	1 (16.7)	2 (2.4)	0.333
<i>Isointensity</i>	0 (0.0)	0 (0.0)	0 (0.0)		5 (5.5)	2 (33.3)	3 (3.5)	
<i>Hyperintensity</i>	61 (95.3)	23 (100.0)	38 (92.7)		83 (91.2)	3 (50.0)	80 (94.1)	
<i>Subcapsular</i>								
<i>No</i>	36 (56.2)	14 (60.9)	22 (53.7)	0.754	40 (44.0)	1 (16.7)	39 (45.9)	0.499
<i>Yes</i>	28 (43.8)	9 (39.1)	19 (46.3)		51 (56.0)	5 (83.3)	46 (54.1)	
<i>Perivascular</i>								
<i>No</i>	53 (82.8)	20 (87.0)	33 (80.5)		77 (84.6)	4 (66.7)	73 (85.9)	
<i>Yes</i>	11 (17.2)	3 (13.0)	8 (19.5)	0.652	14 (15.4)	2 (33.3)	12 (14.1)	0.483
<i>ALT (U/L)</i>								

Characteristics	Matched 2ndTA Group				Matched 2ndSR Group			
	Overall (N=64)	Downgraded from high-risk to low-risk subgroup (N=23)	Remaining high-risk subgroup (N=41)	P value	Overall (N=91)	Downgraded from high-risk to low-risk subgroup (N=6)	Remaining high-risk subgroup (N=85)	P value
≤ 40	48 (75.0)	16 (69.6)	32 (78.0)	0.239	56 (61.5)	5 (83.3)	51 (60.0)	0.668
> 40	16 (25.0)	7 (30.4)	9 (22.0)		35 (38.5)	1 (16.7)	34 (40.0)	
AST (U/L)								
≤ 45	46 (71.9)	14 (60.9)	32 (78.0)	0.124	61 (67.0)	5 (83.3)	56 (65.9)	0.184
> 45	18 (28.1)	9 (39.1)	9 (22.0)		30 (33.0)	1 (16.7)	29 (34.1)	
INR								
≤ 1.15	47 (73.4)	20 (87.0)	27 (65.9)	0.253	61 (67.0)	6 (100.0)	55 (64.7)	0.967
> 1.15	17 (26.6)	3 (13.0)	14 (34.1)		30 (33.0)	0 (0.0)	30 (35.3)	
CEA (ng/mL)								
≤ 5	60 (93.8)	20 (87.0)	40 (97.6)	< 0.001	83 (91.2)	6 (100.0)	77 (90.6)	1.000
> 5	4 (6.2)	3 (13.0)	1 (2.4)		8 (8.8)	0 (0.0)	8 (9.4)	
AFP (ng/mL)								
< 200	54 (84.4)	14 (60.9)	40 (97.6)	0.946	77 (84.6)	5 (83.3)	72 (84.7)	1.000
≥ 200	10 (15.6)	9 (39.1)	1 (2.4)		14 (15.4)	1 (16.7)	13 (15.3)	
CA19-9 (U/ml)								
≤ 37	60 (93.8)	21 (91.3)	39 (95.1)	1.000	89 (97.8)	6 (100.0)	83 (97.6)	1.000
> 37	4 (6.2)	2 (8.7)	2 (4.9)		2 (2.2)	0 (0.0)	2 (2.4)	
PNI								

<i>Characteristics</i>	<i>Matched 2ndTA Group</i>				<i>Matched 2ndSR Group</i>			
	<i>Overall (N=64)</i>	<i>Downgraded from high-risk to low-risk subgroup (N=23)</i>	<i>Remaining high-risk subgroup (N=41)</i>	<i>P value</i>	<i>Overall (N=91)</i>	<i>Downgraded from high-risk to low-risk subgroup (N=6)</i>	<i>Remaining high-risk subgroup (N=85)</i>	<i>P value</i>
≥ 42.8	32 (50.0)	12 (52.2)	20 (48.8)		47 (51.6)	3 (50.0)	44 (51.8)	
< 42.8	32 (50.0)	11 (47.8)	21 (51.2)		44 (48.4)	3 (50.0)	41 (48.2)	

Unless indicated otherwise, data are the number of patients, with percentages in parentheses.

† Data are medians, with interquartile ranges in parentheses.

$P < 0.05$ was considered significant.

TA, thermal ablation; SR, surgical resection; CTP, Child-Turcotte-Pugh; ALC, alcoholic cirrhosis; PBC, primary biliary cholangitis; NAFLD, non-alcoholic fatty liver disease; AFP, alpha-fetoprotein; CA19-9, carbohydrate antigen 19-9; CEA, carcinoembryonic antigen; INR, international normalized ratio; PNI, prognostic nutritional index.

L

(page 53/Figures section of the revised supplementary material)

[

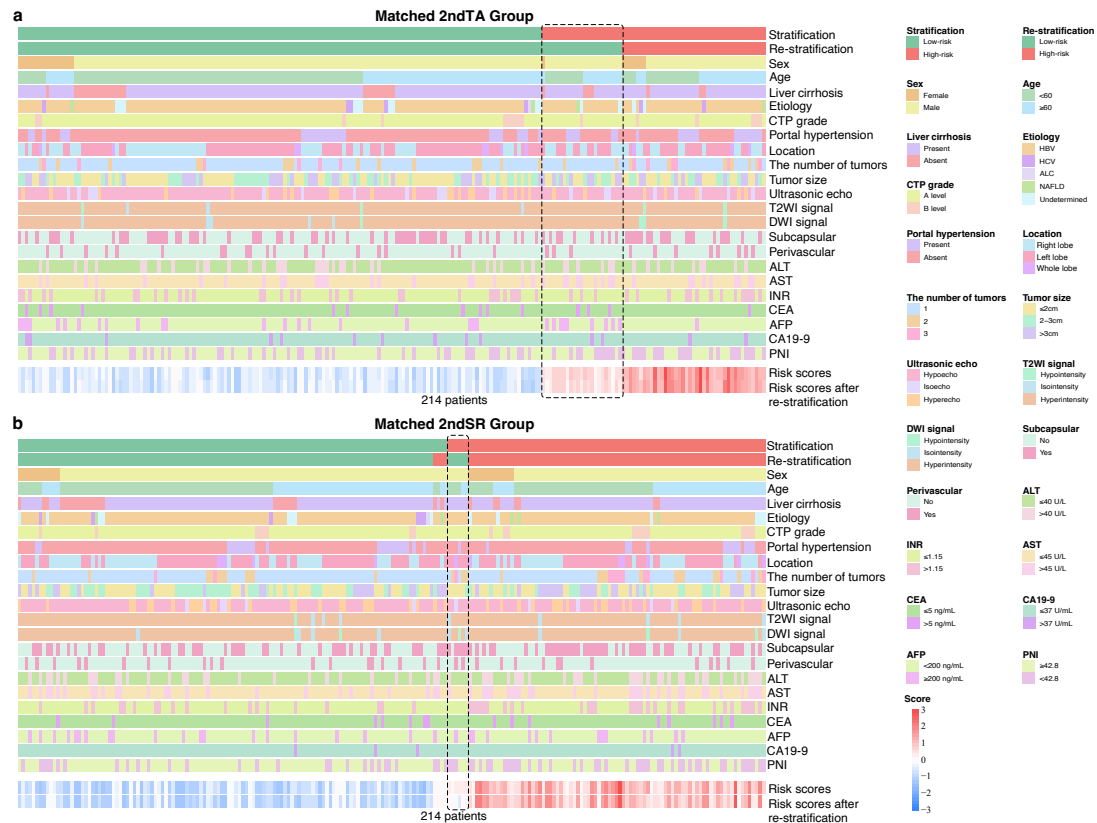


Figure S16. Individual Clinical Variables and Risk Scores in Matched 2ndTA and 2ndSR Groups

The heatmap illustrates the relationship between individual clinical variables and risk scores in the Matched 2ndTA group (a) and Matched 2ndSR group (b). Black dashed boxes highlight the clinical variables and risk scores of patients who were initially categorized as high-risk but were downgraded to low-risk after re-stratification. The color bar represents the normalized continuous values of the risk scores.]

8. [Current staging systems, such as the BCLC, are not designed for recurrent disease. Therefore, it is not surprising that they perform less effectively than the proposed models. This limitation should be explicitly discussed or mentioned in the manuscript.]

Our Response:

Thank you for highlighting this critical point. We agree that current staging systems, including the BCLC system, were not specifically designed for rHCC.

To the best of our knowledge, there are no staging systems or guidelines explicitly tailored for rHCC. While it is acknowledged that this creates an inherent limitation when comparing

our HEROVision model to existing staging systems, such comparisons were included solely to demonstrate the effectiveness of our proposed model. The results of our study show that the AI-based HEROVision model provides superior prognostic performance for patients with early-stage rHCC. This highlights the critical role of HEROVision in addressing a significant gap in clinical practice by offering a tool capable of delivering more precise prognostic information for this specific population.

We have acknowledged this limitation explicitly in the revised manuscript's discussion section. We emphasized that while current staging systems were not specifically designed for recurrent disease and there are no rHCC-specific guidelines, this comparison was intended to validate the effectiveness and potential clinical utility of HEROVision in addressing this critical gap.

Revised text:

(page 9/Discussion section of the revised manuscript)

[... While current staging systems were not specifically designed for recurrent disease and there are no rHCC-specific guidelines, this comparison was intended to validate the effectiveness and potential clinical utility of HEROVision in addressing this critical gap.]

9. [Minor comment: The term "curative-intent" is preferable to "curative" for accuracy and consistency in describing the therapeutic goals.]

Our Response:

Thank you for your suggestion. We have revised the manuscript to use the term “curative-intent” consistently, ensuring greater accuracy and alignment with the therapeutic goals described.

Revised text:

(page 1/Title page of the revised manuscript)

[Vision Transformer-Based Biomarker can Optimize Curative-Intent Treatment for Patients with Recurrent Hepatocellular Carcinoma]

(page 9/Discussion section of the revised manuscript)

[Optimizing curative-intent treatment for patients with rHCC who are at high risk of postoperative early-stage progression is essential for personalized decision-making and improved survival outcomes¹⁵ ... Therefore, our study aimed to develop a multimodal-based DL model, named HEROVision, that can optimize the personalized curative-intent treatment selection between TA and RHR for rHCC patients who initially underwent SR as their first treatment.]

(page 13/Conclusions section of the revised manuscript)

[In summary, by analyzing 1,605 patients with HCC who underwent curative-intent TA and SR, we developed and validated a DL based HEROVision model. It integrates preoperative US and MRI, as well as clinical variables, to predict tumor progression and optimize personalized curative-intent treatment for early-stage rHCC. HEROVision holds promise to complement the existing clinical guidelines and enhance patient care in the management of early-stage rHCC.]

10. [Additionally, CEUS is rarely used as a surveillance tool in clinical practice, whereas CT is more commonly employed. This discrepancy should be addressed in the manuscript, as the lack of discussion on CT may limit the applicability of the study's findings.]

Our Response:

Thank you for raising this important point. We acknowledge that CT is a commonly used imaging modality for liver cancer surveillance in clinical practice.

According to the guidelines of the China Primary Liver Cancer Study Group[1, 2] and the Japanese Society of Hepatology[3], ultrasound is the most commonly used initial imaging modality for liver screening. CT and MRI are preferred for further diagnostic confirmation in cases with abnormal findings on ultrasound and serum AFP testing. However, the guidelines also emphasize that MRI outperforms dynamic contrast-enhanced CT in detecting small nodules (especially ≤ 2.0 cm in diameter), identifying liver cancer, and evaluating tumor invasion and metastasis. Furthermore, due to concerns about radiation exposure, CT is not routinely recommended for follow-up, particularly in China. Instead, CEUS and MRI are preferred in many cases.

We plan to include CT imaging data in future studies as we accumulate more cases, enhancing HEROVision's accuracy and generalizability. This limitation has been explicitly acknowledged in the revised manuscript under the discussion section.

References:

- [1] J. Zhou *et al.*, "Guidelines for Diagnosis and Treatment of Primary Liver Cancer in China (2017 Edition)," (in eng), *Liver Cancer*, vol. 7, no. 3, pp. 235-260, Sep 2018, doi: 10.1159/000488035.
- [2] G. O. o. N. H. Commission, "Standard for diagnosis and treatment of primary liver cancer (2022 edition)," *J Clin Hepatol*, vol. 38, no. 2, pp. 288-303, 2022, doi: 10.3969/j.issn.1001-5256.2022.02.009.
- [3] M. Kudo, O. Matsui, N. Izumi, H. Iijima, M. Kadoya, and Y. Imai, "Surveillance and diagnostic algorithm for hepatocellular carcinoma proposed by the Liver Cancer Study Group of Japan: 2014 update," (in eng), *Oncology*, vol. 87 Suppl 1, pp. 7-21, 2014, doi: 10.1159/000368141.

Revised text:

(page 13/Discussion section of the revised manuscript)

[... Finally, as more cases are accumulated, the use of CT—a commonly employed imaging modality for surveillance in clinical practice—will enhance HEROVision’s accuracy and generalizability.]

Response to the Comments of All Reviewers

Revision Report

Manuscript Title: Vision Transformer-Based Biomarker can Optimize Curative-Intent Treatment for Patients with Recurrent Hepatocellular Carcinoma

Manuscript Number: NCOMMS-24-36298B

Date of this Revision: February 11, 2025

Thank you very much for kindly giving us an opportunity to revise our manuscript. We have carefully revised the manuscript in accordance with the Editorial Policy Checklist and the reviewers' comments. We hope the revision has met the standard of *Nature Communications*. Please feel free to contact us if any further concerns should be addressed.

Table of Contents:

●	REVIEWER COMMENTS	2
●	Response to Reviewer 1	3
●	Response to Reviewer 5	4

● **REVIEWER COMMENTS**

Reviewer #1 (Remarks to the Author):

My comments and suggestions were already addressed in the previous version. In this revision, the authors have also satisfactorily addressed the pending issues raised by Reviewer 2.

I have no further comments.

Reviewer #5 (Remarks to the Author):

I would like to thank the authors for their thoughtful and thorough responses to all my questions and comments during the review process. Their detailed explanations and the revisions made have significantly enhanced the clarity and robustness of the manuscript. I greatly appreciate the time and effort invested in addressing these points. While some limitations persist, the authors have acknowledged these and provided a thoughtful discussion of the most important ones.

● **Response to Reviewer 1**

1. [My comments and suggestions were already addressed in the previous version. In this revision, the authors have also satisfactorily addressed the pending issues raised by Reviewer
2. I have no further comments.]

Our Response:

Thank you very much for your recognition of our work. We greatly appreciate your positive feedback and are glad to hear that the revisions meet your expectations.

- **Response to Reviewer 5**

1. [I would like to thank the authors for their thoughtful and thorough responses to all my questions and comments during the review process. Their detailed explanations and the revisions made have significantly enhanced the clarity and robustness of the manuscript. I greatly appreciate the time and effort invested in addressing these points. While some limitations persist, the authors have acknowledged these and provided a thoughtful discussion of the most important ones.]

Our Response:

Thank you once again for your kind recognition of our work. We sincerely appreciate your thoughtful and constructive feedback, which has been invaluable in improving the clarity and quality of the manuscript.